Summary of the Phase-Field Method for Fracture

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While studying the phase-field method (PFM) for fracture, I summarized various aspects of phase-field research in recent years.

I. Classical Phase-Field Theory of Fracture

1.1 AT2 Phase-Field Model

Consider an elastic domain $\Omega \subseteq \mathbb{R}^{\dim}$ subjected to a body force $\mathbf{b}$, with the Dirichlet boundary $\partial\Omega_u$ subjected to a prescribed displacement $\bar{\mathbf{u}}$ and the Neumann boundary $\partial\Omega_t$ subjected to a stress $\mathbf{t}$. The crack surface Γ is represented diffusely by the phase field d:

$$A_\Gamma = \int_\Omega \gamma(d, \nabla d) d\Omega \tag{1.1}$$

The total potential energy of the system comprises the strain energy, crack surface energy, and potential energy of external forces:

$$\Pi = W_e + W_d + W_p = \int_\Omega (g(d)\Psi^+(\varepsilon) + \Psi^-(\varepsilon)) d\Omega + \int_\Omega G_c \gamma(d, \nabla d) d\Omega - \int_\Omega \mathbf{b}\bullet\mathbf{u} d\Omega - \int_{\partial\Omega_t} \mathbf{t}\bullet\mathbf{u} dS \tag{1.2}$$

Here, g(d) is the degradation function, representing the decay of strain energy density as damage increases. $\Psi^{\pm}(\varepsilon)$ denotes the strain energies that open or close the crack in the undamaged material, and $\gamma(\mathrm{d}, \nabla \mathrm{d})$ is the crack density function, characterizing the area of the diffuse crack.

For the classical AT2 phase-field model, the degradation function and crack density function take the forms

$$g(d) = (1-d)^2 \tag{1.3}$$

$$\gamma(d, \nabla d) = \frac{1}{2l}(d^2 + l^2 \nabla d \bullet \nabla d) \tag{1.4}$$

From the first-order stability condition and the second law of thermodynamics, we obtain

$$\delta\Pi \geq 0 \tag{1.5}$$

$$\dot{W}_d \geq 0 \tag{1.6}$$

The admissible spaces for the displacement field and phase field are:

$$\mathbb{Q}_{\mathbf{u}} = \left\{ \mathbf{u} \in H^1\left(\Omega; \mathbb{R}^{\dim}\right) \middle| \mathbf{u} = \bar{\mathbf{u}} \text{ on } \partial\Omega_u \right\} \tag{1.7}$$

$$\mathbb{Q}_d = \left\{ d \in H^1\left(\Omega; [0,1]\right) \middle| \dot{d}\left(\mathbf{x}\right) \geq 0, \forall \mathbf{x} \in \Omega \right\} \tag{1.8}$$

The admissible spaces for their variations are then:

$$\bar{\mathbb{Q}}_{\mathbf{u}} = \left\{ \mathbf{u} \in H^1\left(\Omega; \mathbb{R}^{\dim}\right) \middle| \mathbf{u} = 0 \text{ on } \partial\Omega_u \right\} \tag{1.9}$$

$$\bar{\mathbb{Q}}_d = \left\{ w \in H^1\left(\Omega\right) \middle| \exists h > 0,\ \forall \zeta \in (0, h],\ s.t.\ d + \zeta w \geq d_{i-1},\ \forall \mathbf{x} \in \Omega \right\} \tag{1.10}$$

First, solve the equilibrium equation with the phase field fixed. Assuming $\delta\mathbf{u} \neq 0,\ \delta d = 0$, the first variation of the total energy of the system can be written as:

$$\begin{aligned} \delta\Pi &= \int_\Omega \left(g(d)\frac{\partial\Psi^+}{\partial\boldsymbol{\varepsilon}} + \frac{\partial\Psi^-}{\partial\boldsymbol{\varepsilon}}\right) : \delta\boldsymbol{\varepsilon} d\Omega - \int_\Omega \mathbf{b}\bullet\delta\mathbf{u} d\Omega - \int_{\partial\Omega_u} \mathbf{t}\bullet\delta\mathbf{u} dS \\ &= -\int_\Omega (\nabla\bullet\boldsymbol{\sigma} + \mathbf{b})\bullet\delta\mathbf{u} d\Omega + \int_{\partial\Omega_u} (\boldsymbol{\sigma}\bullet\mathbf{n} - \mathbf{t})\bullet\delta\mathbf{u} d\Omega \geq 0 \end{aligned} \tag{1.11}$$

Since $\delta\mathbf{u}$ is arbitrary and can be positive or negative, for the variational inequality to hold in all cases, we have:

$$\begin{aligned} &\nabla\bullet\boldsymbol{\sigma} + \mathbf{b} = 0, \quad \text{on } \Omega \\ &\boldsymbol{\sigma}\bullet\mathbf{n} = \mathbf{t}, \quad \text{on } \partial\Omega_t \\ &\mathbf{u} = \bar{\mathbf{u}}, \quad \text{on } \partial\Omega_u \end{aligned} \tag{1.12}$$

This yields the equilibrium equation.

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Next, solve the phase-field equation with the displacement field fixed. Assuming $\delta\mathbf{u}=0,\ \delta d\neq 0$, the first variation of the total energy of the system can be written as:

$$\delta\Pi=\int_\Omega g'(d)\Psi^+(\varepsilon)\delta d\Omega+Gc\int_\Omega(\frac{1}{l}d\delta d-l\Delta d\bullet\delta d)d\Omega\geq 0 \tag{1.13}$$

If $d>d_{i-1}$, since ζ can be arbitrarily small, there always exists a phase-field variation $\delta d<0$ or $\delta d<0$ that satisfies the irreversibility constraint. Therefore, the variational inequality is equivalent to:

$$\begin{aligned} &g'(d)\Psi^+(\varepsilon)+\frac{G_c}{l}(d-l^2\Delta d)=0,\ \text{on }\Omega\\ &\nabla d\bullet\mathbf{n}=0,\ \text{on }\partial\Omega \end{aligned} \tag{1.14}$$

If $d=d_{i-1}$, phase-field irreversibility requires the phase-field variation to satisfy $\delta d\geq 0$. Therefore, the variational inequality is equivalent to:

$$\begin{aligned} &g'(d)\Psi^+(\varepsilon)+\frac{G_c}{l}(d-l^2\Delta d)\geq 0,\ \text{on }\Omega\\ &\nabla d\bullet\mathbf{n}=0,\ \text{on }\partial\Omega \end{aligned} \tag{1.15}$$

Considering these two cases, the phase-field evolution equation can be expressed as:

$$\left[g'(d)\Psi^+(\varepsilon)+\frac{G_c}{l}(d-l^2\Delta d)\right]\dot{d}=0,\ g'(d)\Psi^+(\varepsilon)+\frac{G_c}{l}(d-l^2\Delta d)\geq 0,\dot{d}\geq 0 \tag{1.16}$$

1.2 General Form of the Phase-Field Equation

In general, the displacement-field equation has undergone few modifications and is obtained by varying the strain energy with respect to displacement. Many more approaches have been explored for the phase-field equation. The following is the general form of the phase-field equation given by Jian-Ying Wu

$$\frac{G_c}{c_0 l_0}[\alpha'(d)-2l_0^2\Delta d]=-g'(d)\Psi(\varepsilon) \tag{1.17}$$

Here, α(d) is the crack geometric function, and the crack density function becomes

$$\gamma(d,\nabla d)=\frac{1}{c_0 l_0}[\alpha(d)+l_0^2\nabla d\bullet\nabla d] \tag{1.18}$$

II. Crack Geometric Function

2.1 Scaling Coefficient c0[1]

The scaling coefficient c0 is obtained through crack surface regularization. When the crack is fully softened, i.e., $\sigma=0$, the phase-field equation is

$$\alpha'(d)-2l_0^2\Delta d=0 \tag{2.1}$$

Integrating with respect to the coordinate x and using integration by parts gives

$$\alpha(d)-l_0^2\,|\nabla d|^2=0 \tag{2.2}$$

The crack density function is

$$\gamma(d,\nabla d)=\frac{2}{c_0 l_0}\alpha(d) \tag{2.3}$$

Assuming that the crack is planar and perpendicular to the x-axis, projecting the crack density function onto the x-axis gives

$$\int_\Omega\gamma(d,\nabla d)d\Omega=\frac{4}{c_0 l_0}\int_0^D\alpha(d)dx\bullet A_s=A_d \tag{2.4}$$

Here, A_s is the cross-sectional area and A_d is the crack area calculated from the damage field. Thus, A_s=A_d, and hence

$$c_0=4\int_0^D\alpha(d)\frac{1}{l_0}dx=4\int_0^D\alpha(d)\frac{1}{l_0}\frac{1}{|\nabla d|}dd=4\int_0^1\sqrt{\alpha(\beta)}d\beta \tag{2.5}$$

For phase-field models such as AT1 and AT2 that do not account for a softening relation, calibration of c0 is necessary to ensure normalization of the fracture energy. For cohesive phase-field models, however, the softening law is derived or fitted through other coefficients. Changes in c0 therefore do not alter the softening law or the fracture energy (the area enclosed by the softening curve and the x-axis), so c0 need not

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be calibrated.

2.2 Crack Geometric Function[2]

The crack geometric function determines the crack configuration. In one dimension, the crack profile function satisfies the following equation

$$\begin{aligned} &\alpha'(d) - 2l_0^2 \Delta d = 0 \\ &d(0) = 1 \\ &d'(x^*)\,|_{d(x^*)=0} = 0 \end{aligned} \tag{2.6}$$

In the AT2 phase-field model, $\alpha(d) = d^2$, and the above equation becomes

$$d - l_0^2 d'' = 0 \tag{2.7}$$

The crack profile function is obtained as:

$$d(x) = e^{\frac{-|x|}{l_0}} \tag{2.8}$$

Similarly, different crack geometric functions are used in the AT1 model and cohesive phase-field model

$$\alpha(d) = \xi d + (1-\xi) d^2 \tag{2.9}$$

Table 1 Generic geometric crack function $\alpha(\phi)$ and the resulting crack phase-field $\phi(x)$.

$\alpha(\phi)$	ξ	c_0	$\phi(x)$
ϕ^2	0	2	$\exp\left(-\frac{\lvert x\rvert}{l_0}\right)$
ϕ	1	8/3	$\left(1-\frac{\lvert x\rvert}{2l_0}\right)^2$
$2\phi - \phi^2$	2	π	$1-\sin\left(\frac{\lvert x\rvert}{l_0}\right)$

Figure 1. Crack geometric functions in different phase-field models

These crack profiles are shown below

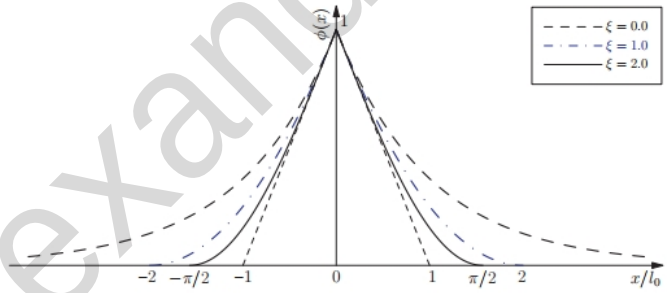


Fig. 11 Crack phase-field $\phi(x)$ for different geometric crack functions $\alpha(\phi)$.

Figure 2. Crack profiles in different models

These models have different half-bandwidths D: ∞ for AT2, $2l_0$ for AT1, and l_0 for the cohesive phase-field model.

Various crack geometric functions have been proposed by many researchers

Table 2 Commonly adopted geometric crack functions $\alpha(\phi)$.

$\alpha(\phi)$	c_0	Authors
ϕ	8/3	Pham et al. (2011)
ϕ^2	2.0	Bourdin et al. (2000)
$1-(1-\phi)^{p/2}$	–	Pham et al. (2011)
$1-(1-\phi)^2$	π	Alessi et al. (2015)
$\xi\phi + (1-\xi)\phi^2$	–	Wu (2017)
$16\phi^2(1-\phi)^2$	8/3	Karma et al. (2001)

Figure 3. Crack geometric functions

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2.3 Analysis of Singular Solutions

From Eq. (1.17), the general form of the phase-field equation can be obtained:

$$\frac{G_c}{c_0 l_0}[\alpha'(d) - 2l_0^2 \Delta d] = -g'(d)\Psi(\varepsilon) \tag{2.10}$$

In general, phase-field problems are often solved using incremental loading, with a very small initial load. At this stage, the strain energy density is 0, the phase field is uniform, and the phase-field gradient is 0. The phase-field equation can be written as:

$$\alpha'(d) = 0 \tag{2.11}$$

The solution of Eq. (2.11) is a singular solution of the phase-field equation. This singular solution deserves attention because it provides an initial value for the phase field in practical simulations. An unfavorable singular solution often prevents the iterations from proceeding.

The singular solution depends only on the crack geometric function. In the AT2 model, $\alpha(d) = d^2$, and the singular solution is d=0. This means that the initial phase field is a uniform field of 0, and the phase field naturally evolves from 0 to 1 without any restrictions. In the AT1 model, $\alpha(d) = d$, and the equation has no singular solution. This leads to a particularly unfavorable situation: the phase field initially blows up numerically, preventing the next iteration, while forcibly constraining d to the range from 0 to 1 prevents the residual from converging to 0. In the unified cohesive phase-field model, $\alpha(d) = 2d - d^2$, and the singular solution is d=1. This means that, under small loading increments, the phase field first converges to 1, indicating complete material damage and likewise preventing correct simulation. Therefore, in practical computations, Jian-Ying Wu constrains the history field to be no less than the strain energy density at the ultimate stress, allowing the initial phase-field value to be 0.

III. Degradation Function

The degradation function characterizes the degradation of material strain energy due to damage and has the following properties

$$\begin{cases} g(0) = 1 \\ g(1) = 0 \\ g'(0) < 0 \\ g'(1) = 0 \end{cases} \tag{3.1}$$

The degradation function g(d) decreases monotonically; d=0 denotes undamaged material, and d=1 denotes fully damaged material

Various degradation functions have been proposed by many researchers

Table 3 Commonly selected energetic degradation functions $g(\phi)$.

$g(\phi)$	Authors
$(1-\phi)^2$	Bourdin et al. (2000)
$3(1-\phi)^2 - 2(1-\phi)^3$	Karma et al. (2001)
$(3-s)(1-\phi)^2 - (2-s)(1-\phi)^3$	Borden et al. (2016)
$4(1-\phi)^3 - 3(1-\phi)^4$	Kuhn et al. (2015)
$\frac{(1-\phi)^2}{(1-\phi)^2 + Q(\phi)}$, $Q(\phi) = a_1\phi + a_1 p\phi^2$	Lorentz, Cuvilliez, and Kazymyrenko (2012), Lorentz, Cuvilliez, and Kazymyrenko (2011), and Lorentz (2017)
$\frac{(1-\phi)^2}{1 + (k-1)Q(\phi)}$, $Q(\phi) = 1 - (1-\phi)^2$	Alessi et al. (2015)
$\frac{(1-\phi)^p}{(1-\phi)^p + Q(\phi)}$, $Q(\phi) = a_1\phi + a_1 a_2\phi^2 + a_1 a_2 a_3\phi^3$	Wu (2017)

Figure 4. Degradation functions[2]

3.1 Control of Crack Initiation Strength[1,3]

In a one-dimensional bar, stress is uniformly distributed along the bar, giving the strain

$$\varepsilon(d) = \frac{\hat{\sigma}}{g(d)E} \tag{3.2}$$

$\hat{\sigma}$ is the effective stress. Substituting (3.2) into (1.17) gives

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$$\frac{G_c}{c_0 l_0}[\alpha'(d) - 2l_0^2 \Delta d] = -g'(d)\frac{E\varepsilon^2}{2} = -\frac{g'(d)}{[g(d)]^2}\frac{\hat{\sigma}^2}{2E} \tag{3.3}$$

At both ends of the bar, the phase field and its gradient are 0. Integrating the above equation gives

$$\begin{aligned} &\hat{\sigma}^2 \Phi(d) - \frac{2EG_c}{c_0 l_0}[\alpha(d) - (l_0 \nabla d)^2] = 0 \\ &\Phi(d) = \int -\frac{g'(d)}{[g(d)]^2} dd = \frac{1}{g(d)} - 1 \\ &\Phi(0) = 0 \end{aligned} \tag{3.4}$$

Assume that the maximum phase-field value occurs at x=0. Then $\nabla d(x=0) = 0$. Writing $d(0) = d^*$, we have

$$\hat{\sigma}(d^*) = \sqrt{\frac{2EG_c}{c_0 l_0}\frac{\alpha(d^*)}{\Phi(d^*)}} \tag{3.5}$$

The crack initiation strength of the material is

$$\sigma_0 = \lim_{d^* \to 0} \hat{\sigma} = \sqrt{\frac{2EG_c}{c_0 l_0}\frac{\alpha(0)}{\Phi(0)}} = \sqrt{\frac{2EG_c}{c_0 l_0}\frac{\alpha'(0)}{\Phi'(0)}} \tag{3.6}$$

In the classical AT2 phase-field model, since $\alpha(d) = d^2$, a linear phase-field equation can be obtained, but the crack initiation strength is 0, meaning that damage begins as soon as loading starts.

For the AT1 phase-field model, $\alpha(d) = d$, and the calculated crack initiation strength is $\sigma_c = \sqrt{\frac{3EG_c}{8l_0}}$

3.2 Control of Ultimate Strength[2,4]

In a one-dimensional bar, assuming that the strain and phase field are uniform, Eq. (1.17) becomes

$$\frac{G_c}{c_0 l_0}\alpha'(d) = -g'(d)\Psi(\varepsilon) = -\frac{g'(d)}{[g(d)]^2}\frac{\hat{\sigma}^2}{2E} \tag{3.7}$$

The equivalent stress is obtained as

$$\hat{\sigma} = \sqrt{-\alpha'(d)\frac{2EG_c}{c_0 l_0}\frac{[g(d)]^2}{g'(d)}} \tag{3.8}$$

The strain is

$$\varepsilon(d) = \frac{\hat{\sigma}}{g(d)E} = \sqrt{-\frac{2G_c}{c_0 l_0 E}\frac{\alpha'(d)}{g'(d)}} \tag{3.9}$$

Combining (3.8) and (3.9) also gives the tangent stiffness expressed in terms of d; see Ref. [4] for the detailed procedure

Combining (3.8) and (3.9) gives the stress-strain curve of the material. Its peak is the ultimate strength of the material, i.e.,

$$\left.\frac{\partial \hat{\sigma}}{\partial \varepsilon}\right|_{\hat{\sigma}=\hat{\sigma}(d^*)} = \left.\frac{\partial \hat{\sigma}}{\partial d}\right|_{d=d^*} = 0 \tag{3.10}$$

d^* satisfies the equation:

$$-\alpha'(d)[g(d)]^2 g''(d) + \alpha''(d)[g(d)]^2 g'(d) + 2\alpha'(d)g(d)[g'(d)]^2 = 0 \tag{3.11}$$

Therefore, the ultimate strength of the material can be obtained

$$\sigma_c = \hat{\sigma}(d^*) = \sqrt{-\alpha'(d^*)\frac{2EG_c}{c_0 l_0}\frac{[g(d^*)]^2}{g'(d^*)}} \tag{3.12}$$

For the AT2 phase-field model, $d^* = \frac{1}{4}$, giving an ultimate strength for the AT2 phase-field model of $\sigma_c = \sqrt{\frac{27EG_c}{256l_0}}$. Since the AT2 phase-field model has a crack initiation strength of 0 and no linear stage, its ultimate strength is reached after damage has already occurred

The AT1 phase-field model, cohesive phase-field model, and similar models have crack initiation strengths other than 0, and Eq. (3.11) gives $d^* = 1$. This indicates that, after damage begins, stress

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gradually decreases as strain increases. The ultimate strength is reached at damage onset, i.e.,

$$\sigma_c = \hat{\sigma}(0) = \sqrt{-\alpha'(0)\frac{2EG_c}{c_0 l_0}\frac{[g(0)]^2}{g'(0)}} \tag{3.13}$$

The ultimate strengths of several phase-field models are given below

Table 5 Peak stress σ_c and the corresponding critical strain ϵ_c for various geometric crack functions $\alpha(\phi)$ and energetic degradation functions $g(\phi)$.

$\alpha(\phi)$	$g(\phi)$	σ_c	ϵ_c
ϕ^2	$(1-\phi)^2$	$\frac{9}{16}\sqrt{\frac{E_0 G_c}{3l_0}}$	$\sqrt{\frac{G_c}{3E_0 l_0}}$
ϕ^2	$3(1-\phi)^2 - 2(1-\phi)^3$	$\frac{27}{250}\sqrt{\frac{30E_0 G_c}{l_0}}$	$\sqrt{\frac{10G_c}{27l_0 E_0}}$
ϕ^2	$4(1-\phi)^3 - 3(1-\phi)^4$	$\frac{128}{243}\sqrt{\frac{2E_0 G_c}{3l_0}}$	$\frac{9}{16}\sqrt{\frac{2G_c}{3l_0 E_0}}$
ϕ	$(1-\phi)^2$	$\sqrt{\frac{3E_0 G_c}{8l_0}}$	$\sqrt{\frac{3G_c}{8E_0 l_0}}$
$2\phi - \phi^2$	$(1-\phi)^2$	$\sqrt{\frac{2E_0 G_c}{\pi l_0}}$	$\sqrt{\frac{2G_c}{\pi E_0 l_0}}$

Figure 5. Ultimate strengths of phase-field models

The ultimate strengths of the AT1 model, cohesive model, and similar models coincide with their crack initiation strengths.

Figure 6 (left) shows the strain-phase-field curves for quadratic, cubic, and quartic degradation functions. The cubic and quartic functions have strain thresholds for increasing damage, whereas the quadratic function does not. Figure 6 (right) shows their stress-strain curves. For the cubic and quartic degradation functions, the slope before the strain threshold is E, the Young's modulus. Just beyond the strain threshold, damage remains small, stiffness degradation is limited, and no obvious change in slope occurs.

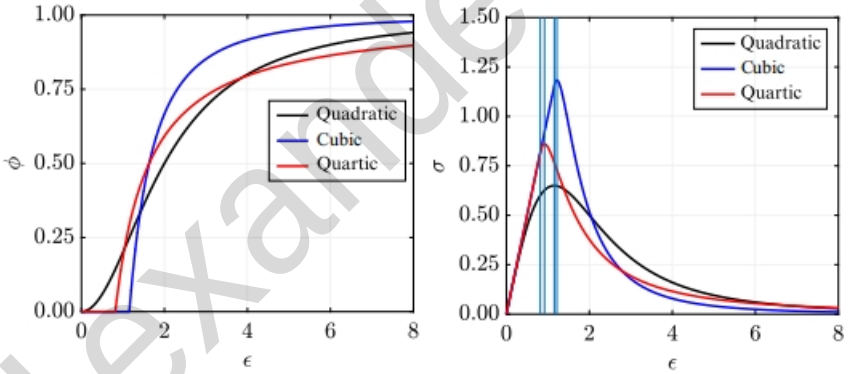


Fig. 20 1-D homogeneous phase-field–strain curves (*left*) and stress–strain curves (*right*): $\alpha = \phi^2$ together with quadratic, cubic, and quartic degradation functions ($l_0 = 0.25$). The *vertical lines* correspond to $\bar{\epsilon}$ and ϵ_σ, respectively.

Figure 6. Strain-phase-field and stress-strain curves

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3.3 Control of Damage Softening Behavior

From (3.9), we obtain

$$\begin{aligned}\frac{d\varepsilon}{dd} &= \sqrt{-\frac{G_c g'(d)}{2Ec_0 l_0 \alpha'(d)}}\frac{\alpha'(d)g''(d) - \alpha''(d)g'(d)}{(g'(d))^2} \\ \frac{d\hat{\sigma}}{dd} &= g'(d)E\varepsilon(d) + g(d)E\frac{d\varepsilon}{dd}\end{aligned} \tag{3.14}$$

The tangent stiffness can be obtained as:

$$\frac{d\hat{\sigma}}{d\varepsilon} = E\frac{g(d)g''(d)\alpha'(d) - 2\alpha'(d)(g'(d))^2 - g(d)g'(d)\alpha''(d)}{\alpha'(d)g''(d) - \alpha''(d)g'(d)} \tag{3.15}$$

For given crack geometric and degradation functions, if there exists $d \in [0,1]$ such that the denominator of (3.15) is 0, the tangent stiffness becomes infinite at that point and snap-back occurs in the stress-strain curve.

To overcome the nonlinearity caused by premature damage in the AT2 phase-field model, Borden[9] et al.

proposed a parameterized cubic degradation function

$$g(d;m)=(m-2)(1-d)^3+(3-m)(1-d)^2 \tag{3.16}$$

Selecting different values of m controls the ultimate strength. This model also has a better linear stage than the AT2 model.

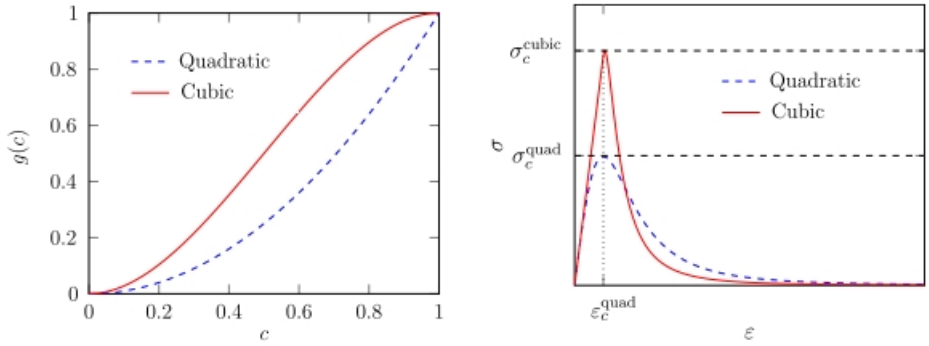


Figure 7. Degradation function and stress-strain relation

Ref. [4] adopts a degradation function of the following form

$$g(d;k,n,w)=(1-w)\frac{1-e^{-k(1-d)^n}}{1-e^{-k}}+wf_c(d) \tag{3.17}$$

When w=0,n=2, Eqs. (3.8) and (3.9) give the stress-strain curve (Figure 6, left)

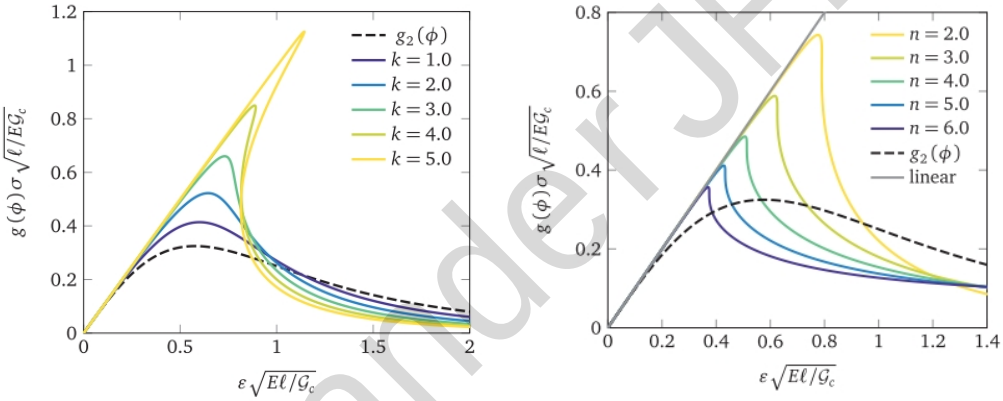


Figure 8. Stress-strain curves

When k is small, the linear segment and stress peak are insufficiently distinct, and the material is quasi-brittle. When k is large, snap-back occurs. Adjusting k gives a stress-strain curve consistent with brittle behavior (Figure 8, right)

The ultimate strength also changes with n, so n can be used to select a suitable degradation function.

w and f_c are correction terms that prevent the linear stage from becoming indistinct when n is very large.

Since the stress extremum in Ref. [4] does not occur at d=0, the stress-strain curve is relatively smooth at the peak rather than turning abruptly.

3.4 Control of Cohesive Softening Behavior[1,2,28]

Building on Section 3.1, stress in a one-dimensional bar is uniform along the bar, and its relation to the maximum damage value d^* is

$$\hat{\sigma}(d^*)=\sqrt{\frac{2EG_c}{c_0 l_0}\frac{\alpha(d^*)}{\Phi(d^*)}} \tag{3.18}$$

From Eq. (3.4), we obtain

$$\frac{d(d)}{dx}=-\frac{1}{l_0}\sqrt{\alpha(d)-\frac{c_0 l_0}{2EG_c}\Phi(d)\hat{\sigma}^2}=-\frac{1}{l_0}\sqrt{\alpha(d)-\frac{\alpha(d^*)}{\Phi(d^*)}\Phi(d)} \tag{3.19}$$

The relation between the damage half-bandwidth D and the maximum damage value d^* can be obtained

$$D(d^*)=\int_0^{d^*}\frac{dx}{d(d)}d(d)=l_0\int_0^{d^*}\frac{1}{\sqrt{\alpha(d)-\frac{\alpha(d^*)}{\Phi(d^*)}\Phi(d)}}d(d) \tag{3.20}$$

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When $d^* = 1$, the half-bandwidth D calculated from Eq. (3.20) is the same as the result in Section 2.2.

The displacement induced by damage is

$$\begin{aligned}\hat{u}(d^*)&=2\int_0^D[\varepsilon(d)-\frac{\hat{\sigma}}{E}]dx=\frac{2\hat{\sigma}}{E}\int_0^D[\frac{1}{g(d)}-1]dx\\&=-\frac{2\hat{\sigma}(d^*)}{E}\int_0^{d^*}\Phi(d)\frac{dx}{d(d)}d(d)\\&=2\sqrt{\frac{2G_cl_0}{Ec_0}}\int_0^{d^*}\sqrt{\frac{\alpha(d^*)}{\Phi(d^*)\alpha(d)-\Phi(d)\alpha(d^*)}}\Phi(d)\,d(d)\end{aligned}\tag{3.21}$$

Combining (3.18) and (3.21) gives the equivalent softening relation of the phase-field model

The unified cohesive phase-field model proposed by Jian-Ying Wu[1] provides specific crack geometric and degradation functions. By setting three parameters in the degradation function, various cohesive models can be simulated.

$$\begin{aligned}\alpha(d)&=2d-d^2\\g(d)&=\frac{(1-d)^p}{(1-d)^p+Q(d)}\\Q(d)&=a_1d+a_1a_2d^2+a_1a_2a_3d^2\end{aligned}\tag{3.22}$$

In the unified cohesive phase-field model[1], parameter a_1 is determined by the ultimate strength (Eq. (3.13)), while parameters p, a_2, and a_3 are determined by optimally fitting sample points on the corresponding softening curves, yielding degradation functions consistent with various theoretical softening laws.

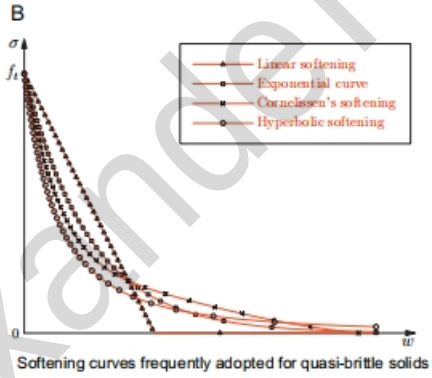


Figure 9. Equivalent softening curves

Note: The area enclosed by the softening curve and the coordinate axes is G_c

In recent years, another cohesive phase-field model derived entirely mathematically has been proposed[28]. It uses a Volterra integral equation of the first kind to provide a theoretical solution for the degradation function.

As in the preceding derivation, in one dimension the model yields a relation among stress $\hat{\sigma}$, maximum phase field d^*, and crack opening displacement $\hat{u}$. Two equations are needed to fully determine their relation. In Ref. [28], the relation between stress and maximum phase field is chosen as:

$$\hat{\sigma}(d^*)=\sqrt{\frac{2EG_c}{c_0l_0}\frac{\alpha(d^*)}{\Phi(d^*)}}=\left(1-d^*\right)\sqrt{\frac{2EG_c}{c_0l_0}}\tag{3.23}$$

This is consistent with the definition of true stress based on the concept of continuity in damage mechanics, although the phase field does not have the same physical meaning as continuity.

We can then define:

$$\hat{u}(d^*)=2\sqrt{\frac{2G_cl_0}{Ec_0}}\int_0^{d^*}\sqrt{\frac{\alpha(d^*)}{\Phi(d^*)\alpha(d)-\Phi(d)\alpha(d^*)}}\Phi(d)d(d)=2\sqrt{\frac{2G_cl_0}{Ec_0}}s\left(d^*\right)\tag{3.24}$$

where:

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$$s(d^*)=\int_0^{d^*}\sqrt{\frac{\Phi(d)\left(1-d^*\right)^2}{(1-d)^2-\left(1-d^*\right)^2}}d(d) \tag{3.25}$$

The Volterra integral equation of the first kind gives:

$$\Phi(d)=\frac{1}{\pi^2}\left(\frac{d}{d\left(d^*\right)}\int_0^d\frac{s\left(d^*\right)}{\sqrt{\left(1-d^*\right)^2-(1-d)^2}}d(d^*)\right)^2 \tag{3.26}$$

For a given softening curve (the $\hat{u}$-$\hat{\sigma}$ relation), combining it with Eq. (3.18) readily gives the $\hat{u}$-d^* relation and thus an expression for $s\left(d^*\right)$. Substitution into Eqs. (3.21) and (3.4) then yields the explicit degradation function (see Ref. [28] for details).

In this model, the geometric function $\alpha(d)$ cannot be chosen arbitrarily; it is determined by Eq. (3.18). Therefore, different softening curves yield different phase-field cross-sectional profiles, as shown below, but all damage half-bandwidths are finite.

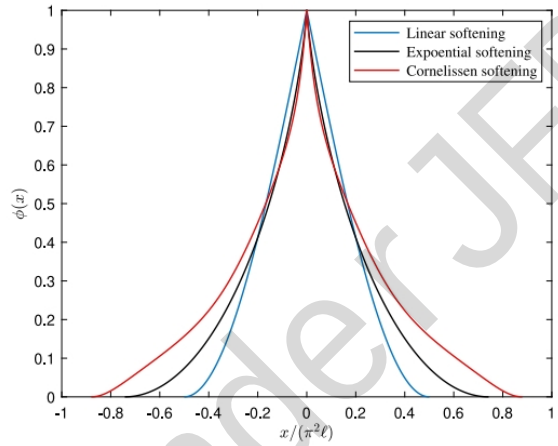


Figure 10. Phase-field cross-sectional profiles for different softening curves (crack profiles)

3.5 Control of Phase-Field Width

In the classical AT1 and AT2 phase-field models, ultimate strength depends on phase-field width (Figure 5). Given material parameters determine the characteristic width lch, and the corresponding characteristic width must be used when simulating material fracture with classical phase-field models.

$$l_{ch}=\frac{EG_c}{\sigma_c^2} \tag{3.27}$$

For concrete, the characteristic width is generally of the order of decimeters. For concrete members of meter-scale dimensions, this characteristic width cannot adequately represent the actual crack width, preventing simulation of fracture behavior[5]. The unified cohesive phase-field model proposed by Jian-Ying Wu decouples phase-field width from ultimate strength through parameter a1 in the degradation function. However, restrictions imposed by Newton-Raphson iteration require the phase-field width used in practice to be smaller than the characteristic width (see Section 5.7). When the phase-field width is below the threshold, different widths give very similar ultimate loads for macroscopic structures of the same size, indicating decoupling of phase-field width from ultimate stress. However, if the macroscopic structure is too large, the maximum allowable phase-field width remains small relative to the structure, requiring mesh refinement and leaving the crack propagation path indistinct.

For common metals, such as aluminum alloys, the characteristic length is of the order of micrometers. For typical macroscopic aluminum alloy components, a phase-field width smaller than the characteristic width inevitably requires an extremely fine mesh and a huge computational cost. A family of crack geometric and degradation functions is therefore needed to allow simulations with phase-field widths greater than the characteristic width.

Lo[8] et al. proposed a new exponential degradation function based on the classical quadratic crack geometric function

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$$\alpha(d) = d$$
$$g(d;s) = s[1-(\frac{s-1}{s})^{d^2}] \tag{3.28}$$

Adjusting parameter s allows a larger phase-field width without changing the ultimate strength. From (3.12), the relation between the actual and original phase-field widths is

$$l_p = (s-1)\ln(\frac{s}{s-1})l_0 \tag{3.29}$$

Although its stress-strain curve exhibits snap-back and it cannot adequately simulate cohesive softening behavior, these issues, which warrant future improvement, have little effect on practical simulations of brittle material behavior.

This model also decouples phase-field width from ultimate stress, but differs from PF-CZM. When a macroscopic structure is scaled proportionally, the phase-field width is scaled by the same factor, and parameter s is adjusted to maintain the same ultimate stress before and after scaling[17]. For precracked cases with pronounced stress concentrations, the simulated ultimate load scales with the square root of the scale factor, consistent with linear elastic fracture mechanics.

For the same macroscopic structure, however, choosing different phase-field widths and adjusting s to keep the ultimate strength constant has little effect on the crack propagation path but does not yield similar ultimate loads. Thus, this decoupling appears to fail when the ratio of macroscopic structural size to phase-field width changes.

An earlier approach to decoupling phase-field width from ultimate strength adds an energy threshold: cracks do not propagate when the energy driving phase-field evolution is below this threshold; see Ref. [34]. However, this approach does not consider a cohesive model and lacks variational consistency. New decoupling approaches have also been proposed in recent years; see Chapter XI.

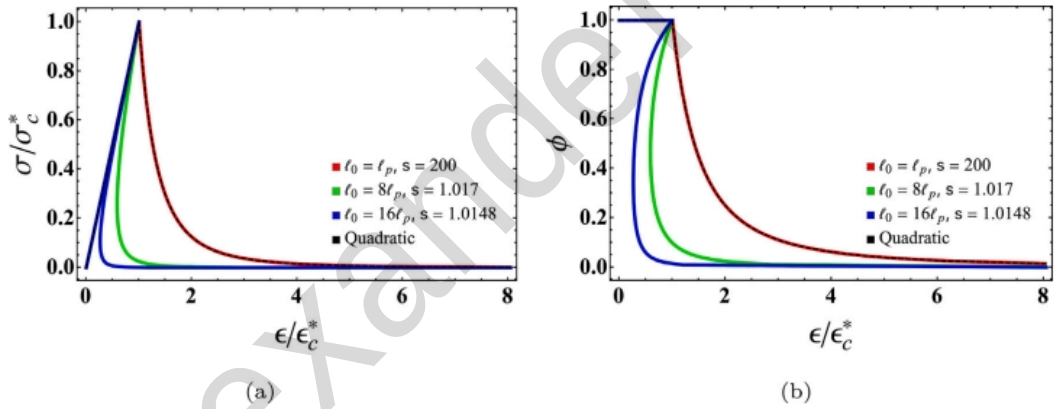


Figure 11. Stress-strain and damage-strain curves

3.6 Half-Bandwidth

Although phase-field width controls crack width, the actual damage band width in a simulation is still of interest. In a one-dimensional bar, assume uniform stress and a maximum phase-field value at x=0, with ∇d(x = 0) = 0. Writing d(0) = d*, from (3.4) we obtain

$$\frac{dd}{dx} = -\frac{1}{l_0}\sqrt{\alpha(d) - \frac{\alpha(d^*)}{\Phi(d^*)}\Phi(d)} \tag{3.30}$$

When the maximum phase-field value

$$D(d^*) = -\int_0^{d^*}\frac{dx}{dd}dd = l_0\int_0^{d^*}\frac{1}{\sqrt{\alpha(d) - \frac{\alpha(d^*)}{\Phi(d^*)}\Phi(d)}}dd \tag{3.31}$$

For the unified cohesive phase-field model, the relation between damage half-bandwidth and maximum damage is shown below. The damage half-bandwidth increases as damage increases.

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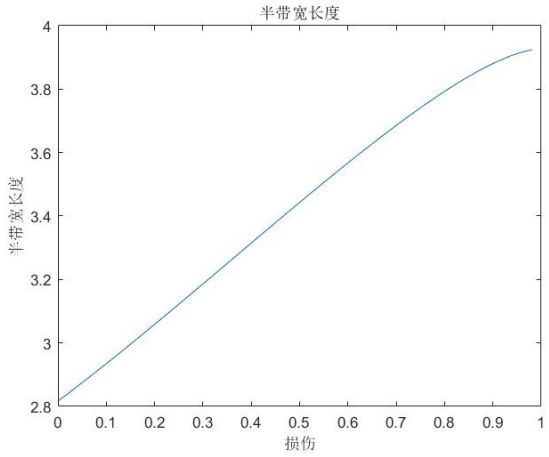


Figure 12. Damage half-bandwidth of the unified cohesive phase-field model

For the phase-field model proposed in (3.28), the relation between damage half-bandwidth and maximum damage is shown below. In both cases, the half-bandwidth decreases as damage increases. This contradicts damage irreversibility and may also cause computational instability.

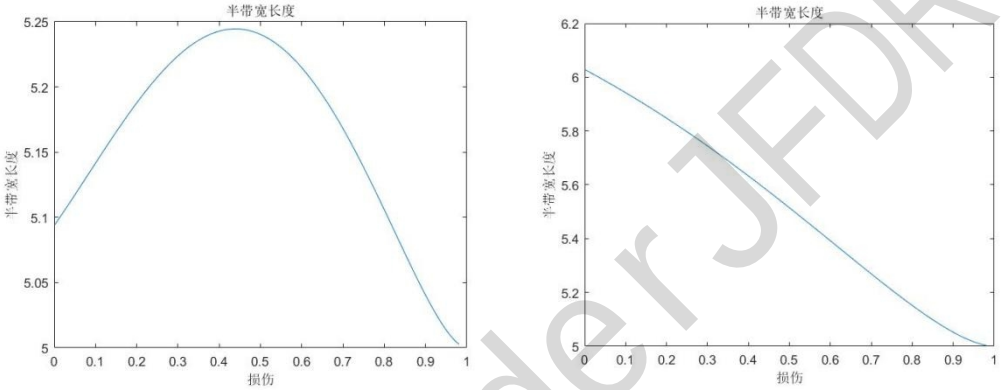


Figure 13. Damage half-bandwidth of the (3.28) phase-field model (left: s=1.1; right: s=1.5)

3.7 Stability Analysis of Solutions

Sections 3.1 and 3.3 above correspond to homogeneous solutions of the phase-field equation, while Sections 3.2 and 3.4 correspond to localized solutions. Both readily satisfy first-order stability of the energy variation. This section discusses second-order stability of phase-field solutions in one dimension. Although Refs. [42] and [47] discuss the stability of homogeneous solutions for their respective phase-field models, a standard framework for discussing homogeneous or localized solutions of the same model is still lacking. Here, the author attempts to analyze the stability of both types of solutions for pfczm.

The total energy of the system is:

$$\Pi = W_e + W_d + W_p = \int_\Omega (g(d)\Psi^+(\varepsilon) + \Psi^-(\varepsilon))d\Omega + \int_\Omega G_c\gamma(d,\nabla d)d\Omega - \int_\Omega \mathbf{b}\bullet\mathbf{u}d\Omega - \int_{\partial\Omega_t} \mathbf{t}\bullet\mathbf{u}dS \tag{3.32}$$

In one dimension, assume a bar of length L with its left end $x=0$ fixed and its right end $x=L$ subjected to a displacement $\bar{u}$. Its energy simplifies to:

$$\Pi(u,d) = \int_0^L L(u,d)\,dx = \int_0^L \frac{g(d)E}{2}(u')^2 + \frac{G_c}{c_0 l_0}\left[\alpha(d) + l_0^2(d')^2\right]dx \tag{3.33}$$

Assuming that the solution is $\{u^*, d^*\}$, the second variation of the energy functional along $\{\hat{u}, \hat{d}\}$ is:

$$\begin{aligned}\delta^2\Pi(u^*,d^*;\hat{u},\hat{d}) &= \int_0^L \frac{\partial^2 L}{\partial u'^2}\hat{u}'^2 + 2\frac{\partial^2 L}{\partial u'\partial d}\hat{u}'\hat{d} + \frac{\partial^2 L}{\partial d^2}\hat{d}^2 + \frac{\partial^2 L}{\partial d'^2}\hat{d}'^2 dx \\ &= \int_0^L g(d^*)E\hat{\varepsilon}^2 + 2g'(d^*)E\varepsilon^*\hat{\varepsilon}\hat{d} + \frac{g''(d^*)E}{2}\varepsilon^{*2}\hat{d}^2 + \frac{G_c}{c_0 l_0}\left[\alpha''(d^*)\hat{d}^2 + 2l_0^2\hat{d}'^2\right]dx\end{aligned} \tag{3.34}$$

If $\{u^*, d^*\}$ is a stable solution, then for every admissible $\{\hat{u}, \hat{d}\}$, we have $\delta^2\Pi(u^*,d^*;\hat{u},\hat{d}) \geq 0$.

The stability of the two solutions is discussed below:

(1) Homogeneous solution

When $d^* = const \in (0,1)$, according to Eq. (3.9), the homogeneous solution is:

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$$\varepsilon^* = \sqrt{-\frac{2G_c}{c_0 l_0 E}\frac{\alpha'(d^*)}{g'(d^*)}},\ d^* = const \tag{3.35}$$

To prove that the homogeneous solution is unstable, it suffices to find a perturbation direction $\left\{\hat{u},\hat{d}\right\}$ such that $\delta^2\Pi\left(u^*,d^*;\hat{u},\hat{d}\right)<0$. This perturbation must satisfy the fixed boundary conditions. We assume a positive phase-field increment, i.e., $d^{(i)} > d^{(i-1)}$, so the phase-field perturbation may be negative. Here, we choose the perturbation direction:

$$\hat{u} = \mu\frac{L}{\pi}\sin\left(\frac{\pi x}{L}\right),\ \hat{d} = \lambda\cos\left(\frac{\pi x}{L}\right) \tag{3.36}$$

Here, λ and μ are arbitrarily small positive numbers. Substituting into the second variation gives:

$$\begin{aligned}
&\delta^2\Pi\left(u^*,d^*;\hat{u},\hat{d}\right) = \int_0^L g(d^*)E\mu^2\cos^2\left(\frac{\pi x}{L}\right) + 2\lambda\mu g'(d^*)E\varepsilon^*\cos^2\left(\frac{\pi x}{L}\right)dx \\
&+\int_0^L \frac{g''(d^*)E}{2}\varepsilon^{*2}\lambda^2\cos^2\left(\frac{\pi x}{L}\right) + \frac{G_c}{c_0 l_0}\left[\alpha''(d^*)\lambda^2\cos^2\left(\frac{\pi x}{L}\right) + 2l_0^2\lambda^2\frac{\pi^2}{L^2}\sin^2\left(\frac{\pi x}{L}\right)\right]dx \\
&= \frac{g(d^*)EL}{2}\mu^2 + \lambda\mu g'(d^*)EL\varepsilon^* + \frac{g''(d^*)EL}{4}\varepsilon^{*2}\lambda^2 + \frac{\lambda^2 G_c}{c_0 l_0}\left(\frac{\alpha''(d^*)L}{2} + \frac{\pi^2 l_0^2}{L}\right)
\end{aligned} \tag{3.37}$$

Let $\tau = \frac{\lambda}{\mu}$; then $\delta^2\Pi\left(u^*,d^*;\hat{u},\hat{d}\right)<0$ is equivalent to:

$$\left[\frac{G_c}{c_0 l_0}\left(\frac{\alpha''(d^*)L}{2} + \frac{\pi^2 l_0^2}{L}\right) + \frac{g''(d^*)EL}{4}\varepsilon^{*2}\right]\tau^2 + g'(d^*)EL\varepsilon^*\tau + \frac{g(d^*)EL}{2} < 0 \tag{3.38}$$

In general, at the critical state, $\varepsilon^* = \frac{\sigma_{cr}}{E}$, $d^* = 0$, and for the degradation function used in pfczm (Eq. (3.22)), we have:

$$g(0) = 1,\ g'(0) = -a_1,\ g''(0) = 2a_1(a_1 - p - a_2) \tag{3.39}$$

Equation (3.38) is then equivalent to:

$$\left[\frac{G_c}{c_0 l_0}\left(-1 + \frac{\pi^2 l_0^2}{L^2}\right) + \frac{a_1(a_1 - p - a_2)}{2E}\sigma_{cr}^2\right]\tau^2 - a_1\sigma_{cr}\tau + \frac{E}{2} < 0 \tag{3.40}$$

If:

$$\frac{G_c}{c_0 l_0}\left(-L + \frac{\pi^2 l_0^2}{L}\right) + \frac{a_1(a_1 - p - a_2)L}{2E}\sigma_{cr}^2 < 0 \tag{3.41}$$

then there necessarily exists a combination of perturbations satisfying $\delta^2\Pi\left(u^*,d^*;\hat{u},\hat{d}\right)<0$, and the homogeneous solution is unstable. Defining the characteristic length $l_{ch} = \frac{EG_c}{\sigma_{cr}^2}$, the above inequality is equivalent to:

$$\frac{l_{ch}}{c_0 l_0}\left(1 - \frac{\pi^2 l_0^2}{L^2}\right) > \frac{a_1(a_1 - p - a_2)}{2} \tag{3.42}$$

Since $a_1 = \frac{4EG_c}{\pi l_0 \sigma_{cr}^2} = \frac{4l_{ch}}{c_0 l_0}$, the above inequality is equivalent to:

$$\frac{1}{2} - \frac{\pi^2 l_0^2}{2L^2} + p + a_2 > \frac{4l_{ch}}{c_0 l_0} \tag{3.43}$$

When the bar length is much greater than the phase-field width, i.e., $L \gg l_0$, and when $l_0 > \frac{4l_{ch}}{c_0\left(\frac{1}{2} + p + a_2\right)}$, a combination of perturbations satisfying $\delta^2\Pi\left(u^*,d^*;\hat{u},\hat{d}\right)<0$ necessarily exists,

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making the homogeneous solution unstable. However, positive definiteness of the iteration matrix requires $l_0 \le \frac{4l_{ch}}{c_0\left(\frac{1}{2}+p+a_2\right)}$ (Eq. (5.34)), which means that the quadratic coefficient in Eq. (3.40) must be positive.

We define Δ as follows:

$$\Delta = a_1^2\sigma_{cr}^2 - 2E\left[\frac{G_c}{c_0 l_0}\left(-1+\frac{\pi^2 l_0^2}{L^2}\right)+\frac{a_1\left(a_1-p-a_2\right)}{2E}\sigma_{cr}^2\right] \tag{3.44}$$

Even if the quadratic coefficient is positive, the linear coefficient is negative. If $\Delta > 0$, an admissible solution with lower energy can still be constructed, i.e.,

$$L > \frac{\pi l_0}{\sqrt{1+2p+2a_2}} \tag{3.45}$$

Therefore, for sufficiently large bar length L, the homogeneous solution is always unstable.

(2) Localized solution

Assuming a maximum phase-field value $d_m \in (0,1)$ and a bar length greater than the phase-field bandwidth, according to Eqs. (3.2), (3.5), and (3.30), the localized solution is:

$$\varepsilon^* = \frac{1}{g(d^*)E}\sqrt{\frac{2EG_c}{c_0 l_0}\frac{\alpha(d_m)}{\Phi(d_m)}},\ \frac{dd^*}{dx} = -\frac{1}{l_0}\sqrt{\alpha(d^*)-\frac{\alpha(d_m)}{\Phi(d_m)}\Phi(d^*)} \tag{3.46}$$

This provides a differential equation for d^*, which can be solved numerically. No analytical solution is provided here; only the analytical approach is presented.

After solving for $\{u^*, d^*\}$, the Legendre-Hadamard condition and coercivity of the second variation provide a sufficient condition for $\{u^*, d^*\}$ to be a minimum, namely:

$$\begin{bmatrix} g(d^*)E & 0 \\ 0 & \frac{2G_c l_0}{c_0} \end{bmatrix}$$

is positive definite, and there exists $\mu > 0$ such that for every $\{\hat{u}, \hat{d}\}$ we have $\delta^2\Pi\left(u^*, d^*; \hat{u}, \hat{d}\right) \ge \mu\int_0^L \|\hat{u}\|_{H^1}^2 + \|\hat{d}\|_{H^1}^2 dx$, then $\{u^*, d^*\}$ is a stable solution.

IV. Energy Decomposition Methods

Tension and compression affect crack evolution differently. During phase-field evolution, strain energy induced by tensile loading degrades, whereas that induced by compression does not. Phase-field models using this energy decomposition are called anisotropic phase-field models. They effectively address the problem of cracks forming under compression in isotropic phase-field models. These anisotropic phase-field models, however, cannot distinguish between Mode I and Mode II fracture.

4.1 Spectral Decomposition

Miehe[15] proposed an energy decomposition based on spectral decomposition. The strain is first transformed to obtain its principal components. For an isotropic linear elastic material, the strain energy density is

$$\Psi_0 = \frac{1}{2}\lambda(\varepsilon_1+\varepsilon_2+\varepsilon_3)^2 + \mu(\varepsilon_1^2+\varepsilon_2^2+\varepsilon_3^2) \tag{4.1}$$

Here, ε_i is the ith eigenvalue of the strain tensor, i.e., the principal strain. The strain energy density is decomposed into tensile and compressive parts

$$\begin{aligned} \Psi^{\pm} &= \frac{1}{2}\lambda\left\langle\varepsilon_1+\varepsilon_2+\varepsilon_3\right\rangle_{\pm}^2 + \mu\left(\left\langle\varepsilon_1\right\rangle_{\pm}^2+\left\langle\varepsilon_2\right\rangle_{\pm}^2+\left\langle\varepsilon_3\right\rangle_{\pm}^2\right) \\ \left\langle x\right\rangle_{\pm} &= (x\pm|x|)/2 \end{aligned} \tag{4.2}$$

The degraded energy is:

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$$\Psi = g(d)\Psi^{+} + \Psi^{-} \tag{4.3}$$

The stress tensor can be written as:

$$\begin{aligned}\sigma &= \frac{\partial \Psi(\varepsilon)}{\partial \varepsilon} = g(d)\frac{\partial \Psi^{+}(\varepsilon)}{\partial \varepsilon} + \frac{\partial \Psi^{-}(\varepsilon)}{\partial \varepsilon} \\ &= g(d)\left[\frac{1}{2}\lambda\frac{\partial\langle tr(\varepsilon)\rangle_{+}^{2}}{\partial \varepsilon} + \mu\frac{\partial\sum_{m}^{3}\langle \varepsilon_m\rangle_{+}^{2}}{\partial \varepsilon}\right] + \frac{1}{2}\lambda\frac{\partial\langle tr(\varepsilon)\rangle_{-}^{2}}{\partial \varepsilon} + \mu\frac{\partial\sum_{m}^{3}\langle \varepsilon_m\rangle_{-}^{2}}{\partial \varepsilon}\end{aligned} \tag{4.4}$$

By the chain rule, we have:

$$\frac{\partial\langle tr(\boldsymbol{\varepsilon})\rangle_{\pm}^{2}}{\partial \boldsymbol{\varepsilon}} = \frac{\partial\langle tr(\boldsymbol{\varepsilon})\rangle_{\pm}^{2}}{\partial\langle tr(\boldsymbol{\varepsilon})\rangle_{\pm}}\frac{\partial\langle tr(\boldsymbol{\varepsilon})\rangle_{\pm}}{\partial tr(\boldsymbol{\varepsilon})}\frac{\partial tr(\boldsymbol{\varepsilon})}{\partial \boldsymbol{\varepsilon}} = 2\langle tr(\boldsymbol{\varepsilon})\rangle_{\pm} H(\pm\langle tr(\boldsymbol{\varepsilon})\rangle_{\pm})\mathbf{I} \tag{4.5}$$

$$\frac{\partial\sum_{m}^{3}\langle \varepsilon_m\rangle_{\pm}^{2}}{\partial \varepsilon} = \frac{\partial\sum_{m}^{3}\langle \varepsilon_m\rangle_{\pm}^{2}}{\partial\langle \varepsilon_n\rangle_{\pm}}\frac{\partial\langle \varepsilon_n\rangle_{\pm}}{\partial \varepsilon} = 2\langle \varepsilon_n\rangle_{\pm}\frac{\partial\langle \varepsilon_n\rangle_{\pm}}{\partial \varepsilon} \tag{4.6}$$

Define tensor M as:

$$\mathbf{M}^{n} = \mathbf{n}^{n} \otimes \mathbf{n}^{n} \tag{4.7}$$

Here, $\mathbf{n}^{n}$ is the nth eigenvector of $\boldsymbol{\varepsilon}$, i.e., $\boldsymbol{\varepsilon} = \sum_{n=1}^{3}\varepsilon_n \mathbf{n}^{n} \otimes \mathbf{n}^{n}$. Then:

$$\frac{\partial\langle \varepsilon_n\rangle_{\pm}}{\partial \varepsilon} = H\left(\pm\langle \varepsilon_n\rangle_{\pm}\right)\mathbf{M}^{n} \tag{4.8}$$

Therefore, the stress can be expressed as:

$$\begin{aligned}\sigma &= g(d)\left[\lambda\langle tr(\boldsymbol{\varepsilon})\rangle_{+} H(\langle tr(\boldsymbol{\varepsilon})\rangle_{+})\mathbf{I} + 2\mu\langle \varepsilon_n\rangle_{+} H\left(\langle \varepsilon_n\rangle_{+}\right)\mathbf{M}^{n}\right] + \lambda\langle tr(\boldsymbol{\varepsilon})\rangle_{-} H(-\langle tr(\boldsymbol{\varepsilon})\rangle_{-})\mathbf{I} + 2\mu\langle \varepsilon_n\rangle_{-} H\left(-\langle \varepsilon_n\rangle_{-}\right)\mathbf{M}^{n} \\ &= g(d)\left[\lambda\langle tr(\boldsymbol{\varepsilon})\rangle_{+} H(\langle tr(\boldsymbol{\varepsilon})\rangle_{+})\mathbf{I} + 2\mu\varepsilon^{+}\right] + \lambda\langle tr(\boldsymbol{\varepsilon})\rangle_{-} H(-\langle tr(\boldsymbol{\varepsilon})\rangle_{-})\mathbf{I} + 2\mu\boldsymbol{\varepsilon}^{-}\end{aligned} \tag{4.9}$$

The equivalent stiffness matrix can be written as:

$$\begin{aligned}C_{ijkl} = \frac{\partial \sigma_{ij}}{\partial \varepsilon_{kl}} &= g(d)\left[\lambda\frac{\partial\langle tr(\boldsymbol{\varepsilon})\rangle_{+}}{\partial \varepsilon_{kl}} H(\langle tr(\boldsymbol{\varepsilon})\rangle_{+})\delta_{ij} + 2\mu\frac{\partial\left(\varepsilon^{+}\right)_{ij}}{\partial \varepsilon_{kl}}\right] + \lambda\frac{\partial\langle tr(\boldsymbol{\varepsilon})\rangle_{-}}{\partial \varepsilon_{kl}} H(-\langle tr(\boldsymbol{\varepsilon})\rangle_{-})\delta_{ij} + 2\mu\frac{\partial\left(\varepsilon^{-}\right)_{ij}}{\partial \varepsilon_{kl}} \\ &= g(d)\left[\lambda H^{2}(\langle tr(\boldsymbol{\varepsilon})\rangle_{+})\delta_{ij}\delta_{kl} + 2\mu\frac{\partial\left(\varepsilon^{+}\right)_{ij}}{\partial \varepsilon_{kl}}\right] + \lambda H^{2}(-\langle tr(\boldsymbol{\varepsilon})\rangle_{-})\delta_{ij}\delta_{kl} + 2\mu\frac{\partial\left(\varepsilon^{-}\right)_{ij}}{\partial \varepsilon_{kl}}\end{aligned} \tag{4.10}$$

We also have:

$$\frac{\partial \varepsilon^{\pm}}{\partial \varepsilon} = \sum_{n=1}^{3}\frac{\partial\langle \varepsilon_n\rangle_{\pm}}{\partial \varepsilon} \otimes \mathbf{M}^{n} + \sum_{n=1}^{3}\langle \varepsilon_n\rangle_{\pm}\frac{\partial \mathbf{M}^{n}}{\partial \varepsilon} \tag{4.11}$$

To derive $\frac{\partial \mathbf{M}^{n}}{\partial \varepsilon}$, we have:

$$\mathbf{I}^{sym} = \frac{\partial \varepsilon}{\partial \varepsilon} = \sum_{n=1}^{3}\frac{\partial \varepsilon_n}{\partial \varepsilon} \otimes \mathbf{M}^{n} + \sum_{n=1}^{3}\varepsilon_n\frac{\partial \mathbf{M}^{n}}{\partial \varepsilon} = \sum_{n=1}^{3}\left(\mathbf{M}^{n} \otimes \mathbf{M}^{n} + \varepsilon_n\frac{\partial \mathbf{M}^{n}}{\partial \varepsilon}\right) \tag{4.12}$$

Here, $\mathbf{I}^{sym}$ is a fourth-order tensor, $\mathbf{I}^{sym} = \delta_{ik}\delta_{jl}$, and:

$$\sum_{n=1}^{3}\mathbf{M}^{n} = \mathbf{I} \tag{4.13}$$

That is:

$$\sum_{n=1}^{3}\frac{\partial \mathbf{M}^{n}}{\partial \varepsilon} = \mathbf{0} \tag{4.14}$$

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In two dimensions, the two equations above form a linear system, and $\frac{\partial \mathbf{M}^n}{\partial \varepsilon}$ can be expressed as:

$$\frac{\partial \mathbf{M}^1}{\partial \varepsilon}=\frac{1}{\varepsilon_1-\varepsilon_2}\left(\mathbf{I}^{sym}-\sum_{n=1}^{3}\mathbf{M}^n\otimes\mathbf{M}^n\right),\quad \frac{\partial \mathbf{M}^2}{\partial \varepsilon}=-\frac{1}{\varepsilon_1-\varepsilon_2}\left(\mathbf{I}^{sym}-\sum_{n=1}^{3}\mathbf{M}^n\otimes\mathbf{M}^n\right) \tag{4.15}$$

To obtain the three-dimensional expression, the following additional expression is needed. From:

$$\boldsymbol{\varepsilon}^2=\sum_{n=1}^{3}\left(\varepsilon_n\right)^2\mathbf{n}^n\otimes\mathbf{n}^n \tag{4.16}$$

we obtain:

$$2\varepsilon\mathbf{I}^{sym}=\frac{\partial\varepsilon^2}{\partial\varepsilon}=\sum_{n=1}^{3}\frac{\partial\left(\varepsilon_n\right)^2}{\partial\varepsilon}\otimes\mathbf{M}^n+\sum_{n=1}^{3}\left(\varepsilon_n\right)^2\frac{\partial\mathbf{M}^n}{\partial\varepsilon}=\sum_{n=1}^{3}\left(2\varepsilon_n\mathbf{M}^n\otimes\mathbf{M}^n+\left(\varepsilon_n\right)^2\frac{\partial\mathbf{M}^n}{\partial\varepsilon}\right) \tag{4.17}$$

where $2\varepsilon\mathbf{I}^{sym}=2\varepsilon_{im}\delta_{km}\delta_{jl}=2\varepsilon_{ik}\delta_{jl}$.

This yields $\frac{\partial\varepsilon^{\pm}}{\partial\varepsilon}$, and hence C_{ijkl}.

Dijk[24] proposed a spectral decomposition for orthotropic materials. Starting from the isotropic strain energy expression:

$$\Psi=\frac{1}{2}\lambda\left[tr\left(\boldsymbol{\varepsilon}\right)\right]^2+\mu tr\left(\boldsymbol{\varepsilon}^2\right)=\Psi_\lambda+\Psi_\mu \tag{4.18}$$

Here, $\boldsymbol{\varepsilon}$ is the strain tensor shown in (4.28). It can be proved that Eqs. (4.1), (4.18), and (4.29) are consistent.

Following the isotropic approach, the strain energy of an anisotropic material can be expressed as:

$$\begin{aligned}&\boldsymbol{\Psi}=\boldsymbol{\Psi}_\lambda+\boldsymbol{\Psi}_\mu\\&\boldsymbol{\Psi}_\lambda=\frac{1}{2}\left(\sqrt{\lambda_{11}}\varepsilon_{11}+\sqrt{\lambda_{22}}\varepsilon_{22}+\sqrt{\lambda_{33}}\varepsilon_{33}\right)^2\\&\boldsymbol{\Psi}_\mu=\left(\sqrt{\mu_{11}}\varepsilon_{11}\right)^2+\left(\sqrt{\mu_{22}}\varepsilon_{22}\right)^2+\left(\sqrt{\mu_{33}}\varepsilon_{33}\right)^2+\left(\sqrt{\mu_{12}}\varepsilon_{12}\right)^2+\left(\sqrt{\mu_{23}}\varepsilon_{23}\right)^2+\left(\sqrt{\mu_{31}}\varepsilon_{31}\right)^2\end{aligned} \tag{4.19}$$

Conventionally, the orthotropic stiffness matrix can be expressed using 9 variables as follows:

$$\mathbf{C}=\begin{bmatrix}\alpha_{11}&\lambda_{12}&\lambda_{31}&&&\\\lambda_{12}&\alpha_{22}&\lambda_{23}&&&\\\lambda_{31}&\lambda_{23}&\alpha_{33}&&&\\&&&\mu_{12}&&\\&&&&\mu_{23}&\\&&&&&\mu_{31}\end{bmatrix} \tag{4.20}$$

The anisotropic stiffness matrix can be expressed using coefficients λ and μ as:

$$\mathbf{C}=\begin{bmatrix}\lambda_{11}+2\mu_{11}&\sqrt{\lambda_{11}\lambda_{22}}&\sqrt{\lambda_{33}\lambda_{11}}&&&\\\sqrt{\lambda_{11}\lambda_{22}}&\lambda_{22}+2\mu_{22}&\sqrt{\lambda_{22}\lambda_{33}}&&&\\\sqrt{\lambda_{33}\lambda_{11}}&\sqrt{\lambda_{22}\lambda_{33}}&\lambda_{33}+2\mu_{33}&&&\\&&&\mu_{12}&&\\&&&&\mu_{23}&\\&&&&&\mu_{31}\end{bmatrix} \tag{4.21}$$

The coefficients satisfy $\mu_{ii}=\frac{1}{2}\left(\alpha_{ii}-\lambda_{ii}\right)$ and $\lambda_{ii}=\frac{\lambda_{ij}\lambda_{ki}}{\lambda_{jk}}$. Similarly, define the variables:

$$\tilde{\boldsymbol{\varepsilon}}_\lambda=\begin{bmatrix}\sqrt{\lambda_{11}}\varepsilon_{11}&\sqrt{\lambda_{12}}\varepsilon_{12}&\sqrt{\lambda_{31}}\varepsilon_{31}\\\sqrt{\lambda_{12}}\varepsilon_{12}&\sqrt{\lambda_{22}}\varepsilon_{22}&\sqrt{\lambda_{23}}\varepsilon_{23}\\\sqrt{\lambda_{31}}\varepsilon_{31}&\sqrt{\lambda_{23}}\varepsilon_{23}&\sqrt{\lambda_{33}}\varepsilon_{33}\end{bmatrix} \tag{4.22}$$

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$$\tilde{\boldsymbol{\varepsilon}}_{\mu}=\begin{bmatrix}\sqrt{\mu_{11}}\varepsilon_{11} & \sqrt{\mu_{12}}\varepsilon_{12} & \sqrt{\mu_{31}}\varepsilon_{31}\\ \sqrt{\mu_{12}}\varepsilon_{12} & \sqrt{\mu_{22}}\varepsilon_{22} & \sqrt{\mu_{23}}\varepsilon_{23}\\ \sqrt{\mu_{31}}\varepsilon_{31} & \sqrt{\mu_{23}}\varepsilon_{23} & \sqrt{\mu_{33}}\varepsilon_{33}\end{bmatrix} \tag{4.23}$$

The strain energy density can be expressed as,

$$\Psi=\frac{1}{2}\left(\tilde{\varepsilon}_{\lambda,11}+\tilde{\varepsilon}_{\lambda,22}+\tilde{\varepsilon}_{\lambda,33}\right)^2+\tilde{\varepsilon}_{\mu,1}^2+\tilde{\varepsilon}_{\mu,2}^2+\tilde{\varepsilon}_{\mu,3}^2 \tag{4.24}$$

Here, $\tilde{\varepsilon}_{\lambda,ii}$ is the element in row i and column i of $\tilde{\boldsymbol{\varepsilon}}_{\lambda}$, and $\tilde{\varepsilon}_{\mu,i}$ is the ith eigenvalue of $\tilde{\varepsilon}_{\mu}$. The positive and negative energies can be decomposed as follows:

$$\Psi^{\pm}=\frac{1}{2}\left\langle\tilde{\varepsilon}_{\lambda,11}+\tilde{\varepsilon}_{\lambda,22}+\tilde{\varepsilon}_{\lambda,33}\right\rangle_{\pm}^2+\left\langle\tilde{\varepsilon}_{\mu,1}\right\rangle_{\pm}^2+\left\langle\tilde{\varepsilon}_{\mu,2}\right\rangle_{\pm}^2+\left\langle\tilde{\varepsilon}_{\mu,3}\right\rangle_{\pm}^2 \tag{4.25}$$

The stiffness matrix for spectral decomposition can be obtained from the definition,

$$C_{ijkl}=\frac{\partial^2\psi}{\partial\varepsilon_{ij}\partial\varepsilon_{kl}}=\frac{\partial\sigma_{ij}}{\partial\varepsilon_{kl}} \tag{4.26}$$

However, obtaining the stiffness matrix is not always straightforward, so spectral decomposition is often implemented together with a hybrid phase-field approach.

4.2 Volumetric-Deviatoric Decomposition

Amor[16] proposed a decomposition into volumetric and deviatoric strain tensors. First, strain is decomposed into volumetric and deviatoric parts

$$\varepsilon=\varepsilon_S+\varepsilon_D \tag{4.27}$$

In component form,

$$\begin{bmatrix}\varepsilon_x & \frac{1}{2}\gamma_{xy} & \frac{1}{2}\gamma_{xz}\\ \frac{1}{2}\gamma_{yx} & \varepsilon_y & \frac{1}{2}\gamma_{yz}\\ \frac{1}{2}\gamma_{zx} & \frac{1}{2}\gamma_{zy} & \varepsilon_z\end{bmatrix}=\begin{bmatrix}\varepsilon_0 & 0 & 0\\ 0 & \varepsilon_0 & 0\\ 0 & 0 & \varepsilon_0\end{bmatrix}+\begin{bmatrix}\varepsilon_x-\varepsilon_0 & \frac{1}{2}\gamma_{xy} & \frac{1}{2}\gamma_{xz}\\ \frac{1}{2}\gamma_{yx} & \varepsilon_y-\varepsilon_0 & \frac{1}{2}\gamma_{yz}\\ \frac{1}{2}\gamma_{zx} & \frac{1}{2}\gamma_{zy} & \varepsilon_z-\varepsilon_0\end{bmatrix} \tag{4.28}$$

$$\varepsilon_0=(\varepsilon_x+\varepsilon_y+\varepsilon_z)/3=tr(\varepsilon)/3$$

The strain energy density is

$$\Psi_0=\frac{1}{2}(\lambda+\frac{2\mu}{3})[tr(\varepsilon)]^2+\mu\varepsilon_D:\varepsilon_D \tag{4.29}$$

The strain energy is decomposed into tensile and compressive parts

$$\begin{aligned}&\Psi=g(d)\Psi^{+}+\Psi^{-}\\ &\Psi^{+}=\frac{1}{2}(\lambda+\frac{2\mu}{3})\left\langle tr(\varepsilon)\right\rangle_{+}^2+\mu\varepsilon_D:\varepsilon_D\\ &\Psi^{-}=\frac{1}{2}(\lambda+\frac{2\mu}{3})\left\langle tr(\varepsilon)\right\rangle_{-}^2\\ &\left\langle x\right\rangle_{\pm}=(x\pm|x|)/2\end{aligned} \tag{4.30}$$

The stress tensor can be written as:

$$\begin{aligned}\sigma&=\frac{\partial\Psi(\varepsilon)}{\partial\varepsilon}=g(d)\frac{\partial\Psi^{+}(\varepsilon)}{\partial\varepsilon}+\frac{\partial\Psi^{-}(\varepsilon)}{\partial\varepsilon}\\ &=g(d)\left[\frac{1}{2}(\lambda+\frac{2\mu}{3})\frac{\partial\left\langle tr(\varepsilon)\right\rangle_{+}^2}{\partial\varepsilon}+\mu\frac{\partial\varepsilon_D:\varepsilon_D}{\partial\varepsilon}\right]+\frac{1}{2}(\lambda+\frac{2\mu}{3})\frac{\partial\left\langle tr(\varepsilon)\right\rangle_{-}^2}{\partial\varepsilon}\end{aligned} \tag{4.31}$$

By the chain rule, we have:

$$\frac{\partial\left\langle tr(\boldsymbol{\varepsilon})\right\rangle_{\pm}^2}{\partial\boldsymbol{\varepsilon}}=\frac{\partial\left\langle tr(\boldsymbol{\varepsilon})\right\rangle_{\pm}^2}{\partial\left\langle tr(\boldsymbol{\varepsilon})\right\rangle_{\pm}}\frac{\partial\left\langle tr(\boldsymbol{\varepsilon})\right\rangle_{\pm}}{\partial tr(\boldsymbol{\varepsilon})}\frac{\partial tr(\boldsymbol{\varepsilon})}{\partial\boldsymbol{\varepsilon}}=2\left\langle tr(\boldsymbol{\varepsilon})\right\rangle_{\pm}H(\pm\left\langle tr(\boldsymbol{\varepsilon})\right\rangle_{\pm})\mathbf{I} \tag{4.32}$$

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$$\frac{\partial \varepsilon_D : \varepsilon_D}{\partial \varepsilon} = 2(\varepsilon_D)_{mn} \frac{\partial (\varepsilon_D)_{mn}}{\partial \varepsilon_{ij}} \tag{4.33}$$

Therefore, the stress can be expressed as:

$$\sigma = g(d)\left[(\lambda + \frac{2\mu}{3}) \langle tr(\boldsymbol{\varepsilon}) \rangle_+ H(\langle tr(\boldsymbol{\varepsilon}) \rangle_+) \mathbf{I} + 2\mu (\varepsilon_D)_{mn} \frac{\partial (\varepsilon_D)_{mn}}{\partial \varepsilon} \right] + (\lambda + \frac{2\mu}{3}) \langle tr(\boldsymbol{\varepsilon}) \rangle_- H(-\langle tr(\boldsymbol{\varepsilon}) \rangle_-) \mathbf{I} \tag{4.34}$$

The equivalent stiffness matrix can be written as:

$$C_{ijkl} = \frac{\partial \sigma_{ij}}{\partial \varepsilon_{kl}} = g(d)\left[(\lambda + \frac{2\mu}{3}) \frac{\partial \langle tr(\boldsymbol{\varepsilon}) \rangle_+}{\partial \varepsilon_{kl}} H(\langle tr(\boldsymbol{\varepsilon}) \rangle_+) \delta_{ij} + 2\mu \frac{\partial (\varepsilon_D)_{mn}}{\partial \varepsilon_{kl}} \frac{\partial (\varepsilon_D)_{mn}}{\partial \varepsilon_{ij}} \right] + (\lambda + \frac{2\mu}{3}) \frac{\partial \langle tr(\boldsymbol{\varepsilon}) \rangle_-}{\partial \varepsilon_{kl}} H(-\langle tr(\boldsymbol{\varepsilon}) \rangle_-) \delta_{ij}$$

$$= g(d)\left[(\lambda + \frac{2\mu}{3}) H^2(\langle tr(\boldsymbol{\varepsilon}) \rangle_+ \delta_{ij} \delta_{kl} + 2\mu \frac{\partial (\varepsilon_D)_{mn}}{\partial \varepsilon_{kl}} \frac{\partial (\varepsilon_D)_{mn}}{\partial \varepsilon_{ij}} \right] + (\lambda + \frac{2\mu}{3}) H^2 - \langle tr(\boldsymbol{\varepsilon}) \rangle_- \delta_{ij} \delta_{kl} \tag{4.35}$$

The degraded stiffness matrix can be obtained as above. In practice, however, it is commonly replaced by a uniformly degraded stiffness matrix, following the idea of hybrid phase field[14].

Nguyen also applied volumetric-deviatoric decomposition to anisotropic materials[22], with the following energy decomposition:

$$\Psi = \Psi^+ + \Psi^-$$

$$\begin{cases} \Psi = \frac{1}{2} g(d) (\boldsymbol{\varepsilon} : \mathbf{C}_0 : \boldsymbol{\varepsilon}), \ \mathrm{tr}(\boldsymbol{\varepsilon}) \geq 0 \\ \Psi = \frac{1}{2} \boldsymbol{\varepsilon}_S : \mathbf{C}_0 : \boldsymbol{\varepsilon}_S + g(d) (\boldsymbol{\varepsilon}_D : \mathbf{C}_0 : \boldsymbol{\varepsilon}_D), \ \mathrm{tr}(\boldsymbol{\varepsilon}) < 0 \end{cases} \tag{4.36}$$

This approach is questionable because, for anisotropic materials, $\Psi = \Psi^0$ is not satisfied when d=0, whereas it is satisfied for isotropic materials, in which case the formulation reduces to the VD decomposition proposed by Amor.

4.3 Projection-Based Energy Decomposition[20,21]

Positive-negative projection decomposes stress or strain and distinguishes strain energies induced by tensile and compressive loads according to the signs of the stress or strain components. We first introduce the classical positive-negative projection algorithm,

$$\begin{cases} \boldsymbol{\sigma} = \boldsymbol{\sigma}^+ + \boldsymbol{\sigma}^- \\ \boldsymbol{\sigma}^+ \in L^+ \\ \boldsymbol{\sigma}^- \in L^- \\ \boldsymbol{\sigma}^+ : \boldsymbol{\sigma}^- = \boldsymbol{\sigma}^- : \boldsymbol{\sigma}^+ = 0 \end{cases} \tag{4.37}$$

The above expression means that the stress tensor can be decomposed into positive and negative parts. The positive stress tensor is positive semidefinite, and the negative stress tensor is negative semidefinite. Thus, the projection of the positive stress tensor in any direction is nonnegative, and that of the negative stress tensor in any direction is nonpositive. The positive and negative stress tensors are also orthogonal.

This decomposition is implemented as follows:

$$\begin{aligned} \boldsymbol{\sigma}^\pm &= \sum_i \sigma_i^\pm \, \mathbf{p}_i \otimes \mathbf{p}_i \\ \sigma_i^\pm &= \langle \sigma_i \rangle_\pm \end{aligned} \tag{4.38}$$

Here, σ_i is a principal stress and $\mathbf{p}_i$ is its principal direction. In the principal coordinate system, the stress matrix is diagonal and can be decomposed according to the signs of its diagonal elements. The positive and negative stress tensors are then readily shown to be orthogonal. When stress is transformed to any orientation, the transformation matrix is an orthonormal matrix (a Hermitian matrix), so the positive and negative stress tensors remain orthogonal.

Similarly, for an isotropic constitutive relation, positive and negative energies can be defined,

$$\Psi^\pm = \frac{1}{2} \boldsymbol{\sigma}^\pm : \mathbf{S} : \boldsymbol{\sigma}^\pm \tag{4.39}$$

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Here, $\mathbf{S}$ is the compliance matrix. Although the energy decompositions in (4.39) and (4.2) are similar, they do not satisfy $\boldsymbol{\Psi}=\boldsymbol{\Psi}^{+}+\boldsymbol{\Psi}^{-}$ and therefore cannot be used in a phase-field model.

To address this issue, Zhang[23] proposed another definition of positive and negative energies:

$$\Psi^{\pm}=\frac{1}{2}\boldsymbol{\sigma}^{\pm}:\boldsymbol{\varepsilon} \tag{4.40}$$

This satisfies $\boldsymbol{\Psi}=\boldsymbol{\Psi}^{+}+\boldsymbol{\Psi}^{-}$ and is also applicable to anisotropic cases. The degraded stiffness matrix can be obtained from Eq. (4.10).

Building on this approach, Jian-Ying Wu proposed an improved stress tensor decomposition, defined as follows:

$$\begin{cases}\boldsymbol{\sigma}=\boldsymbol{\sigma}^{+}+\boldsymbol{\sigma}^{-}\\ \boldsymbol{\sigma}^{+}\in L^{+}\\ \boldsymbol{\sigma}^{-}\in L^{-}\\ \boldsymbol{\sigma}^{+}:\mathbf{S}:\boldsymbol{\sigma}^{-}=\boldsymbol{\sigma}^{-}:\mathbf{S}:\boldsymbol{\sigma}^{+}=0\end{cases} \tag{4.41}$$

Compared with the preceding approach, Jian-Ying Wu requires the positive and negative stress tensors to be orthogonal with respect to the compliance matrix. Under the definition in (4.41), it is readily proved that $\boldsymbol{\Psi}=\boldsymbol{\Psi}^{+}+\boldsymbol{\Psi}^{-}$ is satisfied. To decompose the stress tensor according to the definition in (4.41), first consider the following stress-strain relations:

$$\begin{aligned}\boldsymbol{\varepsilon}^{\pm}&=\mathbf{S}:\boldsymbol{\sigma}^{\pm}\\ \boldsymbol{\sigma}^{-}:\boldsymbol{\varepsilon}^{+}&=0\end{aligned} \tag{4.42}$$

Positive and negative strain tensors are readily defined from the positive and negative stress tensors, and orthogonality also exists between the positive and negative stress and strain tensors. For isotropic materials, the corresponding principal stress and strain directions can be shown to coincide. Both positive and negative stress and strain tensors can therefore be transformed into the principal coordinate system, where orthogonality is expressed as:

$$\sum_{n}\boldsymbol{\sigma}_{n}^{-}\boldsymbol{\varepsilon}_{n}^{+}=0 \tag{4.43}$$

Because an orthotropic stiffness matrix may contain tension-shear coupling coefficients after coordinate rotation, the principal stress and strain directions do not coincide, and the relation (4.43) cannot be obtained. This is also a limitation of the current theory.

From (4.43), the following equality can be obtained:

$$\begin{cases}\dfrac{1}{E}\sigma_{1}^{-}\left[\sigma_{1}^{+}-v\left(\sigma_{2}^{+}+\sigma_{3}^{+}\right)\right]=0\\ \dfrac{1}{E}\sigma_{2}^{-}\left[\sigma_{2}^{+}-v\left(\sigma_{1}^{+}+\sigma_{3}^{+}\right)\right]=0\\ \dfrac{1}{E}\sigma_{3}^{-}\left[\sigma_{3}^{+}-v\left(\sigma_{1}^{+}+\sigma_{2}^{+}\right)\right]=0\end{cases} \tag{4.44}$$

Combining this with $\sigma_{n}^{+}=\sigma_{n}-\sigma_{n}^{-}$ and first assuming

$$\sigma_{m}>\sigma_{n},\ m<n \tag{4.45}$$

the positive and negative stresses can be defined as,

$$\begin{cases}\sigma_{1}^{+}=\left\langle\sigma_{1}\right\rangle_{+}\\ \sigma_{2}^{+}=\left\langle\max\left(\sigma_{2},\tilde{v}\sigma_{1}\right)\right\rangle_{+}\\ \sigma_{3}^{+}=\left\langle\max\left[\max\left(\sigma_{3},v\left(\sigma_{1}+\sigma_{2}\right)\right),\tilde{v}\sigma_{1}\right]\right\rangle_{+}\end{cases} \tag{4.46}$$

Here, $\tilde{v}=\dfrac{v}{1-v}$ can be defined in two dimensions as:

$$\begin{cases}\sigma_{1}^{+}=\left\langle\sigma_{1}\right\rangle_{+}\\ \sigma_{2}^{+}=\left\langle\max\left(\sigma_{2},v\sigma_{1}\right)\right\rangle_{+}\end{cases} \tag{4.47}$$

See Appendix B of Ref. [21] for the detailed derivation and case-by-case verification of orthogonality.

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The positive and negative stress tensors can be defined from their principal stresses using Eq. (4.38). By symmetry, the damage envelope resulting from this energy decomposition is obtained as shown below

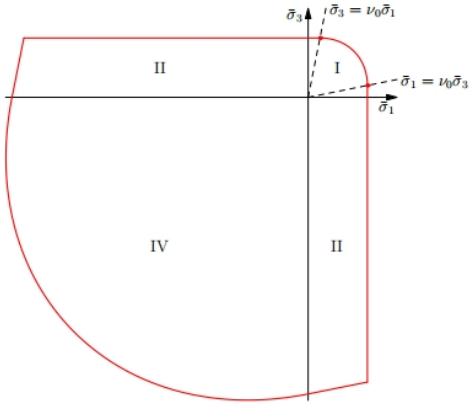


Fig. 2. Elastic domain in the 2-D plane stress condition.

Figure 13. Damage envelope

In phase-field problems, a damage envelope exists only when at least one of the first and third principal stresses is positive; no damage envelope exists in the third quadrant. This is itself a fracture criterion that distinguishes tension from compression through energy decomposition. It is similar to the Tsai-Hill strength criterion for composites, although its underlying principle differs.

Earlier pfczm models[1,19] use a hybrid approach in simulations and thus lack a unified energy functional. This decomposition was proposed to resolve that inconsistency, meaning that the decomposed and degraded stiffness matrix must be used when solving for displacement.

In gradient damage or phase-field models, the relation between effective stress $\bar{\boldsymbol{\sigma}}$ and basic stress $\boldsymbol{\sigma}$ is:

$$\bar{\boldsymbol{\sigma}} = \left(1-d^{+}\right)\boldsymbol{\sigma}^{+} + \left(1-d^{-}\right)\boldsymbol{\sigma}^{-} \tag{4.48}$$

$$\bar{\boldsymbol{\sigma}} = g\left(d\right)\boldsymbol{\sigma}^{+} + \boldsymbol{\sigma}^{-} \tag{4.49}$$

Defining matrix $\mathbf{D}$, the definition of $\mathbf{D}$ is readily obtained from the following relation:

$$\bar{\boldsymbol{\sigma}} = \left(\mathbf{I}-\mathbf{D}\right):\boldsymbol{\sigma} \tag{4.50}$$

Here, $\mathbf{I}$ is the identity matrix. According to Ref. [21], the specific value of $\mathbf{D}$ must be determined case by case. The following relations then hold:

$$\begin{aligned}\bar{\boldsymbol{\sigma}} &= \left(\mathbf{I}-\mathbf{D}\right):\boldsymbol{\sigma} = \left(\mathbf{I}-\mathbf{D}\right):\mathbf{C}_0:\boldsymbol{\varepsilon} \\ &= \left(\mathbf{I}-\mathbf{D}\right):\mathbf{S}_0^{-1}:\mathbf{S}:\bar{\boldsymbol{\sigma}} \\ \bar{\boldsymbol{\sigma}} &= \mathbf{C}:\boldsymbol{\varepsilon} = \mathbf{S}^{-1}:\boldsymbol{\varepsilon}\end{aligned} \tag{4.51}$$

Here, $\mathbf{C}_0$ and $\mathbf{S}_0$ are the undamaged stiffness and compliance matrices, while $\mathbf{C}$ and $\mathbf{S}$ are the corresponding matrices accounting for damage. From Eq. (4.51), the expressions for $\mathbf{C}$ and $\mathbf{S}$ are:

$$\begin{aligned}\mathbf{S} &= \mathbf{S}_0:\left(\mathbf{I}-\mathbf{D}\right)^{-1} \\ \mathbf{C} &= \mathbf{S}^{-1}\end{aligned} \tag{4.52}$$

This approach yields the equivalent stiffness matrix accounting for energy degradation. Although it is expressed in the principal coordinate system, rotation matrices can transform it to the global coordinate system for finite element solution.

4.4 Energy Decomposition for Anisotropic Materials

The projection-based energy decomposition in Section 4.3 is difficult to apply to anisotropic materials. This section mainly introduces orthogonal projection methods suitable for anisotropic materials.

First, define the equivalent strain tensor:

$$\tilde{\boldsymbol{\varepsilon}} = \mathbf{C}^{1/2}:\boldsymbol{\varepsilon} \tag{4.53}$$

The square root of the stiffness matrix, $\mathbf{C}^{1/2}$, can be obtained as follows[25],

$$\begin{aligned}\mathbf{C} &= \sum_i \Lambda_i \boldsymbol{\omega}_i \otimes \boldsymbol{\omega}_i \\ \mathbf{C}^{1/2} &= \sum_i \Lambda_i^{1/2} \boldsymbol{\omega}_i \otimes \boldsymbol{\omega}_i\end{aligned} \tag{4.54}$$

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Here, Λ_i is an eigenvalue of the stiffness matrix, and $\boldsymbol{\omega}_i$ is the corresponding eigenvector. Since the stiffness matrix is symmetric positive definite, it can be diagonalized by a similarity transformation, making the operation in Eq. (4.54) possible. The strain energy density can then be expressed as,

$$\Psi = \frac{1}{2}\boldsymbol{\varepsilon} : \mathbf{C} : \boldsymbol{\varepsilon} = \frac{1}{2}\tilde{\boldsymbol{\varepsilon}} : \tilde{\boldsymbol{\varepsilon}} \tag{4.55}$$

Ref. [26] proposed an energy decomposition algorithm based on orthogonal projection. The positive and negative equivalent strains can be decomposed as follows,

$$\tilde{\boldsymbol{\varepsilon}}^{\pm} = \sum_i \left\langle \tilde{\varepsilon}_i \right\rangle_{\pm} \mathbf{p}_i \otimes \mathbf{p}_i \tag{4.56}$$

Here, $\tilde{\varepsilon}_i$ is the ith eigenvalue of the equivalent strain, and $\mathbf{p}_i$ is the corresponding eigenvector. This decomposition satisfies the following conditions,

$$\begin{cases} \tilde{\boldsymbol{\varepsilon}} = \tilde{\boldsymbol{\varepsilon}}^{+} + \tilde{\boldsymbol{\varepsilon}}^{-} \\ \tilde{\boldsymbol{\varepsilon}}^{+} : \tilde{\boldsymbol{\varepsilon}}^{-} = \tilde{\boldsymbol{\varepsilon}}^{-} : \tilde{\boldsymbol{\varepsilon}}^{+} = 0 \end{cases} \tag{4.57}$$

The energy decomposition also satisfies:

$$\begin{cases} \Psi^{\pm} = \tilde{\boldsymbol{\varepsilon}}^{\pm} : \tilde{\boldsymbol{\varepsilon}}^{\pm} \\ \Psi = \Psi^{+} + \Psi^{-} \end{cases} \tag{4.58}$$

Define the projection operators between the positive and negative equivalent strains and the equivalent strain:

$$\tilde{\boldsymbol{\varepsilon}}^{\pm} = \tilde{\mathbf{P}}^{\pm} : \tilde{\boldsymbol{\varepsilon}} \tag{4.59}$$

The projection operator $\tilde{\mathbf{P}}^{\pm} = f\left(\tilde{\varepsilon}_i, \mathbf{p}_i\right)$ is readily expressed in the principal coordinate system, but must be transformed back using a rotation matrix. Thus, the projection operators between positive and negative strains and strain can be expressed as,

$$\begin{aligned} \boldsymbol{\varepsilon}^{\pm} &= \mathbf{P}^{\pm} : \boldsymbol{\varepsilon} \\ \mathbf{P}^{\pm} &= \mathbf{C}^{-1/2} : \tilde{\mathbf{P}}^{\pm} : \mathbf{C}^{1/2} \end{aligned} \tag{4.60}$$

The strain and equivalent stiffness matrix are defined as,

$$\begin{aligned} \boldsymbol{\sigma}^{\pm} &= \mathbf{C} : \boldsymbol{\varepsilon}^{\pm} = \mathbf{C} : \mathbf{P}^{\pm} : \boldsymbol{\varepsilon} = \mathbf{C}^{1/2} : \tilde{\mathbf{P}}^{\pm} : \mathbf{C}^{1/2} : \boldsymbol{\varepsilon} = \mathbf{C}^{\pm} : \boldsymbol{\varepsilon} \\ \mathbf{C}^{\pm} &= \mathbf{C}^{1/2} : \tilde{\mathbf{P}}^{\pm} : \mathbf{C}^{1/2} \end{aligned} \tag{4.61}$$

The equivalent stiffness matrix can also be obtained from the strain energy density:

$$\begin{aligned} \Psi^{\pm} &= \frac{1}{2}\boldsymbol{\varepsilon}^{\pm} : \mathbf{C} : \boldsymbol{\varepsilon}^{\pm} = \frac{1}{2}\left(\mathbf{P}^{\pm} : \boldsymbol{\varepsilon}\right)^{T} \mathbf{C} \left(\mathbf{P}^{\pm} : \boldsymbol{\varepsilon}\right) = \boldsymbol{\varepsilon}^{\pm} : \left(\mathbf{C}^{1/2} \tilde{\mathbf{P}}^{\pm} \tilde{\mathbf{P}}^{\pm} \mathbf{C}^{1/2}\right) : \boldsymbol{\varepsilon}^{\pm} \\ \mathbf{C}^{\pm} &= \frac{\partial^2 \Psi^{\pm}}{\partial \boldsymbol{\varepsilon}^2} = \mathbf{C}^{1/2} \tilde{\mathbf{P}}^{\pm} \tilde{\mathbf{P}}^{\pm} \mathbf{C}^{1/2} \end{aligned} \tag{4.62}$$

Since the projection operator is idempotent, i.e., $\tilde{\mathbf{P}}^{\pm}\tilde{\mathbf{P}}^{\pm} = \tilde{\mathbf{P}}^{\pm}$, the equivalent stiffness matrices given by Eqs. (4.61) and (4.62) are identical.

Ref. [27] proposed an energy decomposition algorithm combining equivalent strain with volumetric-deviatoric decomposition. First, the equivalent strain is decomposed into volumetric and deviatoric parts,

$$\begin{aligned} \tilde{\boldsymbol{\varepsilon}} &= \tilde{\boldsymbol{\varepsilon}}_S + \tilde{\boldsymbol{\varepsilon}}_D = \begin{bmatrix} \tilde{\varepsilon}_0 \\ \tilde{\varepsilon}_0 \\ 0 \end{bmatrix} + \begin{bmatrix} \tilde{\varepsilon}_x - \tilde{\varepsilon}_0 \\ \tilde{\varepsilon}_y - \tilde{\varepsilon}_0 \\ \tilde{\gamma}_{xy} \end{bmatrix} \\ \tilde{\varepsilon}_0 &= \frac{1}{2} tr\left(\tilde{\boldsymbol{\varepsilon}}\right) \end{aligned} \tag{4.63}$$

Clearly, $\tilde{\boldsymbol{\varepsilon}}_S$ and $\tilde{\boldsymbol{\varepsilon}}_D$ are orthogonal. The positive and negative equivalent strains are defined as,

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$$\begin{cases}\tilde{\boldsymbol{\varepsilon}}^{+}=\left\langle\tilde{\boldsymbol{\varepsilon}}_{S}\right\rangle_{+}+\tilde{\boldsymbol{\varepsilon}}_{D}\\ \tilde{\boldsymbol{\varepsilon}}^{-}=\left\langle\tilde{\boldsymbol{\varepsilon}}_{S}\right\rangle_{-}\end{cases}$$
$$\left\langle\tilde{\boldsymbol{\varepsilon}}_{S}\right\rangle_{\pm}=\begin{bmatrix}\left\langle\tilde{\varepsilon}_{0}\right\rangle_{\pm}\\ \left\langle\tilde{\varepsilon}_{0}\right\rangle_{\pm}\\ 0\end{bmatrix} \tag{4.64}$$
$$\tilde{\boldsymbol{\varepsilon}}=\tilde{\boldsymbol{\varepsilon}}^{+}+\tilde{\boldsymbol{\varepsilon}}^{-}$$

Similarly, the projection operator $\mathbf{P}^{\pm}$ can be defined:

$$\tilde{\boldsymbol{\varepsilon}}^{\pm}=\tilde{\mathbf{P}}^{\pm}:\tilde{\boldsymbol{\varepsilon}} \tag{4.65}$$

The projection operator $\mathbf{P}^{\pm}$ for volumetric-deviatoric decomposition is also idempotent. Following the discussion above, the strain and equivalent stiffness matrix can likewise be expressed as,

$$\begin{aligned}&\boldsymbol{\sigma}^{\pm}=\mathbf{C}:\boldsymbol{\varepsilon}^{\pm}=\mathbf{C}^{1/2}:\tilde{\mathbf{P}}^{\pm}:\mathbf{C}^{1/2}:\boldsymbol{\varepsilon}=\mathbf{C}^{\pm}:\boldsymbol{\varepsilon}\\ &\mathbf{C}^{\pm}=\mathbf{C}^{1/2}:\tilde{\mathbf{P}}^{\pm}:\mathbf{C}^{1/2}\end{aligned} \tag{4.66}$$

4.5 Maximum History Variable

Miehe[15] first introduced the maximum history variable in AT2 phase-field theory. Replacing strain energy with the maximum history variable H gives the phase-field equation,

$$\begin{aligned}&\frac{G_c}{c_0 l_0}[\alpha'(d)-2l_0^2\Delta d]=-g'(d)H\\ &H=\max_{\tau\in[0,t]}\left\{\Psi\left(\varepsilon,\tau\right)\right\}\end{aligned} \tag{4.67}$$

This maximum history variable ensures that the phase-field variable d increases monotonically and does not heal during unloading.

The maximum history variable can also constrain the phase-field variable to the range from 0 to 1. In the AT2 phase-field model, d naturally satisfies this constraint. As shown in Section 2.3, under incremental loading, the initial AT2 phase field is 0, whereas other phase-field models require treatment to ensure an initial value of 0. The PF-CZM model is discussed here.

When the crack geometric function is $\alpha(d)=2d-d^2$, the initial phase field under incremental loading is 1. If we define the history variable H as:

$$H=\max_{\tau\in[0,t]}\left\{-\frac{G_c}{c_0 l_0}\frac{\alpha'(0)}{g'(0)},\Psi\left(\varepsilon,\tau\right)\right\} \tag{4.68}$$

At the beginning of loading, when the load is small, substituting the history variable $H=-\frac{G_c}{c_0 l_0}\frac{\alpha'(0)}{g'(0)}=\frac{2G_c}{\pi l_0 a_1}$ into the phase-field equation readily gives d=0. When the derivative of the degradation function $g'(d)$ decreases monotonically, d increases monotonically with H, allowing the phase-field variable to increase monotonically from 0 to 1. Related details are presented in Ref. [18].

Jian-Ying Wu uses the following history variable in Ref. [5],

$$H=\max_{\tau\in[0,t]}\left\{\frac{f_t^2}{2E},\Psi\left(\varepsilon,\tau\right)\right\} \tag{4.69}$$

Here, f_t is the ultimate material strength. In PF-CZM theory, substituting parameter $a_1=\frac{4EG_c}{\pi f_t^2 l_0}$ into Eq. (4.68) also yields the form of Eq. (4.69).

When the maximum history variable is used to enforce phase-field irreversibility, the following condition must be satisfied:

$$g'\left(d\right)<0,g''\left(d\right)>0 \tag{4.70}$$

V. Solution Methods

Assume that the finite element discretizations of the displacement and phase fields are:

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$$
\begin{aligned}
u(x) &= \sum_I N_I(x) u_I = \mathbf{N}\mathbf{u} \\
\varepsilon(x) &= \nabla u(x) = \sum_I B_I(x) u_I = \mathbf{B}\mathbf{u} \\
d(x) &= \sum_I \bar{N}_I(x) d_I = \bar{\mathbf{N}}\mathbf{d} \\
\nabla d(x) &= \sum_I \bar{B}_I(x) d_I = \bar{\mathbf{B}}\mathbf{d}
\end{aligned}
\tag{5.1}
$$

The Galerkin variational principle gives the residuals:

$$
\begin{aligned}
r^u &= f - \int_\Omega \mathbf{B}^T \sigma d\Omega = 0 \\
r^d &= -\int_\Omega [\bar{\mathbf{N}}^T (g'(d) H + \frac{G_c}{c_0 l_0} \alpha'(d)) + \frac{2 G_c l_0}{c_0} \bar{\mathbf{B}}^T \nabla d] d\Omega = 0 \\
\sigma &= g(d)\sigma^+ + \sigma^-
\end{aligned}
\tag{5.2}
$$

5.1 Monolithic Solution Method

The displacement and phase-field equations form a system of nonlinear partial differential equations that can be solved using Newton iteration. The finite element iteration equations are:

$$
\begin{bmatrix} K^{uu} & K^{ud} \\ K^{du} & K^{dd} \end{bmatrix} \begin{bmatrix} du \\ dd \end{bmatrix} = \begin{bmatrix} r^u \\ r^d \end{bmatrix}
$$

$$
\begin{aligned}
K^{uu} &= -\frac{\partial r^u}{\partial u} = \int_\Omega \mathbf{B}^T \left(\frac{\partial \sigma}{\partial \varepsilon} \right) \mathbf{B} d\Omega \\
K^{ud} &= -\frac{\partial r^u}{\partial d} = \int_\Omega \mathbf{B}^T \left(\frac{\partial \sigma}{\partial d} \right) \bar{\mathbf{N}} d\Omega \\
K^{du} &= -\frac{\partial r^d}{\partial u} = \int_\Omega \bar{\mathbf{N}}^T \left(g'(d) \frac{\partial H}{\partial \varepsilon} \right) \mathbf{B} d\Omega \\
K^{dd} &= -\frac{\partial r^d}{\partial d} = \int_\Omega \left[\bar{\mathbf{N}}^T \left(g''(d) H + \frac{G_c}{c_0 l_0} \alpha''(d) \right) \bar{\mathbf{N}} + \frac{2 G_c l_0}{c_0} \bar{\mathbf{B}}^T \bar{\mathbf{B}} \right] d\Omega
\end{aligned}
\tag{5.3}
$$

This coupled iteration scheme is nonconvex and difficult to converge, motivating the development of a staggered solution strategy

5.2 Staggered Solution Method

For weakly nonlinear systems, a staggered solution method can be used. First proposed by Bourdin[6] et al., its iteration equations are:

$$
\begin{aligned}
K^{uu} du &= r^u \\
K^{dd} dd &= r^d \\
K^{uu} &= -\frac{\partial r^u}{\partial u} = \int_\Omega \mathbf{B}^T \left(\frac{\partial \sigma}{\partial \varepsilon} \right) \mathbf{B} d\Omega \\
K^{dd} &= -\frac{\partial r^d}{\partial d} = \int_\Omega \left[\bar{\mathbf{N}}^T \left(g''(d) H + \frac{G_c}{c_0 l_0} \alpha''(d) \right) \bar{\mathbf{N}} + \frac{2 G_c l_0}{c_0} \bar{\mathbf{B}}^T \bar{\mathbf{B}} \right] d\Omega
\end{aligned}
\tag{5.4}
$$

The staggered method neglects the coupling terms between the displacement and phase fields, making each subproblem convex and providing strong algorithmic robustness.

The staggered method generally uses small loading increments, with multiple iterations in each increment. In each iteration, displacement is first solved using (5.4)-1, followed by the phase field using (5.4)-2. Repeated iterations bring both fields to convergence. Convergence is usually determined using residual or increment thresholds, or only one iteration is performed[5].

5.3 Solution Using the BFGS Algorithm

BFGS is a rank-2 quasi-Newton algorithm. Although it generally requires more iterations to converge than Newton's method, it updates the stiffness matrix in a simple manner, reducing computational cost. Ref.

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[5] describes the BFGS procedure.

For the nonlinear equation:

$$g(z)=0 \tag{5.5}$$

the iteration equation is:

$$\begin{aligned}&\tilde{K}\delta z=\delta g\\&\delta z=z^{(k)}-z^{(k-1)}\\&\delta g=g^{(k)}-g^{(k-1)}\end{aligned} \tag{5.6}$$

The tangent stiffness matrix is updated as:

$$\tilde{K}=\tilde{K}^{(k-1)}-\frac{(\tilde{K}^{(k-1)}\delta z)(\tilde{K}^{(k-1)}\delta z)^T}{\delta z^T\tilde{K}^{(k-1)}\delta z}+\frac{\delta g\delta g^T}{\delta z^T\delta g} \tag{5.7}$$

For the inverse BFGS algorithm, the inverse tangent stiffness matrix is updated as:

$$\tilde{K}^{-1}=(I-\frac{\delta z\delta g^T}{\delta z^T\delta g})(\tilde{K}^{(k-1)})^{-1}(I-\frac{\delta z\delta g^T}{\delta z^T\delta g})^T+\frac{\delta z\delta z^T}{\delta z^T\delta g} \tag{5.8}$$

The initial tangent stiffness matrix can be replaced by its initial form with coupling terms neglected, ensuring that both the initial and updated tangent stiffness matrices are symmetric positive definite.

$$\tilde{K}^{(0)}=\begin{bmatrix}K_{uu}&0\\0&K_{dd}\end{bmatrix} \tag{5.9}$$

5.4 Linearized Solution

For phase-field models that cannot be solved iteratively, the governing partial differential equations are usually linearized and solved in total form.

The linear elastic displacement equation is already linear and requires no additional treatment; the phase-field equation must be linearized. Terms associated with the degradation and crack geometric functions that cause nonlinearity are generally replaced by their values at the end of the previous increment and treated as constants, thereby linearizing the nonlinear equation. From (5.2), the linearized phase-field equation is

$$\int_\Omega\frac{2G_cl_0}{c_0}\overline{\mathbf{B}}^T\nabla d d\Omega=-\int_\Omega\overline{\mathbf{N}}^T(g'(d_{\text{t-1}})H+\frac{G_c}{c_0l_0}\alpha'(d_{t-1}))d\Omega \tag{5.10}$$

However, this approach substantially modifies the coefficient matrix and generally cannot provide accurate results. When the crack geometric function is quadratic, freezing only the degradation-function term also yields a linearized equation with a smaller solution error.

$$\begin{aligned}&\alpha(d)=d^2\\&\int_\Omega(\frac{G_c}{l_0}\overline{\mathbf{N}}^Td+G_cl_0\overline{\mathbf{B}}^T\nabla d)d\Omega=-\int_\Omega\overline{\mathbf{N}}^Tg'(d_{\text{t-1}})Hd\Omega\end{aligned} \tag{5.11}$$

5.5 Nonlinear Optimization Algorithm (Reduced-Space Active-Set Newton Method)

From the energy variational principle and phase-field irreversibility constraint, the KKT conditions yield the phase-field evolution equation:

$$\left[g'(d)H-\frac{G_c}{c_0l_0}\left(\alpha'(d)-2l_0^2\Delta d\right)\right]\dot{d}=0,\ g'(d)H-\frac{G_c}{c_0l_0}\left(\alpha'(d)-2l_0^2\Delta d\right)\le 0,\ \dot{d}\ge 0 \tag{5.12}$$

The phase field must satisfy irreversibility, generally enforced through a history variable, although this has limitations. The phase field must also lie in [0,1]. Usually, an initial value of 0 combined with irreversibility satisfies the lower bound. For the upper bound, the quadratic degradation function of the AT2 model makes the phase-field driving force $g'(d)H=0$ at d=1, preventing further growth. Other degradation functions, such as that of PF-CZM, require additional constraints.

When the reduced-space active-set Newton method[29] is used, first define the active set:

$$\left\{d_i\,\middle|\,d_i^{n+1}=d_i^n,r_i^d<0\right\}\cup\left\{d_i\,\middle|\,d_i^{n+1}=1,r_i^d>0\right\} \tag{5.13}$$

Fix the nodes in the active set and apply Newton iteration in the subspace of inactive nodes to solve the phase-field equation. Then check whether the solution exceeds the bounds and project it onto the boundary, i.e., if $d_i^{n+1}>1$, then $d_i^{n+1}=1$. Update the active set and repeat until convergence.

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5.6 Path-Following Strategies[29,30]

Phase-field fracture problems are generally solved using fixed displacement or load increments. For quasi-static fracture of brittle materials, however, this loading scheme may cause a crack to suddenly propagate over a long distance at the critical load, violating the quasi-static assumption. In a quasi-static model, the crack should generally advance only a short, finite distance per increment. A path-following strategy is therefore needed to scale the load and maintain a prescribed crack advance. Assume that, at increment n:

$$\mathbf{f}^{ext} = \lambda \hat{\mathbf{f}}^{ext} + \mathbf{f}_0^{ext} \tag{5.14}$$

$$\mathbf{u} = \mathbf{L}\mathbf{u}_f + \lambda \hat{\mathbf{u}} + \mathbf{u}_0 \tag{5.15}$$

$$\mathbf{d} = \bar{\mathbf{L}}\mathbf{d}_f + \mathbf{d}_0 \tag{5.16}$$

Here, λ is the scaling factor; $\hat{\mathbf{u}}$ and $\hat{\mathbf{f}}^{ext}$ are the prescribed reference displacement or external load increments for this step; $\mathbf{f}_0^{ext}$, $\mathbf{u}_0$, and $\mathbf{d}_0$ denote the nodes with fixed external forces, displacements, and phase fields in this step; $\mathbf{u}_f$ and $\mathbf{d}_f$ denote the displacement and phase-field nodes to be solved; and $\mathbf{L}$ is a general linear constraint of the form:

$$\mathbf{u} = \begin{bmatrix} \mathbf{u}_f \\ \mathbf{u}_p \end{bmatrix} + \lambda \hat{\mathbf{u}} = \begin{bmatrix} \mathbf{I} \\ \mathbf{L}_f \end{bmatrix} \mathbf{u}_f + \begin{bmatrix} \mathbf{0} \\ \mathbf{u}_0 \end{bmatrix} + \lambda \begin{bmatrix} \hat{\mathbf{u}} \\ \mathbf{0} \end{bmatrix} = \mathbf{L}\mathbf{u}_f + \lambda \hat{\mathbf{u}} + \mathbf{u}_0 \tag{5.17}$$

Introduce an additional constraint equation:

$$r^{\lambda} := \Delta l - f\left(\Delta \mathbf{u}, \Delta \mathbf{d}, \Delta \lambda\right) = 0 \tag{5.18}$$

The Lagrange multiplier method gives the monolithic Newton-Raphson iteration scheme:

$$\begin{bmatrix} K^{uu} & K^{ud} & K^{u\lambda} \\ K^{du} & K^{dd} & K^{d\lambda} \\ K^{\lambda u} & K^{\lambda d} & K^{\lambda\lambda} \end{bmatrix} \begin{bmatrix} \delta \mathbf{u} \\ \delta \mathbf{d} \\ \delta \lambda \end{bmatrix} = \begin{bmatrix} r^u \\ r^d \\ r^{\lambda} \end{bmatrix}$$

$$K^{uu} = -\frac{\partial r^u}{\partial u} = \int_{\Omega} \mathbf{B}^T \left(\frac{\partial \sigma}{\partial \varepsilon} \right) \mathbf{B} d\Omega,\ K^{ud} = -\frac{\partial r^u}{\partial d} = \int_{\Omega} \mathbf{B}^T \left(\frac{\partial \sigma}{\partial d} \right) \bar{\mathbf{N}} d\Omega$$

$$K^{u\lambda} = -\frac{\partial r^u}{\partial \lambda} = -\hat{\mathbf{f}}^{ext},\ K^{du} = -\frac{\partial r^d}{\partial u} = \int_{\Omega} \bar{\mathbf{N}}^T \left(g'(d) \frac{\partial H}{\partial \varepsilon} \right) \mathbf{B} d\Omega \tag{5.19}$$

$$K^{dd} = -\frac{\partial r^d}{\partial d} = \int_{\Omega} \left[\bar{\mathbf{N}}^T \left(g''(d) H + \frac{G_c}{c_0 l_0} \alpha''(d) \right) \bar{\mathbf{N}} + \frac{2 G_c l_0}{c_0} \bar{\mathbf{B}}^T \bar{\mathbf{B}} \right] d\Omega$$

$$K^{d\lambda} = -\frac{\partial r^d}{\partial \lambda} = 0,\ K^{\lambda u} = -\frac{\partial r^{\lambda}}{\partial u},\ K^{\lambda d} = -\frac{\partial r^{\lambda}}{\partial d},\ K^{\lambda\lambda} = -\frac{\partial r^{\lambda}}{\partial \lambda}$$

A change of variables gives:

$$\begin{bmatrix} \delta \mathbf{u} \\ \delta \mathbf{d} \\ \delta \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{L} & \mathbf{0} & \hat{\mathbf{u}} \\ \mathbf{0} & \bar{\mathbf{L}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \delta \mathbf{u}_f \\ \delta \mathbf{d}_f \\ \delta \lambda \end{bmatrix} \tag{5.20}$$

Then:

$$\begin{bmatrix} \mathbf{L}^T & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \bar{\mathbf{L}}^T & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} K^{uu} & K^{ud} & K^{u\lambda} \\ K^{du} & K^{dd} & K^{d\lambda} \\ K^{\lambda u} & K^{\lambda d} & K^{\lambda\lambda} \end{bmatrix} \begin{bmatrix} \mathbf{L} & \mathbf{0} & \hat{\mathbf{u}} \\ \mathbf{0} & \bar{\mathbf{L}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \delta \mathbf{u}_f \\ \delta \mathbf{d}_f \\ \delta \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{L}^T & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \bar{\mathbf{L}}^T & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} r^u \\ r^d \\ r^{\lambda} \end{bmatrix} \tag{5.21}$$

That is:

$$\begin{bmatrix} \mathbf{L}^T K^{uu} \mathbf{L} & \mathbf{L}^T K^{ud} \bar{\mathbf{L}} & \mathbf{L}^T \left(K^{uu} \hat{\mathbf{u}} + K^{u\lambda} \right) \\ \bar{\mathbf{L}}^T K^{du} \mathbf{L} & \bar{\mathbf{L}}^T K^{dd} \bar{\mathbf{L}} & \bar{\mathbf{L}}^T K^{du} \hat{\mathbf{u}} \\ K^{\lambda u} \mathbf{L} & K^{\lambda d} \bar{\mathbf{L}} & K^{\lambda u} \hat{\mathbf{u}} + K^{\lambda\lambda} \end{bmatrix} \begin{bmatrix} \delta \mathbf{u}_f \\ \delta \mathbf{d}_f \\ \delta \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{L}^T r^u \\ \bar{\mathbf{L}}^T r^d \\ r^{\lambda} \end{bmatrix} \tag{5.22}$$

Several typical constraints are given here:

1. Crack surface control, by controlling the crack surface increment in the current step:

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$$r^{\lambda}\left(\mathbf{d}\right) \coloneqq \Delta l - \Delta A_d = \Delta l + A_d^n - A_d\left(\mathbf{d}\right) = 0 \tag{5.23}$$

2. Displacement control, by controlling the displacement increment at a specified node or the relative displacement increment between nodes (e.g., COD):

$$r^{\lambda}\left(\mathbf{u}\right) \coloneqq \Delta l - \mathbf{S}\Delta\mathbf{u} = 0 \tag{5.24}$$

Here, $\mathbf{S}$ is a selection vector. For COD, for example, $\mathbf{S} = \begin{bmatrix} 0 & \cdots & 0 & 1 & 0 & \cdots & 0 & -1 & 0 & \cdots & 0 \end{bmatrix}$ can specify relative displacement in a given direction between two nodes. See Ref. [30] for further control methods.

As discussed above, monolithic iteration is nonconvex, so a staggered scheme is generally adopted. With crack surface control, displacement is usually fixed first while the coupled phase-field and scaling-factor equations are solved, followed by solving for displacement with the phase field fixed. With displacement control, the phase field is usually fixed first while the coupled displacement and scaling-factor equations are solved, followed by solving for the phase field with displacement fixed. The latter case is detailed below:

We have the coupled equations for displacement and the scaling factor:

$$\begin{bmatrix} \mathbf{L}^T K^{uu}\mathbf{L} & \mathbf{L}^T\left(K^{uu}\hat{\mathbf{u}} + K^{u\lambda}\right) \\ K^{\lambda u}\mathbf{L} & K^{\lambda u}\hat{\mathbf{u}} + K^{\lambda\lambda} \end{bmatrix} \begin{bmatrix} \delta\mathbf{u}_f \\ \delta\lambda \end{bmatrix} = \begin{bmatrix} \mathbf{L}^T r^u \\ r^{\lambda} \end{bmatrix} \tag{5.25}$$

We can directly solve for $\delta\mathbf{u}_f$ and $\delta\lambda$, then update $\delta\mathbf{u}$:

$$\delta\mathbf{u} = \mathbf{L}\delta\mathbf{u}_f + \delta\lambda\hat{\mathbf{u}} \tag{5.26}$$

To reduce the size of the system to be solved, introduce the transformation:

$$\delta\mathbf{u}_f = \mathbf{u}_f^I\delta\lambda + \delta\mathbf{u}_f^{II} \tag{5.27}$$

Then:

$$\begin{cases} \mathbf{L}^T K^{uu}\mathbf{L}\left(\mathbf{u}_f^I\delta\lambda + \delta\mathbf{u}_f^{II}\right) + \delta\lambda\mathbf{L}^T\left(K^{uu}\hat{\mathbf{u}} + K^{u\lambda}\right) = \mathbf{L}^T r^u \\ K^{\lambda u}\mathbf{L}\left(\mathbf{u}_f^I\delta\lambda + \delta\mathbf{u}_f^{II}\right) + \delta\lambda\left(K^{\lambda u}\hat{\mathbf{u}} + K^{\lambda\lambda}\right) = r^{\lambda} \end{cases} \tag{5.28}$$

Let:

$$\begin{aligned} K^{uu}\mathbf{L}\mathbf{u}_f^I &= -K^{u\lambda} - K^{uu}\hat{\mathbf{u}} = \hat{\mathbf{f}}^{ext} - K^{uu}\hat{\mathbf{u}} \\ K^{uu}\mathbf{L}\delta\mathbf{u}_f^{II} &= r^u \end{aligned} \tag{5.29}$$

and

$$\begin{aligned} \mathbf{u}^I &= \mathbf{L}\mathbf{u}_f^I \\ \delta\mathbf{u}^{II} &= \mathbf{L}\delta\mathbf{u}_f^{II} \end{aligned} \tag{5.30}$$

Then:

$$\begin{aligned} K^{uu}\mathbf{u}^I &= \hat{\mathbf{f}}^{ext} - K^{uu}\hat{\mathbf{u}} \\ K^{uu}\delta\mathbf{u}^{II} &= r^u \end{aligned} \tag{5.31}$$

After obtaining $\mathbf{u}^I$ and $\delta\mathbf{u}^{II}$, we can calculate $\delta\mathbf{u}$:

$$\delta\mathbf{u} = \delta\lambda\mathbf{u}^I + \delta\mathbf{u}^{II} + \delta\lambda\hat{\mathbf{u}} \tag{5.32}$$

If the transformed system is still too large, an unscaled solution ($\lambda = 1$) can first be obtained. Then λ is adjusted according to the relation between the constraint equation and load, and the system is solved once more to obtain the increment result, reducing computational cost at the expense of control accuracy; see Ref. [30].

5.7 Restrictions on the Degradation Function

Phase-field fracture is a nonlinear problem requiring nonlinear iteration. The iteration matrix must have a spectral radius less than 1 and be positive definite.

To ensure convergence in a phase-field model, the iteration matrix must be symmetric positive definite. From (5.4)-4, the following condition must hold[5]:

$$g''(d)H + \frac{G_c}{c_0 l_0}\alpha''(d) \geq 0 \tag{5.33}$$

For the classical AT2 phase-field model ($g\left(d\right) = \left(1-d\right)^2$, $\alpha\left(d\right) = d^2$), Eq. (5.33) is always satisfied

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at d=0. For other degradation functions $g(d)$ and crack geometric functions $\alpha(d)$, however, Eq. (5.33) may not hold. The crack geometric function and initial load must therefore be selected appropriately (the initial load should be neither too large nor too small), so that the phase field converges to the undamaged homogeneous solution at the start of iteration.

In the unified cohesive method, to satisfy (5.33), the phase-field width should meet:

$$l_0 \le \frac{4}{(a_2 + p + 0.5)\pi} l_{ch}$$
$$l_{ch} = \frac{EG_c}{\sigma_0^2} \tag{5.34}$$

For other crack geometric and degradation functions that may not satisfy (5.33), Ref. [4] proposes a solution. Using a Taylor expansion, the second derivative of the degradation function is replaced by:

$$g''(d) \approx -\frac{g'(d)}{1-d} \tag{5.35}$$

This ensures a positive second derivative of the degradation function and satisfies Eq. (5.33). Although the approximation in (5.35) may seem unreasonable, it provides a convergent iteration matrix. Compared with the Newton-Raphson iteration matrix, this approximate matrix requires more iterations to converge but is more robust.

Lo[8] et al. proposed crack geometric and degradation functions in which, although the derivative of the degradation function at d=0 is close to 0 to accommodate a larger phase-field width, its second derivative at d=0 is also sufficiently close to 0 to satisfy Eq. (5.33). Iterative solution is therefore possible during the initial stage of phase-field evolution. Whether Eq. (5.33) remains satisfied during the middle stage requires numerical verification; the method in Section 5.4 is generally used.

5.8 Iteration Termination Criteria

Karlo[13] et al. use the residual as the convergence criterion: convergence is assumed when the residual norm falls below a threshold

$$R(u,d) \le Tol \tag{5.36}$$

Bourdin[6] use the phase-field increment as the iteration convergence criterion

$$\delta d = | d^k - d^{k-1} | \le Tol \tag{5.37}$$

Ambati[14] use the system energy as the convergence criterion

$$f(E_k) \le Tol$$
$$f(E_k) := \frac{E_k - E_{k+1}}{E_{k+1}} \times 100\% \tag{5.38}$$
$$f(E_k) := \arctan(E_k - E_{k+1}) \times \frac{180^\circ}{\pi}$$

VI. Adaptive Mesh Refinement

Most adaptive phase-field models refine the mesh by adding hanging nodes, usually within open-source AMR frameworks such as deal.ll. FreeFEM++ does not support these hanging nodes, but provides another adaptive refinement strategy through the adaptivemesh function; see https://github.com/urbainvaes/cahn-hilliard.git for details.

VI. Computational Software

7.1 FreeFEM++

Phase-field problems can be solved using the open-source finite element software FreeFEM++.

The main solution steps are parameter and function definitions, finite element meshing, weak-form definition, iterative solution, and postprocessing. FreeFem++ supports parallel computation through the PETSc numerical analysis library.

See the corresponding code and FreeFEM user guide for detailed solution procedures and FreeFEM++ syntax.

Examples are provided in the Freefem folder, using the unified phase-field model as a framework. Users

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may choose the unified phase-field model, AT2 model, or other models.

Other examples:

Thermomechanically coupled phase-field fracture model: https://github.com/AlexanderJFDR/Thermo-Mechanic-PhaseField-Model.git.

7.2 Abaqus

Abaqus generally solves phase-field problems using subroutines, mainly UMAT or UEL implementations.

For phase-field fracture solution using abaqus subroutines, see the engineering examples on the github page: https://github.com/AlexanderJFDR?tab=repositories.

7.2.1 UMAT Subroutine Implementation

In a UMAT subroutine, the phase-field problem is solved by making it equivalent to a coupled displacement-temperature problem.

The governing equations for the steady-state coupled displacement-temperature problem are:

$$\begin{cases} \nabla \bullet \sigma + b = 0 \\ k\Delta T + r = 0 \end{cases} \tag{7.1}$$

Here, k is the thermal conductivity and r(x,T) is the heat source function. When the heat source depends on temperature, this steady-state coupled displacement-temperature problem is also nonlinear.

The hybrid phase-field model equations are:

$$\begin{cases} \nabla \bullet \sigma + b = 0 \\ \dfrac{G_c}{c_0 l_0}[\alpha'(d) - 2l_0^2 \Delta d] = -g'(d)\Psi(\varepsilon) \end{cases} \tag{7.2}$$

Comparing the two sets of equations shows that their displacement equations are identical. The phase-field equation must be transformed into:

$$\Delta d - \frac{1}{2l_0^2}\alpha'(d) - \frac{c_0}{2G_c l_0} g'(d)\Psi(\varepsilon) = 0 \tag{7.3}$$

In UMAT, set the material thermal conductivity k=1 and the heat source function to

$$r = -\frac{1}{2l_0^2}\alpha'(d) - \frac{c_0}{2G_c l_0} g'(d)\Psi(\varepsilon) \tag{7.4}$$

A nonlinear solution then allows the temperature field to represent the phase field.

Many UMAT examples are available on github, so the author provides none here.

7.2.2 UEL Subroutine Implementation

Thanks to Jian-Ying Wu for providing the PF-CZM code. Based on this code, we analyze the techniques for solving phase-field problems using UEL subroutines. The Github address is: https://github.com/jianyingwu/pfczm-abaqus.

Jian-Ying Wu uses 4-node quadrilateral elements. The corresponding Abaqus mesh type is CPS4, a 4-node plane-stress quadrilateral element.

The element is illustrated below:

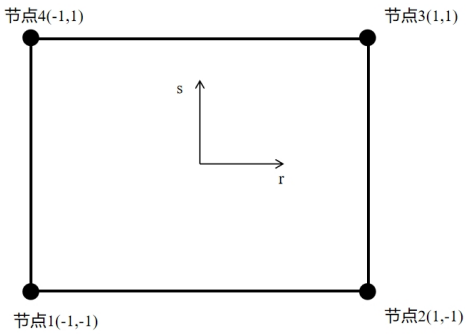


Figure 14. Schematic of a four-node quadrilateral element

Here, r and s are the local element coordinates. At any point within the element, the global coordinates x and y can be expressed as:

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$$\begin{cases} x(r,s)=\sum_{i=1}^{4} x_i N_i(r,s) \\ y(r,s)=\sum_{i=1}^{4} y_i N_i(r,s) \end{cases} \tag{7.5}$$

The displacement and phase fields within the element are expressed as:

$$\begin{cases} u(r,s)=\sum_{i=1}^{4} u_i N_i(r,s) \\ v(r,s)=\sum_{i=1}^{4} v_i N_i(r,s) \\ d(r,s)=\sum_{i=1}^{4} d_i N_i(r,s) \end{cases} \tag{7.6}$$

Here, a staggered iteration method is used: the displacement and phase fields are solved separately, then assembled into the same linear system for solution in Abaqus.

$N_i(r,s)$ is a shape function equal to 1 at node i and 0 at the other nodes. The shape functions are:

$$\begin{cases} N_1(r,s)=\frac{1}{4}(1-r)(1-s) \\ N_2(r,s)=\frac{1}{4}(1+r)(1-s) \\ N_3(r,s)=\frac{1}{4}(1+r)(1+s) \\ N_4(r,s)=\frac{1}{4}(1-r)(1+s) \end{cases} \tag{7.7}$$

The Jacobian matrix J for the transformation between local coordinates r,s and global coordinates x,y is:

$$J=\frac{\partial(x,y)}{\partial(r,s)}=\begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial s} & \frac{\partial y}{\partial s} \end{bmatrix}=\begin{bmatrix} \frac{\partial N_1}{\partial r} & \frac{\partial N_2}{\partial r} & \frac{\partial N_3}{\partial r} & \frac{\partial N_4}{\partial r} \\ \frac{\partial N_1}{\partial s} & \frac{\partial N_2}{\partial s} & \frac{\partial N_3}{\partial s} & \frac{\partial N_4}{\partial s} \end{bmatrix}\begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_4 & y_4 \end{bmatrix} \tag{7.8}$$

Then:

$$\begin{bmatrix} \frac{\partial N_i}{\partial x} \\ \frac{\partial N_i}{\partial y} \end{bmatrix}=\begin{bmatrix} \frac{\partial r}{\partial x} & \frac{\partial s}{\partial x} \\ \frac{\partial r}{\partial y} & \frac{\partial s}{\partial y} \end{bmatrix}\begin{bmatrix} \frac{\partial N_i}{\partial r} \\ \frac{\partial N_i}{\partial s} \end{bmatrix}=\frac{\partial(r,s)}{\partial(x,y)}\begin{bmatrix} \frac{\partial N_i}{\partial r} \\ \frac{\partial N_i}{\partial s} \end{bmatrix}=J^{-1}\begin{bmatrix} \frac{\partial N_i}{\partial r} \\ \frac{\partial N_i}{\partial s} \end{bmatrix} \tag{7.9}$$

For the displacement field, write the nodal variables in the following vector form:

$$\mathbf{u}=\begin{bmatrix} u_1 & v_1 & u_2 & v_2 & u_3 & v_3 & u_4 & v_4 \end{bmatrix}^T \tag{7.10}$$

To obtain the strain, we have:

$$\begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix}=\begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & \frac{\partial N_2}{\partial x} & 0 & \frac{\partial N_3}{\partial x} & 0 & \frac{\partial N_4}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \frac{\partial N_2}{\partial y} & 0 & \frac{\partial N_3}{\partial y} & 0 & \frac{\partial N_4}{\partial y} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial x} & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial x} & \frac{\partial N_4}{\partial y} & \frac{\partial N_4}{\partial x} \end{bmatrix}\begin{bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \end{bmatrix}=B_u\,\mathbf{u} \tag{7.11}$$

From the displacement-field residual and iteration matrix, using the hybrid phase-field model and neglecting internal and external forces, we obtain

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$$r^u = \int_\Omega \mathbf{B}^T \sigma d\Omega = 0$$
$$K^{uu} = \int_\Omega \mathbf{B}^T \hat{C}\mathbf{B} d\Omega \tag{7.12}$$

The element stiffness matrix and residual are:

$$K^{uu} = \int_\Omega \mathbf{B}^T \hat{C}\mathbf{B} dxdy = \int_{-1}^{1}\int_{-1}^{1} \mathbf{B}^T \hat{C}\mathbf{B} J drds$$
$$r^u = K^{uu}\mathbf{u} \tag{7.13}$$

Integration of the shape functions within the element is generally approximated using Gaussian quadrature. In the quadrilateral element domain, Gaussian quadrature gives:

$$\int_{-1}^{1}\int_{-1}^{1} f(x,y)\,dxdy = \sum_{i=1}^{m}\sum_{j=1}^{n} w_i w_j f\left(x_i, y_j\right) \tag{7.14}$$

The two-dimensional second-order Gaussian quadrature formula gives:

$$K^{uu} = \sum_{i=1}^{2}\sum_{j=1}^{2} w_i w_j \mathbf{B}^T \hat{C}\mathbf{B} J$$
$$B = B\left(r_i, s_j\right), J = J\left(r_i, s_j\right)$$
$$w_1, w_2 = 1 \tag{7.15}$$
$$r_1 = s_1 = -\frac{\sqrt{3}}{3}, r_2 = s_2 = \frac{\sqrt{3}}{3}$$

The displacement residual can be substituted directly for solution.

For the phase field, write the nodal variables in the following vector form:

$$\mathbf{d} = \begin{bmatrix} d_1 & d_2 & d_3 & d_4 \end{bmatrix}^T \tag{7.16}$$

For the phase-field gradient, we have:

$$\nabla d = \begin{bmatrix} \frac{\partial d}{\partial x} \\ \frac{\partial d}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial x} & \frac{\partial N_3}{\partial x} & \frac{\partial N_4}{\partial x} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_2}{\partial y} & \frac{\partial N_3}{\partial y} & \frac{\partial N_4}{\partial y} \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{bmatrix} = B_d \mathbf{d} \tag{7.17}$$

From the phase-field residual and iteration matrix, we obtain

$$r^d = -\int_\Omega [\bar{\mathbf{N}}^T (g'(d)H + \frac{G_c}{c_0 l_0}\alpha'(d)) + \frac{2G_c l_0}{c_0}\bar{\mathbf{B}}^T \nabla d] d\Omega = 0$$
$$K^{dd} = -\frac{\partial r^d}{\partial d} = \int_\Omega \left[\bar{\mathbf{N}}^T \left(g''(d)H + \frac{G_c}{c_0 l_0}\alpha''(d) \right) \bar{\mathbf{N}} + \frac{2G_c l_0}{c_0}\bar{\mathbf{B}}^T \bar{\mathbf{B}} \right] d\Omega \tag{7.18}$$

The element stiffness matrix and residual are:

$$r^d = -\int_{-1}^{1}\int_{-1}^{1} [\bar{\mathbf{N}}^T (g'(d)H + \frac{G_c}{c_0 l_0}\alpha'(d)) + \frac{2G_c l_0}{c_0}\bar{\mathbf{B}}^T \nabla d] J drds$$
$$K^{dd} = \int_{-1}^{1}\int_{-1}^{1} \left[\bar{\mathbf{N}}^T \left(g''(d)H + \frac{G_c}{c_0 l_0}\alpha''(d) \right) \bar{\mathbf{N}} + \frac{2G_c l_0}{c_0}\bar{\mathbf{B}}^T \bar{\mathbf{B}} \right] J drds \tag{7.19}$$

The two-dimensional second-order Gaussian quadrature formula similarly yields the element stiffness matrix and residual for the phase field.

Finally, in Abaqus, the displacement and phase-field variables must be combined. Denote the complete nodal variable vector as:

$$\mathbf{X} = \begin{bmatrix} u_1 & v_1 & d_1 & u_2 & v_2 & d_2 & u_3 & v_3 & d_3 & u_4 & v_4 & d_4 \end{bmatrix}^T \tag{7.20}$$

The element iteration equation is:

$$K\mathbf{X} = R \tag{7.21}$$

Ensure that the components of the tangent stiffness matrix K and residual vector are positioned correctly.

For an 8-node quadrilateral element, the element and node layout are shown below:

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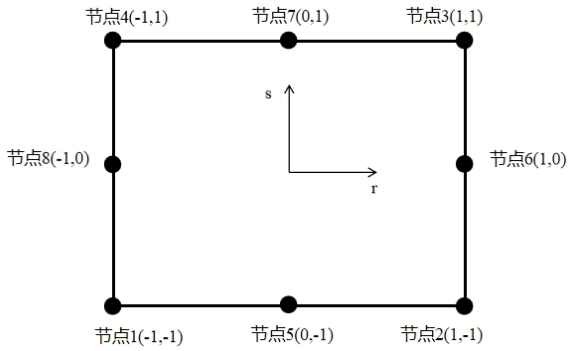


Figure 15. Schematic of an eight-node quadrilateral element

The shape functions are:

$$\begin{cases} N_1(r,s)=\frac{1}{4}(1-r)(1-s)-\frac{1}{4}(1-r^2)(1-s)-\frac{1}{4}(1-r)(1-s^2) \\ N_2(r,s)=\frac{1}{4}(1+r)(1-s)-\frac{1}{4}(1-r^2)(1-s)-\frac{1}{4}(1+r)(1-s^2) \\ N_3(r,s)=\frac{1}{4}(1+r)(1+s)-\frac{1}{4}(1+r)(1-s^2)-\frac{1}{4}(1-r^2)(1+s) \\ N_4(r,s)=\frac{1}{4}(1-r)(1+s)-\frac{1}{4}(1-r^2)(1+s)-\frac{1}{4}(1-r)(1-s^2) \\ N_5(r,s)=\frac{1}{2}(1-r^2)(1-s) \\ N_6(r,s)=\frac{1}{2}(1+r)(1-s^2) \\ N_7(r,s)=\frac{1}{2}(1-r^2)(1+s) \\ N_8(r,s)=\frac{1}{2}(1-r)(1-s^2) \end{cases} \tag{7.22}$$

Their derivatives of various orders are:

$$\begin{cases} \frac{\partial N_1}{\partial r}=\frac{1}{4}(1-s)(2r+s), \frac{\partial N_1}{\partial s}=\frac{1}{4}(1-r)(r+2s) \\ \frac{\partial N_2}{\partial r}=\frac{1}{4}(1-s)(2r-s), \frac{\partial N_2}{\partial s}=-\frac{1}{4}(1+r)(r-2s) \\ \frac{\partial N_3}{\partial r}=\frac{1}{4}(1+s)(2r+s), \frac{\partial N_3}{\partial s}=\frac{1}{4}(1+r)(r+2s) \\ \frac{\partial N_4}{\partial r}=\frac{1}{4}(1+s)(2r-s), \frac{\partial N_4}{\partial s}=-\frac{1}{4}(1-r)(r-2s) \\ \frac{\partial N_5}{\partial r}=-r(1-s), \frac{\partial N_5}{\partial s}=\frac{1}{2}(r^2-1) \\ \frac{\partial N_6}{\partial r}=\frac{1}{2}(1-s^2), \frac{\partial N_6}{\partial s}=-s(1+r) \\ \frac{\partial N_7}{\partial r}=-r(1+s), \frac{\partial N_7}{\partial s}=\frac{1}{2}(1-r^2) \\ \frac{\partial N_8}{\partial r}=\frac{1}{2}(s^2-1), \frac{\partial N_5}{\partial s}=-s(1-r) \end{cases} \tag{7.23}$$

Obtaining the strain gradient requires the second derivatives of the shape functions with respect to global coordinates x and y. The following relations hold:

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$$\frac{\partial N}{\partial \xi}=\frac{\partial N}{\partial X}\frac{\partial X}{\partial \xi}+\frac{\partial N}{\partial Y}\frac{\partial Y}{\partial \xi}$$
$$\frac{\partial N}{\partial \eta}=\frac{\partial N}{\partial X}\frac{\partial X}{\partial \eta}+\frac{\partial N}{\partial Y}\frac{\partial Y}{\partial \eta}$$
$$\frac{\partial^2 N}{\partial \xi^2}=\frac{\partial^2 N}{\partial X^2}\left(\frac{\partial X}{\partial \xi}\right)^2+\frac{\partial N}{\partial X}\frac{\partial^2 X}{\partial \xi^2}+\frac{\partial^2 N}{\partial Y^2}\left(\frac{\partial Y}{\partial \xi}\right)^2+\frac{\partial N}{\partial Y}\frac{\partial^2 Y}{\partial \xi^2}+2\frac{\partial^2 N}{\partial X\partial Y}\frac{\partial X}{\partial \xi}\frac{\partial Y}{\partial \xi} \tag{7.24}$$
$$\frac{\partial^2 N}{\partial \eta^2}=\frac{\partial^2 N}{\partial X^2}\left(\frac{\partial X}{\partial \eta}\right)^2+\frac{\partial N}{\partial X}\frac{\partial^2 X}{\partial \eta^2}+\frac{\partial^2 N}{\partial Y^2}\left(\frac{\partial Y}{\partial \eta}\right)^2+\frac{\partial N}{\partial Y}\frac{\partial^2 Y}{\partial \eta^2}+2\frac{\partial^2 N}{\partial X\partial Y}\frac{\partial X}{\partial \eta}\frac{\partial Y}{\partial \eta}$$
$$\frac{\partial^2 N}{\partial \xi\partial \eta}=\frac{\partial^2 N}{\partial X^2}\frac{\partial X}{\partial \xi}\frac{\partial X}{\partial \eta}+\frac{\partial N}{\partial X}\frac{\partial^2 X}{\partial \xi\partial \eta}+\frac{\partial^2 N}{\partial Y^2}\frac{\partial Y}{\partial \xi}\frac{\partial Y}{\partial \eta}+\frac{\partial N}{\partial Y}\frac{\partial^2 Y}{\partial \xi\partial \eta}+\frac{\partial^2 N}{\partial X\partial Y}\frac{\partial X}{\partial \xi}\frac{\partial Y}{\partial \eta}+\frac{\partial^2 N}{\partial X\partial Y}\frac{\partial X}{\partial \eta}\frac{\partial Y}{\partial \xi}$$

Thus, in matrix form:

$$\begin{bmatrix}\frac{\partial N}{\partial X}\\ \frac{\partial N}{\partial Y}\\ \frac{\partial^2 N}{\partial X^2}\\ \frac{\partial^2 N}{\partial Y^2}\\ \frac{\partial^2 N}{\partial X\partial Y}\end{bmatrix}=\begin{bmatrix}\frac{\partial X}{\partial \xi} & \frac{\partial Y}{\partial \xi} & 0 & 0 & 0\\ \frac{\partial X}{\partial \eta} & \frac{\partial Y}{\partial \eta} & 0 & 0 & 0\\ \frac{\partial^2 X}{\partial \xi^2} & \frac{\partial^2 Y}{\partial \xi^2} & \left(\frac{\partial X}{\partial \xi}\right)^2 & \left(\frac{\partial Y}{\partial \xi}\right)^2 & 2\frac{\partial X}{\partial \xi}\frac{\partial Y}{\partial \xi}\\ \frac{\partial^2 X}{\partial \eta^2} & \frac{\partial^2 Y}{\partial \eta^2} & \left(\frac{\partial X}{\partial \eta}\right)^2 & \left(\frac{\partial Y}{\partial \eta}\right)^2 & 2\frac{\partial X}{\partial \eta}\frac{\partial Y}{\partial \eta}\\ \frac{\partial^2 X}{\partial \xi\partial \eta} & \frac{\partial^2 Y}{\partial \xi\partial \eta} & \frac{\partial X}{\partial \xi}\frac{\partial X}{\partial \eta} & \frac{\partial Y}{\partial \xi}\frac{\partial Y}{\partial \eta} & \frac{\partial X}{\partial \xi}\frac{\partial Y}{\partial \eta}+\frac{\partial X}{\partial \eta}\frac{\partial Y}{\partial \xi}\end{bmatrix}^{-1}\begin{bmatrix}\frac{\partial N}{\partial \xi}\\ \frac{\partial N}{\partial \eta}\\ \frac{\partial^2 N}{\partial \xi^2}\\ \frac{\partial^2 N}{\partial \eta^2}\\ \frac{\partial^2 N}{\partial \xi\partial \eta}\end{bmatrix} \tag{7.25}$$

7.2.3 Considerations for Solving Phase-Field Problems in abaqus

For Abaqus subroutine programming conventions, see the author's other open-source github project. Only considerations specific to phase-field problems are discussed here.

Mesh size:

As recommended by Jian-Ying Wu, the mesh size in the phase-field propagation region should preferably be less than one-third of the phase-field width to ensure solution accuracy.

Elements:

CPS3 or CPS4 elements are generally used to solve the phase field. These linear elements ensure a positive phase field at every integration point and allow convergence. Higher-order elements do not necessarily give correct phase-field results; the author has not identified the reason due to limited expertise. Higher-order elements can be used for displacement, and are required when strain gradients are needed. In that case, mixed-element techniques can be used to solve the displacement and phase fields.

Variable settings:

In two-dimensional phase-field problems, variables 1 and 2 are generally assigned to u and v. Variable 11 represents temperature in abaqus (symbol NT11) and is used here to store the phase-field results.

Convergence settings:

The exponential and concrete softening laws of the pfczm phase-field model account for cohesive effects and yield relatively gradual post-peak curves; see the results in Ref. [1]. This model therefore converges more readily, and default abaqus convergence criteria can be used, as in Jian-Ying Wu's code.

The linear softening case of pfczm[19] and the AT2 model[15], however, exhibit brittle fracture behavior: the external force drops rapidly after reaching its peak, making convergence difficult. Simply increasing the number of iterations may be insufficient, and convergence may remain impossible. Miehe[15] therefore uses very small increments with only one iteration per increment. When solving these brittle phase-field models in abaqus, the convergence criteria generally need to be relaxed to allow computation to proceed. The AT2 model calculated with default convergence criteria is shown in Figure 16. A single increment can cause the crack to pass through the specimen, necessarily involving many iterations and producing an almost vertical post-peak curve.

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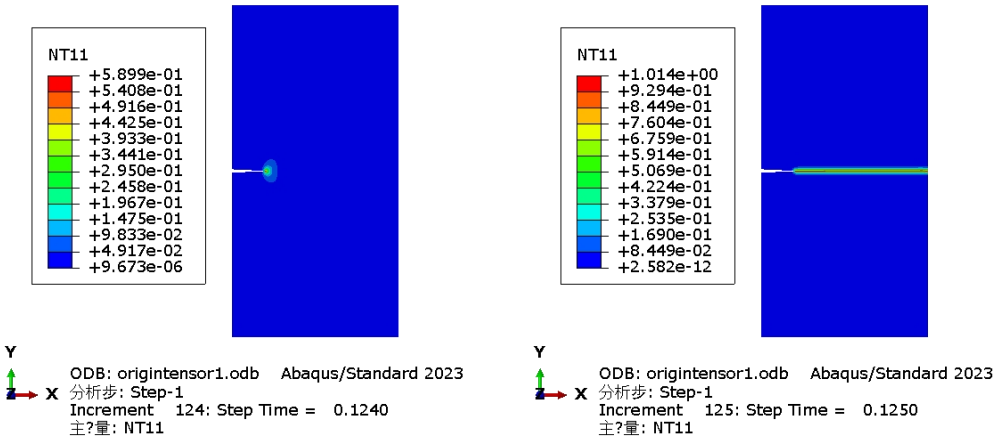


Figure 16. AT2 phase-field model with default convergence criteria. The left and right images differ by only one increment, during which the crack rapidly propagates through the specimen and multiple iterations occur

VIII. Applications to Anisotropic Materials

Anisotropy may refer to stiffness anisotropy, fracture of different constituents in heterogeneous materials, or anisotropy of fracture toughness. This section introduces phase-field models for these forms of anisotropy, of which fracture toughness anisotropy is the most widely studied.

8.1 Tensorial Damage Variable[7]

Anisotropic materials have direction-dependent stiffness. Modifying the material stiffness matrix in a conventional phase-field model can approximately describe their fracture behavior. However, this causes all directional stiffnesses to degrade simultaneously as damage increases, eventually reaching zero. In reality, material failure is not isotropic: stiffness often becomes 0 in one direction while remaining nonzero in others.

To address this issue, PETRINI[7] et al. proposed a fourth-order degradation tensor G for an anisotropic damage phase-field model. The crack density function is defined as:

$$\gamma(G,\nabla G)=\frac{1}{2l_0}[G::G+l_0^2\nabla G::\nabla G] \tag{8.1}$$

The equivalent stress is defined as:

$$\hat{\sigma}=(G:C^0:G):\varepsilon \tag{8.2}$$

If G reduces to a one-dimensional scalar, the model simplifies to AT2, except that G=0 denotes complete damage and G=1 denotes the undamaged state.

The governing partial differential equations are:

$$\begin{cases}\nabla\bullet\hat{\sigma}+b=0\\ \hat{\sigma}=(G:C^0:G):\varepsilon\\ \varepsilon\otimes(C^0:G:\varepsilon)+\dfrac{G_c}{l}(G-l^2\Delta G)=0\end{cases} \tag{8.3}$$

In this model, the fourth-order damage tensor characterizes the degradation of the fourth-order stiffness tensor. For anisotropic material stiffness, different loads lead to different directional stiffness degradation and crack paths.

8.2 Damage Distinguishing Matrix and Fiber Fracture

A composite lamina is a typical anisotropic material. Song[12] proposed a phase-field model that distinguishes fiber and matrix fracture to simulate lamina fracture. Based on AT2, it introduces two phase-field variables to represent fiber and matrix fracture separately.

$$\begin{aligned}&\nabla\bullet\sigma+b=0,\ \text{in}\,\Omega\\ &g'(d_1)H_L+\frac{G_c^L}{l}(d_1-l^2\Delta d_1)=0,\ \text{in}\Omega\\ &g'(d_2)H_T+\frac{G_c^T}{l}(d_2-l^2\Delta d_2)=0,\ \text{in}\Omega\end{aligned} \tag{8.4}$$

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Here, d_1 and d_2 denote the axial fiber fracture and transverse matrix cracking modes, respectively. H_L and H_T denote the strain energies driving these two modes, obtained by decomposing strain in the principal material coordinate system.

8.3 Modified Anisotropic Phase-Field Model

Ref. [3] introduces a modified anisotropic phase-field model for composite lamina fracture. Based on conventional AT2, the crack density function is defined as

$$\gamma(d,\nabla d)=\frac{1}{2l}d^2+\frac{l}{2}\left|\nabla d\right|^2\left[1+\beta\cos^2\left\langle N,\nabla d\right\rangle\right] \tag{8.5}$$

Here, $\cos\left\langle N,\nabla d\right\rangle$ is the cosine of the angle between the fiber direction and the phase-field gradient.

In the principal material coordinate system, strain energy is decomposed into fiber fracture, matrix Mode I failure, and matrix Mode II failure

$$\begin{aligned}
\Psi^{+}_{fiber}&=\frac{1}{2}\left\langle\sigma_{11}\right\rangle^{+}\left\langle\varepsilon_{11}\right\rangle^{+}\\
\Psi^{+}_{matrixI}&=\frac{1}{2}\left\langle\sigma_{22}\right\rangle^{+}\left\langle\varepsilon_{22}\right\rangle^{+}+\frac{1}{2}\left\langle\sigma_{33}\right\rangle^{+}\left\langle\varepsilon_{33}\right\rangle^{+}\\
\Psi^{+}_{matrixII}&=\frac{1}{2}\tau_{12}\gamma_{12}+\frac{1}{2}\tau_{23}\gamma_{23}+\frac{1}{2}\tau_{13}\gamma_{13}\\
\left\langle x\right\rangle^{+}&=(x\pm\left|x\right|)/2
\end{aligned} \tag{8.6}$$

The phase-field evolution equation is modified to

$$\begin{aligned}
&g'(d)H+\frac{1}{l}[d-l^2\Delta d(1+\beta\cos^2\left\langle N,\nabla d\right\rangle)]=0\\
&H=\max_{\tau\in[0,t]}\left\{\frac{\Psi^{+}_{fiber}}{\bar{G}_f}\right\}+\max_{\tau\in[0,t]}\left\{\frac{\Psi^{+}_{matrixI}}{G_{mI}}\right\}+\max_{\tau\in[0,t]}\left\{\frac{\Psi^{+}_{matrixII}}{G_{mII}}\right\}
\end{aligned} \tag{8.7}$$

Here, G_{mI} and G_{mII} are the critical energy release rates for matrix Mode I and Mode II failure, respectively; $\bar{G}_f$ is the modified critical fiber energy release rate, used to eliminate the effect of β on the critical energy release rate in the fiber direction.

8.4 Structural Tensor

In second-order phase-field models, a structural tensor $\mathbf{L}$ is commonly introduced to represent anisotropic fracture toughness. The phase-field equation can be written as:

$$g'(d)H+\frac{G_c}{c_0 l_0}\left[\alpha'(d)-l^2\nabla\left(\mathbf{L}\bullet\nabla d\right)\right]=0 \tag{8.8}$$

In two dimensions, consider an infinite domain containing an infinitely long straight crack inclined to the global coordinate system $\mathrm{O}-x_1x_2$ at an angle θ, as shown in Figure 17. Assume that the global coordinate system is aligned with the principal directions of tensor $\mathbf{L}$. In global coordinates, the tensor is therefore:

$$\mathbf{L}=\begin{bmatrix} l_1 & 0\\ 0 & l_2\end{bmatrix} \tag{8.9}$$

Here, l_1 and l_2 are its eigenvalues, with $l_1\le l_2$ assumed. In the local coordinate system $\mathrm{O}'-x_1'x_2'$ aligned with the crack, $\frac{\partial d^0}{\partial x_1'}=0$. The function therefore reduces to a single-variable function $\tilde{d}\left(x_2'\right)$. The crack surface energy in the shaded region can thus be expressed as:

$$W_{frac}=\Delta a\int_{-\infty}^{+\infty}\frac{\bar{G}_c}{2l}\left(\left(\tilde{d}\right)^2+l^2L_{22}'\left(\frac{\partial\tilde{d}}{\partial x_2'}\right)^2\right)dx_2' \tag{8.10}$$

where $L_{22}'=l_1\sin^2\theta+l_2\cos^2\theta$. The corresponding phase-field equation follows from the variational principle:

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$$\tilde{d}\left(x_2'\right)=Arg\inf_{d\in\mathbb{K}}\left\{W_{frac}\right\},\ \mathbb{K}=\left\{d\,\middle|\,d\left(0\right)=1,d\left(\pm\infty\right)=0\right\} \tag{8.11}$$

From the above equation, the phase-field expression is:

$$\tilde{d}\left(x_2'\right)=e^{-\frac{|x|}{l\sqrt{L_{22}'}}} \tag{8.12}$$

Correspondingly, the crack surface energy can be expressed as:

$$W_{frac}=\sqrt{L_{22}'}\bar{G}_c\Delta a=\sqrt{\frac{l_1+l_2}{2}-\frac{l_1-l_2}{2}\cos 2\theta}\bar{G}_c\Delta a \tag{8.13}$$

The equivalent critical energy release rate $\tilde{G}_c\left(\theta\right)$ for different propagation directions is therefore:

$$\tilde{G}_c\left(\theta\right)=\sqrt{\frac{l_1+l_2}{2}-\frac{l_1-l_2}{2}\cos 2\theta}\bar{G}_c \tag{8.14}$$

Within $\left[0,\pi\right]$, the equivalent critical energy release rate has one maximum and one minimum. For crack propagation along x_1, it is $\sqrt{l_2}\bar{G}_c$; for propagation along x_2, it is $\sqrt{l_1}\bar{G}_c$. These represent the maximum and minimum fracture energies for crack propagation in different directions and characterize fracture anisotropy.

Ref. [31] provides supplementary material on this topic.

The directional tensor in a second-order phase-field model is second order, which limits the fracture toughness periodicity to π. In fourth-order phase-field models, a fourth-order directional tensor allows a period of $\pi/2$, representing strong anisotropy and simulating zigzag crack propagation; see Chapter XIV.

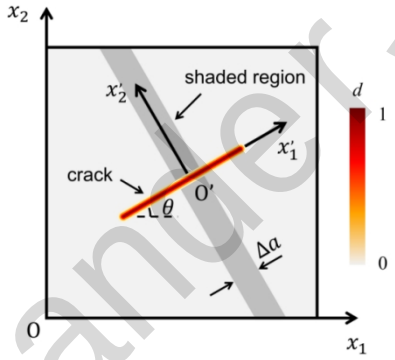


Figure 17. Phase-field representation of a straight crack topology in an infinite domain

IX. Phase-Field Fracture for Mixed-Mode Fracture

Mixed-mode fracture generally considers Mode I and Mode II fracture (separating tension-compression and shear), general failure criteria (separating tension, compression, and shear), or composite failure (separating fiber tension, matrix shear, interface tension and shear, etc.). Such problems are generally addressed by modifying the crack driving force or using two phase fields. Mixed-mode phase-field models generally lack variational consistency because their complex crack driving forces make explicit derivation of tangent stiffness difficult; they are therefore often combined with Hybrid phase field[14]. Ref. [32] provides a variationally consistent mixed-mode phase-field model.

9.1 Dual-Phase-Field Model for Mixed-Mode Fracture in Rock

Fan[10] et al. proposed a dual-phase-field model based on the unified cohesive phase-field model that distinguishes opening and sliding cracks in rock.

Using a mixed-mode fracture criterion similar to the F-criterion, they first determine the crack orientation under loading θ

$$\begin{aligned}&\theta=\arg\max_{\theta}[F(\theta)]\Big|_{\varepsilon}\\&F(\theta):=\frac{H_I(\varepsilon,\theta)}{G_I}+\frac{H_{II}(\varepsilon,\theta)}{G_{II}}\\&H_I=\frac{1}{2E}\bar{\sigma}_{nn}^2,\ \ H_{II}=\frac{1}{2G}\bar{\tau}^2\end{aligned} \tag{9.1}$$

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The sign of the normal stress in that direction then determines whether the point is in tension or compression, allowing the Mode I and Mode II strain energies to be calculated. Based on PF-CZM theory, the dual-phase-field formulation is

$$\begin{aligned} &\nabla \bullet \sigma + b = 0, \quad \text{in}\Omega \\ &-g_I'(d_I)H_I^+ + \frac{G_I}{\pi L}(2L^2\Delta d_I - 2 + 2d_I) = 0, \quad in\Omega \\ &-g_{II}'(d_{II})H_{II}^+ + \frac{G_{II}}{\pi L}(2L^2\Delta d_{II} - 2 + 2d_{II}) = 0, \quad in\Omega \end{aligned} \tag{9.2}$$

In simulations of concrete splitting tests, the method produces tensile and frictional shear crack modes

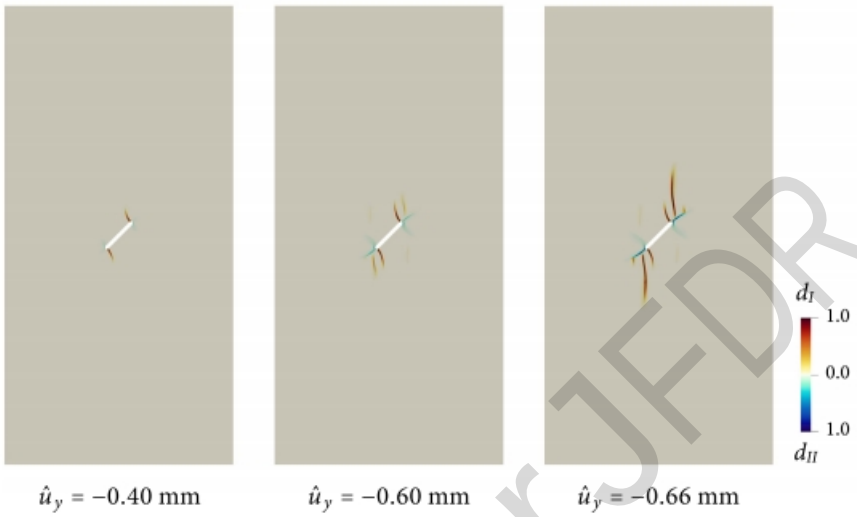


Figure 18. Concrete splitting test

9.2 Mixed-Mode Power-Law Phase-Field Model

Shen[11] proposed a mixed-mode power-law phase-field model to distinguish material fracture behavior with different Mode I and Mode II energy release rates. Starting from classical AT2, the phase-field equation is equivalently written as

$$lg'(d)\frac{\Psi^+(\varepsilon)}{G_c} + d - l^2\Delta d = 0 \tag{9.3}$$

Combining this with the power-law form of the mixed-mode fracture criterion

$$F = (\frac{G_I(\theta)}{G_I^c})^\alpha + (\frac{G_{II}(\theta)}{G_{II}^c})^\beta \tag{9.4}$$

Here, G_I and G_{II} are the Mode I and Mode II energy release rates, respectively, and α and β are the control parameters of the fracture criterion. By analogy with (9.3), the mixed-mode power-law phase-field formulation is

$$lg'(d)[(\frac{H_I}{G_I^c})^\alpha + (\frac{H_{II}}{G_{II}^c})^\beta] + d - l^2\Delta d = 0 \tag{9.5}$$

The history variables H_I and H_{II} correspond to tensile and shear strain energies, respectively, and can be obtained by volumetric-deviatoric decomposition or decomposition in principal strain space.

Simulations of single-edge-notched shear tests show that varying the ratio of Mode I to Mode II energy release rates changes the crack orientation, with a gradual transition from opening to sliding cracks.

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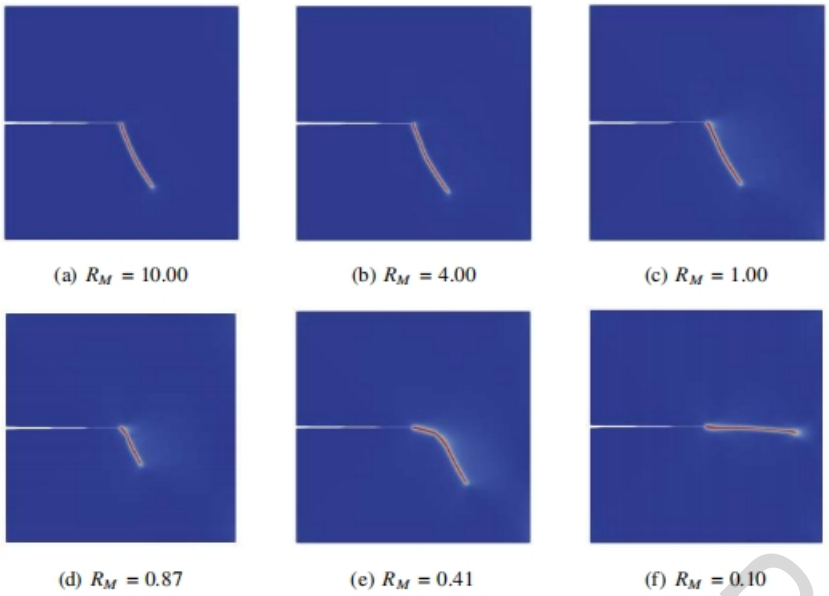


Figure 19. Shear test simulation

9.3 Mixed-Mode Phase-Field Fracture Model Satisfying General Damage Criteria

Ref. [33] is used as an example. It considers several concrete failure and damage criteria and presents a representative approach.

Consider the following quadratic damage criterion:

$$F(\zeta,\rho,\theta)=\left(A_f\rho\right)^2+m\left(B_f\rho r(\theta,e)+C_f\zeta\right)-1=0$$
$$\zeta=\frac{1}{\sqrt{3}}\bar{I}_1,\ \rho=\sqrt{2\bar{J}_2},\ \cos(3\theta)=\frac{3\sqrt{3}}{2}\frac{\bar{J}_3}{\sqrt{\bar{J}_2^3}} \tag{9.6}$$

Here, A_f, B_f, and C_f are coefficients containing the ultimate strength σ_c, while m and e are coefficients independent of the ultimate strength σ_c. Following Section 3.2, replacing the equivalent stress in the strain energy with the following form incorporates this quadratic damage criterion into the phase-field model:

$$\hat{\sigma}_{eq}=c_1\rho r(\theta,e)+c_2\zeta+\sqrt{\left(c_1\rho r(\theta,e)+c_2\zeta\right)^2+c_3\rho^2} \tag{9.7}$$

With this equivalent stress, the energy decomposition becomes:

$$\Psi=g(d)\Psi^{+}+\Psi^{-}=g(d)\left(\frac{\hat{\sigma}_{eq}^2}{2E}\right)+\left(\frac{1}{2}\boldsymbol{\varepsilon}:\mathbf{C}:\boldsymbol{\varepsilon}-\frac{\hat{\sigma}_{eq}^2}{2E}\right) \tag{9.8}$$

The tangent stiffness is then difficult to derive theoretically, so a hybrid phase-field formulation is generally used.

X. Dynamic Phase-Field Fracture

10.1 Implicit Dynamic Phase Field

The coupled phase-field model equations are:

$$\tilde{\sigma}_{ij,j}+f_i=\rho\ddot{u}_i,\ \text{ in }\Omega\times[0,T] \tag{10.1}$$
$$\frac{G_c}{c_0 l_0}[\alpha'(d)-2l_0^2\Delta d]=-g'(d)H \tag{10.2}$$

Boundary conditions:

$$u_i^0=\bar{u}_i^0,\qquad \text{on }\partial\Omega_u\cup C\times[0,T]$$
$$\tilde{\sigma}_{ij}^0 n_j=p_i,\qquad \text{on }\partial\Omega_p\times[0,T] \tag{10.3}$$
$$L_{ij}\frac{\partial d^0}{\partial x_i}n_j=0,\ \text{on }\partial\Omega\times[0,T]$$

Initial conditions:

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$$\mathbf{u}^0(\mathbf{x},0)=\mathbf{u}_0,\ \text{ in } \Omega$$
$$\dot{\mathbf{u}}^0(\mathbf{x},0)=\dot{\mathbf{u}}_0,\ \text{ in } \Omega \tag{10.4}$$
$$d^0(\mathbf{x},0)=d_0,\ \text{ in } \Omega$$

where

$$\alpha(d)=2d-d^2 \tag{10.5}$$

$$g(d)=\frac{(1-d)^2}{(1-d)^2+Q(d)} \tag{10.6}$$

$$Q(d)=a_1 d+a_1 a_2 d^2+a_1 a_2 a_3 d^2$$

Different choices of a1, a2, and a3 fit linear, exponential, and hyperbolic softening curves. (See *J.-Y. Wu, A unified phase-field theory for the mechanics of damage and quasi-brittle failure, J. Mech. Phys. Solids 103 (2017) 72–99.* for details.)

After finite element discretization, the residuals can be expressed as:

$$\mathbf{r}:=\mathbf{f}^{ext}-\int_\Omega\left(\mathbf{N}^u\right)^T\rho\mathbf{u}d\Omega-\int_\Omega\left(\mathbf{B}^\varepsilon\right)^T\tilde{\boldsymbol{\sigma}}^h d\Omega \tag{10.7}$$

$$\bar{\mathbf{r}}=-\int_\Omega\left\{\left(\bar{\mathbf{N}}^d\right)^T\left[g'\left(d^h\right)H+\frac{G_c}{c_0 l_0}\alpha'(d^h)\right]+G_c l\left(\bar{\mathbf{B}}^d\right)^T\nabla d^h\right\}d\Omega \tag{10.8}$$

The Newton-Raphson iteration scheme is:

$$\mathbf{M}_{uu}\delta\ddot{\mathbf{u}}+\mathbf{K}_{uu}\delta\mathbf{u}=\mathbf{r} \tag{10.9}$$
$$\mathbf{K}_{dd}\delta\mathbf{d}=\bar{\mathbf{r}} \tag{10.10}$$

where,

$$\mathbf{M}_{uu}=\int_\Omega\left(\mathbf{N}^u\right)^T\rho\mathbf{N}^u d\Omega$$
$$\mathbf{K}_{uu}=\int_\Omega\left(\mathbf{B}^\varepsilon\right)^T\frac{\partial\tilde{\boldsymbol{\sigma}}^h}{\partial\boldsymbol{\varepsilon}^h}\mathbf{B}^\varepsilon d\Omega\approx g(d)\int_\Omega\left(\mathbf{B}^\varepsilon\right)^T\mathbf{C}\mathbf{B}^\varepsilon d\Omega$$
$$\mathbf{f}^{ext}=\int_\Omega\left(\mathbf{N}^u\right)^T\mathbf{f}d\Omega+\int_{\partial\Omega_p}\left(\mathbf{N}^u\right)^T\mathbf{p}dS$$
$$\mathbf{K}_{dd}=\int_\Omega\left\{\left(\bar{\mathbf{N}}^d\right)^T\left[g''\left(d^h\right)H+\frac{G_c}{c_0 l_0}\alpha''(d^h)\right]\bar{\mathbf{N}}^d+\bar{G}_c l\left(\bar{\mathbf{B}}^d\right)^T\mathbf{L}\bar{\mathbf{B}}^d\right\}d\Omega \tag{10.11}$$

At time t_{n+1}, the equilibrium equation can be written using the HHT implicit dynamic algorithm as:

$$\bar{\mathbf{K}}_{uu}^{n+1}\delta\mathbf{u}^{n+1}=\tilde{\mathbf{r}}^{n+1} \tag{10.12}$$

$$\bar{\mathbf{K}}_{uu}^{n+1}=\frac{1}{\beta\Delta t^2}\mathbf{M}_{uu}^{n+1}+\frac{\gamma(1+\alpha)}{\beta\Delta t}\mathbf{C}_{uu}^{n+1}+(1+\alpha)\mathbf{K}_{uu}^{n+1} \tag{10.13}$$

$$\tilde{\mathbf{r}}^{n+1}=-\mathbf{M}_{uu}^{n+1}\mathbf{a}^n-(1+\alpha)\mathbf{C}_{uu}^{n+1}\mathbf{v}^n-(1+\alpha)\mathbf{K}_{uu}^{n+1}\mathbf{u}^n+\alpha\left(\mathbf{C}_{uu}^n\mathbf{v}^{n-1}+\mathbf{K}_{uu}^n\mathbf{u}^{n-1}\right) \tag{10.14}$$

The weakly coupled tangent stiffness dynamic equation is:

$$\begin{bmatrix}\bar{\mathbf{K}}_{uu}^{n+1} & \\ & \mathbf{K}_{dd}\end{bmatrix}\begin{bmatrix}\delta\mathbf{u}^{n+1}\\ \delta\mathbf{d}^{n+1}\end{bmatrix}=\begin{bmatrix}\tilde{\mathbf{r}}^{n+1}\\ \bar{\mathbf{r}}^{n+1}\end{bmatrix} \tag{10.15}$$

In the abaqus UEL subroutine, only the tangent stiffness matrix and residual need to be provided.

10.2 Explicit Dynamic Phase Field

To obtain an explicit phase-field equation, a viscous term is introduced:

$$\tilde{\sigma}_{ij,j}+f_i=\rho\ddot{u}_i,\ \text{ in } \Omega\times[0,T] \tag{10.16}$$

$$\eta\dot{d}+\frac{G_c}{c_0 l_0}[\alpha'(d)-2l_0^2\Delta d]=-g'(d)H \tag{10.17}$$

Finite element discretization gives:

$$\mathbf{M}_{uu}\ddot{\mathbf{u}}=\int_\Omega\left(\mathbf{N}^u\right)^T\mathbf{f}d\Omega+\int_{\partial\Omega_p}\left(\mathbf{N}^u\right)^T\mathbf{p}dS-\int_\Omega\left(\mathbf{B}^\varepsilon\right)^T\tilde{\boldsymbol{\sigma}}^h d\Omega \tag{10.18}$$

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$$\mathbf{C}_{dd}\dot{\mathbf{d}} = -\int_{\Omega}\left\{\left(\bar{\mathbf{N}}^{d}\right)^{T}\left[g'\left(d^{h}\right)H + \frac{G_c}{c_0 l_0}\alpha'(d^{h})\right] + G_c l\left(\bar{\mathbf{B}}^{d}\right)^{T}\nabla d^{h}\right\}d\Omega \tag{10.19}$$

where:

$$\mathbf{C}_{dd} = \int_{\Omega}\left(\bar{\mathbf{N}}^{d}\right)^{T}\eta\bar{\mathbf{N}}^{d}d\Omega \tag{10.20}$$

The central difference method is used for the equilibrium equation:

$$\dot{\mathbf{u}}^{n+1/2} = \dot{\mathbf{u}}^{n-1/2} + \frac{\left(\Delta t_n + \Delta t_{n+1}\right)}{2}\ddot{\mathbf{u}}^{n},\ \ \mathbf{u}^{n+1} = \mathbf{u}^{n} + \Delta t_{n+1}\dot{\mathbf{u}}^{n+1/2} \tag{10.21}$$

$$\Delta t_u = \frac{h_{\min}}{c_d},\ \ c_d = \sqrt{\frac{\lambda + 2\mu}{\rho}} \tag{10.22}$$

The forward difference method is used for the phase-field equation:

$$\mathbf{d}^{n+1} = \mathbf{d}^{n} + \Delta t_{n+1}\dot{\mathbf{d}}^{n} \tag{10.23}$$

$$\Delta t_d \approx \frac{h_{\min}^{2}}{2\alpha},\ \ \alpha = \frac{2G_c l_0}{c_0\eta} \tag{10.24}$$

In the abaqus VUEL subroutine, displacement is updated using central differences, requiring the mass matrix (M) and right-hand side. The phase field (temperature) is updated using forward differences, requiring the heat capacity matrix (C); a thermal conduction Jacobian matrix may also be provided to improve convergence. The corresponding displacement and temperature degrees of freedom are updated automatically. Both mass and heat capacity matrices are lumped and diagonalized.

The code uses a specific procedure to constrain the phase-field range. The explicit algorithm supports parallel computation.

For a thermomechanically coupled explicit phase-field fracture model, see: https://github.com/AlexanderJFDR/Thermo-Mechanic-PhaseField-Model.git.

For further examples, see: https://github.com/AlexanderJFDR/spall_phase_field_VUEL.git.

10.3 Considerations for Dynamic Phase-Field Fracture

(1) Inconsistent stress wave speeds

Studies of nonlinear fracture at large deformation show that degrading stiffness while keeping density unchanged in a phase-field model reduces the stress wave speed near the crack tip. This limits the conversion of sufficient strain energy into fracture energy, preventing sufficiently rapid crack propagation. Oscillatory instability of the crack path cannot then be observed in nonlinear dynamic fracture. By applying the same degradation to mass, researchers maintain the wave speed near the crack tip and successfully reproduce crack-tip instability[37].

(2) Widening of the phase-field region

Researchers[38] showed that, without mass degradation, additional elastic waves are continually generated in the damaged region, causing persistent fracture dissipation in the diffuse phase-field region and widening the phase field. They prevent continued growth of the damaged region by degrading mass to maintain a constant stress wave speed. Another group[39] analyzed rarefaction-wave dispersion and phase-field widening using wave impedance theory. They pointed out that mass degradation violates mass conservation and cannot ensure normal propagation of compressive waves through damaged regions. They therefore recommend a phase-field model without stiffness degradation, as introduced in Section 11.3.

(3) Abnormally high fracture energy (the Kalthoff-Winkler dynamic shear test)

The author first became interested in dynamic crack modeling after observing abnormally high fracture energies in phase-field simulations of the Kalthoff-Winkler dynamic shear test. Most studies use the same geometry and material parameters. At complete fracture, the crack length is approximately 80mm, giving a theoretical fracture energy of about 1800J. However, Borden[40] reported approximately 3500J, while Nguyen and Wu[41] predicted approximately 2460J. The author surveyed existing dynamic phase-field models for this example and found that their integrated fracture energies were all substantially greater than their geometric fracture energies.

The author's investigation identified the following factors causing abnormally high integrated fracture energy:

I. Mesh size (**): In finite elements, the phase-field maximum cannot be represented exactly as a straight line. Instead, a plateau of maxima formed by connected elements increases the integrated fracture

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energy.

II. Mass degradation (*****): Without simultaneous mass degradation in dynamic phase-field fracture simulations, the wave speed decreases in the damaged region, and changes in wave impedance cause fracture dissipation.

III. Irreversibility constraint (*): In some phase-field models, such as [8], the damage half-bandwidth decreases as the phase field increases. Irreversibility prevents the phase-field profile from recovering its theoretical topology, increasing the integrated fracture energy.

IV. Γ-convergence (*): As the phase-field width varies, the fracture energy Γ-converges to the sharp crack surface energy.

V. History variable (****): The history variable changes the phase-field distribution, preventing its profile from maintaining the minimum-energy topology.

The author experimentally verified these factors and ultimately obtained a geometric fracture energy of 1808J and an integrated fracture energy of 1968J, only 8.8% higher. The code is available at https://github.com/AlexanderJFDR/KW_fracture_PFM.git.

XI. Length-Scale-Insensitive Phase-Field Models[35,36]

In classical AT1 and AT2 models, phase-field width affects ultimate strength. Cohesive phase-field models decouple them by introducing a width-dependent coefficient into the degradation function. However, convergence requires a positive-definite iteration matrix, restricting the phase-field width to below a certain limit; see Section 5.7. When the model size is much larger than the characteristic phase-field width, maintaining accuracy requires a very fine mesh and an extremely large computational problem.

In essence, a phase-field model must represent the total strain before and after damage. Consider a one-dimensional phase-field damage problem:

$$\varepsilon = \varepsilon_e + \varepsilon_c \tag{11.1}$$

Here, $\varepsilon_e = \frac{\sigma}{E}$ is the elastic strain and ε_c is the eigenstrain induced by cracking. Two approaches are currently used:

(1) Reduce stiffness and increase compliance

Rewrite (11.1) as:

$$\varepsilon = \frac{\sigma}{E} + \frac{\sigma}{f(d)E} = \frac{1+f(d)}{f(d)E}\sigma = \frac{\sigma}{E'} \tag{11.2}$$

Here, E' is the degraded stiffness, and E' decreases as the phase field increases. Previous models (AT1, AT2, pf-czm), as well as the work in Sections 11.1 and 11.2 below, follow this approach.

(2) Introduce an additional independent eigenstrain

In this model, eigenstrain ε_c is introduced as an independent parameter, similarly to plastic strain. It satisfies the nonnegativity condition $\varepsilon_c \geq 0$ and represents the additional deformation caused by crack opening. The work in Section 11.3 follows this approach.

11.1 Classical Length-Scale-Insensitive Phase-Field Model

As discussed in Sections 3.2 and 3.3, phase-field width can be decoupled from ultimate strength by modifying the history variable as well as the degradation function. In addition to adding an energy threshold[34], an extra variable can modify the history variable. First proposed in Ref. [35], this method divides the system energy into four parts:

$$\Pi = \int_\Omega \Psi^e(\varepsilon,\omega)d\Omega + \int_\Omega \Psi^d(d)d\Omega + \int_\Omega \Psi^\omega(\omega,d)d\Omega - \int_\Omega \mathbf{b}\cdot\mathbf{u}d\Omega - \int_{\partial\Omega_t} \mathbf{t}\cdot\mathbf{u}dS \tag{11.3}$$

where:

$$\Psi^e = \Psi_0^e + \left(\frac{1}{\omega+1}-1\right)\bar{\Psi}_0^e = \frac{1}{2}\boldsymbol{\varepsilon}:\mathbf{C}^0:\boldsymbol{\varepsilon} + \left(\frac{1}{\omega+1}-1\right)\Psi_0^{e+} \tag{11.4}$$

$$\Psi^d = \frac{G_c}{2l_0}[d^2 + l_0^2\nabla d\cdot\nabla d] \tag{11.5}$$

$$\Psi^\omega = (1-d)^2\Psi_c\omega \tag{11.6}$$

Here, Ψ_0^{e+} is the positive part of the energy decomposition without degradation, and Ψ_c is the critical

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fracture energy, expressed in one dimension as $\Psi_c = \frac{\sigma_c^2}{2E}$.

The variational principle gives the evolution equations:

$$\nabla \bullet \boldsymbol{\sigma} + \mathbf{b} = 0$$

$$\boldsymbol{\sigma} = \frac{\partial \Psi^e}{\partial \boldsymbol{\varepsilon}} \tag{11.7}$$

$$\left[-2(1-d)\Psi_c \omega + \frac{G_c}{l_0}(d - l_0^2 \Delta d) \right] \dot{d} = 0, -2(1-d)\Psi_c \omega + \frac{G_c}{l_0}(d - l_0^2 \Delta d) \geq 0, \dot{d} \geq 0 \tag{11.8}$$

$$\left[-\frac{1}{(\omega+1)^2}\Psi_0^{e+} + (1-d)^2 \Psi_c \right] \dot{\omega} = 0, -\frac{1}{(\omega+1)^2}\Psi_0^{e+} + (1-d)^2 \Psi_c \geq 0, \dot{\omega} \geq 0 \tag{11.9}$$

In one dimension, the above equations give:

$$\omega = \begin{cases} \dfrac{\varepsilon - (1-d)\varepsilon_c}{(1-d)\varepsilon_c}, \varepsilon > \varepsilon_c \\ 0, \varepsilon \leq \varepsilon_c \end{cases} \tag{11.10}$$

$$\sigma = \frac{1}{\omega+1} E\varepsilon = \begin{cases} (1-d) E\varepsilon_c, \varepsilon > \varepsilon_c \\ E\varepsilon, \varepsilon \leq \varepsilon_c \end{cases} \tag{11.11}$$

Since the homogeneous form of the phase-field equation is:

$$(d-1) E\varepsilon_c^2 \omega + \frac{G_c}{l} d = 0 \tag{11.12}$$

the phase-field solution is:

$$d = \begin{cases} 1, \varepsilon \geq \dfrac{G_c}{E\varepsilon_c l_0} \\ \dfrac{E\varepsilon_c^2 - E\varepsilon_c \varepsilon}{E\varepsilon_c^2 - \dfrac{G_c}{l_0}}, \varepsilon_c < \varepsilon < \dfrac{G_c}{E\varepsilon_c l_0} \\ 0, \varepsilon \leq \varepsilon_c \end{cases} \tag{11.13}$$

The stress-strain relation is therefore:

$$\sigma = \begin{cases} 0, \varepsilon \geq \dfrac{G_c}{E\varepsilon_c l_0} \\ E\varepsilon_c \dfrac{\dfrac{G_c}{l_0} - E\varepsilon_c \varepsilon}{E\varepsilon_c^2 - \dfrac{G_c}{l_0}}, \varepsilon_c < \varepsilon < \dfrac{G_c}{E\varepsilon_c l_0} \\ E\varepsilon, \varepsilon \leq \varepsilon_c \end{cases} \tag{11.14}$$

This analysis shows that the model gives AT2 a crack initiation strain and a linear stress-strain relation. A problem remains, however: when $l_0 > \frac{G_c}{E\varepsilon_c^2}$, the stress-strain curve exhibits snap-back, so Newton-Raphson iteration cannot be used and an arc-length method is required. This is indeed the approach used in Ref. [34].

11.2 Improved Length-Scale-Insensitive Phase-Field Model

Ref. [36] improved this approach to accommodate different softening curves. The total energy of the system is:

$$\Pi = \int_\Omega \Psi^e(\varepsilon, \omega) d\Omega + \int_\Omega \Psi^d(d) d\Omega - \int_\Omega \mathbf{b} \bullet \mathbf{u} d\Omega - \int_{\partial\Omega_t} \mathbf{t} \bullet \mathbf{u} dS \tag{11.15}$$

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where:

$$\Psi^e = \frac{1}{2}\boldsymbol{\varepsilon}:\mathbf{C}^0:\boldsymbol{\varepsilon} - \left[1-g_1(\omega)\right]\frac{\langle\bar{\sigma}\rangle^2}{2E} - \left[1-g_2(\omega)\right]\frac{\bar{\tau}^2}{2G} \tag{11.16}$$

$$\Psi^d = \frac{G_c}{2l_0}[\Omega(d)^2 + l_0^2\nabla d\bullet\nabla d] + \frac{G_c}{l_0}(1-d)^2\omega \tag{11.17}$$

where:

$$g_1(\omega) = \frac{1}{1+c_1\omega}, g_2(\omega) = \frac{1}{1+c_2\omega} \tag{11.18}$$

$$c_1 = \frac{G_c}{l_0}\frac{2E}{\sigma_{cr}^2}, c_2 = \frac{G_c}{l_0}\frac{2G}{\tau_{cr}^2} \tag{11.19}$$

The variational principle gives the evolution equations:

$$\nabla\bullet\boldsymbol{\sigma} + \mathbf{b} = 0$$

$$\boldsymbol{\sigma} = \frac{\partial\Psi^e}{\partial\boldsymbol{\varepsilon}} \tag{11.20}$$

$$\left[-2(1-d)\frac{G_c}{l_0}\omega + \frac{G_c}{l_0}(\Omega(d)\Omega'(d) - l_0^2\Delta d)\right]\dot{d} = 0, -2(1-d)\frac{G_c}{l_0}\omega + \frac{G_c}{l_0}(\Omega(d)\Omega'(d) - l_0^2\Delta d) \geq 0, \dot{d} \geq 0 \tag{11.21}$$

$$\left[g_1'(\omega)\frac{\langle\bar{\sigma}\rangle^2}{2E} + g_2'(\omega)\frac{\bar{\tau}^2}{2G} + \frac{G_c}{l_0}(1-d)^2\right]\dot{\omega} = 0, g_1'(\omega)\frac{\langle\bar{\sigma}\rangle^2}{2E} + g_2'(\omega)\frac{\bar{\tau}^2}{2G} + \frac{G_c}{l_0}(1-d)^2 \geq 0, \dot{\omega} \geq 0 \tag{11.22}$$

In one dimension, the stress can be expressed as:

$$\sigma = g_1(\omega)E\varepsilon \tag{11.23}$$

and:

$$\begin{aligned} &g_1'(\omega)\frac{\bar{\sigma}^2}{2E} + \frac{G_c}{l_0}(1-d)^2 = g_1'(\omega)\frac{E\varepsilon^2}{2} + \frac{G_c}{l_0}(1-d)^2 \\ &= -c_1 g_1^2(\omega)\frac{E\varepsilon^2}{2} + \frac{G_c}{l_0}(1-d)^2 \geq 0 \end{aligned} \tag{11.24}$$

That is:

$$\left[\sigma - (1-d)\sigma_{cr}\right]\dot{\omega} = 0, \sigma - (1-d)\sigma_{cr} \geq 0, \dot{\omega} \geq 0 \tag{11.25}$$

The strain can be expressed as:

$$\varepsilon = (1+c_1\omega)\frac{\sigma}{E} = \frac{\sigma}{E} + \varepsilon^+, \varepsilon^+ = \frac{c_1\omega\sigma}{E} \tag{11.26}$$

Define the crack opening displacement of a one-dimensional bar of length L as:

$$\Delta = \int_0^L \varepsilon^+ dx \tag{11.27}$$

In the limit of phase-field width $l_0 \to 0$, the crack strain can be represented by a Dirac function

$$\varepsilon^+ = \Delta\delta(x - x^*) \tag{11.28}$$

We also have:

$$\omega = \frac{\hat{\Delta} l_0}{1-d^*}\delta(x-x^*), \hat{\Delta} = \frac{\Delta}{\Delta_{ch}}, \Delta_{ch} = \frac{2G_c}{\sigma_{cr}} \tag{11.29}$$

When $l_0 \to 0$, the phase-field equations are therefore equivalent to:

$$\begin{cases} -2l_0\hat{\Delta}\delta(x-x^*) + \Omega(d^*)\Omega'(d^*) - l_0^2 d_{,xx} = 0, x = x^* \\ \Omega(d)\Omega'(d) - l_0^2 d_{,xx} = 0, x \in [0,L]/\{x^*\} \end{cases} \tag{11.30}$$

Equation (2) is equivalent to:

$$\left[\Omega(d)^2 - l_0^2(d_{,x})^2\right]_{,x} = 0, x \in [0,L]/\{x^*\} \tag{11.31}$$

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Integrating over the intervals $\left[0,x^*\right)$ and $\left(x^*,L\right]$, respectively, and considering $d(0)=d(L)=d_{,x}(0)=d_{,x}(L)=0$, we obtain:

$$\Omega(d)^2-l_0^2\left(d_{,x}\right)^2=\Omega(0)^2,x\in[0,L]/\{x^*\} \tag{11.32}$$

Since $\lim_{x\to x^*}d(x)=d\left(x^*\right)>0$, and assuming $\Omega(d)>\Omega(0)$, we have:

$$\begin{cases} d_{,x}=\dfrac{1}{l_0}\sqrt{\Omega(d)^2-\Omega(0)^2},x\in\left[0,x^*\right) \\ d_{,x}=-\dfrac{1}{l_0}\sqrt{\Omega(d)^2-\Omega(0)^2},x\in\left(x^*,L\right] \end{cases} \tag{11.33}$$

Therefore:

$$\begin{cases} d_{,xx}=-\dfrac{2}{l_0}\sqrt{\Omega\left(d^*\right)^2-\Omega(0)^2}\delta\left(x-x^*\right),x=x^* \\ d_{,xx}=-\dfrac{2\Omega(d)\Omega'(d)}{l_0^2},x\in\left[0,x^*\right) \\ d_{,xx}=\dfrac{2\Omega(d)\Omega'(d)}{l_0^2},x\in\left(x^*,L\right] \end{cases} \tag{11.34}$$

When $d^*\neq1$, $\Omega\left(d^*\right)$ and $\Omega'\left(d^*\right)$ are bounded. Around x^*, substituting the above expression into the phase-field equation and integrating over the neighborhood $\left(x^*-\varepsilon,x^*+\varepsilon\right)$ gives:

$$\begin{aligned} &\lim_{\varepsilon\to0}\left[\int_{x^*-\varepsilon}^{x^*-\tau}\left(\Omega(d)\Omega'(d)-l_0^2d_{,xx}\right)dx+\int_{x^*+\tau}^{x^*+\varepsilon}\left(\Omega(d)\Omega'(d)-l_0^2d_{,xx}\right)dx\right. \\ &\left.+\int_{x^*-\tau}^{x^*+\tau}\left(-2l_0\hat{\Delta}\delta\left(x-x^*\right)+\Omega\left(d^*\right)\Omega'\left(d^*\right)-l_0^2d_{,xx}\right)dx\right] \\ &=\lim_{\tau\to0}\left[\int_{x^*-\tau}^{x^*+\tau}\left(-2l_0\hat{\Delta}\delta\left(x-x^*\right)+2l_0\sqrt{\Omega\left(d^*\right)^2-\Omega(0)^2}\delta\left(x-x^*\right)\right)dx\right] \\ &=-2l_0\hat{\Delta}+2l_0\sqrt{\Omega\left(d^*\right)^2-\Omega(0)^2}=0 \end{aligned} \tag{11.35}$$

Thus, the parametric equations of the softening law are:

$$\begin{cases} \sigma=\left(1-d^*\right)\sigma_{cr} \\ \Delta=\Delta_{ch}\sqrt{\Omega\left(d^*\right)^2-\Omega(0)^2} \end{cases} \tag{11.36}$$

This softening law is obtained for $l_0\to0$ and $d^*\neq1$, with $\Omega\left(d^*\right)$ and $\Omega'\left(d^*\right)$ bounded, unlike the derivation at finite phase-field width in Sections 3.3 and 3.4.

We next discuss the model's independence from phase-field width.

From the governing equation for ω, we obtain:

$$\omega=\frac{\left(\bar{\sigma}-\sigma\right)\sigma_{cr}l_0}{2EG_c\left(1-d\right)},\bar{\sigma}=E\varepsilon \tag{11.37}$$

Substituting into the phase-field equation gives:

$$-\frac{\left(\bar{\sigma}-\sigma\right)\sigma_{cr}}{E}+\frac{G_c}{l_0}\left[\Omega(d)\Omega'(d)-l_0^2\Delta d\right]=0 \tag{11.38}$$

At the phase-field maximum:

$$-\frac{\bar{\sigma}\sigma_{cr}}{E}+\frac{(1-d)\sigma_{cr}^2}{E}+\frac{G_c}{l_0}\Omega(d)\Omega'(d)=0 \tag{11.39}$$

At damage onset, $d=0$, and hence $\bar{\sigma}=\sigma=\sigma_{cr}$. The initiation stress is therefore independent of

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phase-field width. If the initiation stress is also the maximum stress, the latter is likewise independent of the phase field.

The stress-strain relation

$$\sigma = g_1(\omega) E\varepsilon \tag{11.40}$$

also ensures that snap-back does not occur.

11.3 Length-Scale-Insensitive Phase-Field Model with Eigenstrain

Ref. [42] introduces eigenstrain $\boldsymbol{\varepsilon}_c$ to represent the additional displacement caused by crack opening. The total system energy is:

$$\Pi = \int_\Omega \Psi^e(\varepsilon, \varepsilon_c) d\Omega + \int_\Omega \Psi^c(\varepsilon_c, d) d\Omega + \int_\Omega \Psi^d(d) d\Omega - \int_\Omega \mathbf{b}\bullet\mathbf{u} d\Omega - \int_{\partial\Omega_t} \mathbf{t}\bullet\mathbf{u} dS \tag{11.41}$$

where:

$$\Psi^e = \frac{1}{2}(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_c) : \mathbf{C}^0 : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_c) \tag{11.42}$$

$$\Psi^d = \frac{G_c}{2l_0}[d^2 + l_0^2 \nabla d \bullet \nabla d] \tag{11.43}$$

$$\Psi^c = (1-d)^2 \boldsymbol{\sigma}_0 : \boldsymbol{\varepsilon}_c \tag{11.44}$$

Let $\{\mathbf{u}^*, \boldsymbol{\varepsilon}_c^*, d^*\}$ be a solution satisfying first-order stability. It then satisfies the sufficient condition:

$$\delta_{\mathbf{u}}\Pi = 0, \delta_{\boldsymbol{\varepsilon}_c}\Pi = 0, \delta_d \Pi = 0 \tag{11.45}$$

That is

$$\mathbf{u}^* = Arg \min_{\mathbf{u}} \{\Pi\} = Arg \min_{\mathbf{u}} \{\Psi^e\} \tag{11.46}$$

$$\boldsymbol{\varepsilon}_c^* = Arg \min_{\boldsymbol{\varepsilon}_c \geq 0} \{\Pi\} = Arg \min_{\boldsymbol{\varepsilon}_c} \{\Psi^e + \Psi^c\} \tag{11.47}$$

$$d^* = Arg \min_{\dot{d} \geq 0} \{\Pi\} = Arg \min_{d} \{\Psi^c + \Psi^d\} \tag{11.48}$$

The equilibrium equation is obtained as:

$$\begin{gathered} \nabla \bullet \boldsymbol{\sigma} + \mathbf{b} = 0 \\ \boldsymbol{\sigma} = \frac{\partial \Psi^e}{\partial \boldsymbol{\varepsilon}} = \mathbf{C}^0 : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_c) \end{gathered} \tag{11.49}$$

The KKT conditions give the eigenstrain evolution equation:

$$\boldsymbol{\varepsilon}_c \geq 0,\ \boldsymbol{\lambda} \geq 0,\ \boldsymbol{\lambda} : \boldsymbol{\varepsilon}_c = 0,\ \frac{\partial \Psi^e}{\partial \boldsymbol{\varepsilon}_c} + \frac{\partial \Psi^c}{\partial \boldsymbol{\varepsilon}_c} - \boldsymbol{\lambda} = 0 \tag{11.50}$$

That is:

$$\boldsymbol{\varepsilon}_c \geq 0,\ (1-d)^2 \boldsymbol{\sigma}_0 - \mathbf{C}^0 : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_c) \geq 0,\ \left[(1-d)^2 \boldsymbol{\sigma}_0 - \mathbf{C}^0 : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_c)\right] : \boldsymbol{\varepsilon}_c = 0 \tag{11.51}$$

The KKT conditions give the phase-field evolution equation:

$$\left[-2(1-d)\boldsymbol{\sigma}_0 : \boldsymbol{\varepsilon}_c + \frac{G_c}{l_0}(d - l_0^2 \Delta d)\right] \dot{d} = 0, -2(1-d)\boldsymbol{\sigma}_0 : \boldsymbol{\varepsilon}_c + \frac{G_c}{l_0}(d - l_0^2 \Delta d) \geq 0, \dot{d} \geq 0 \tag{11.52}$$

For a one-dimensional bar in tension, with length L and relative end displacement u:

$$\varepsilon = \frac{u}{L},\ \sigma = E(\varepsilon - \varepsilon_c) \tag{11.53}$$

$$\varepsilon_c = \begin{cases} 0, & \varepsilon \leq \dfrac{\sigma_0}{E} \\ \varepsilon - (1-d)^2 \dfrac{\sigma_0^2}{E}, & \varepsilon > \dfrac{\sigma_0}{E} \end{cases} \tag{11.54}$$

Phase-field evolution begins when $\varepsilon_c > 0$. The crack initiation strength is then σ_0, independent of phase-field width. Furthermore, when the phase field equals 1, $\varepsilon_c = \varepsilon$, $\sigma = 0$, meaning that the bar cannot carry any load.

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By defining $\boldsymbol{\sigma}_0$, strength surfaces of arbitrary shape can be obtained.

XII. Phase-Field Fracture Models with Interfaces

XIII. Phase-Field Models for Fatigue Fracture

XIV. Fourth-Order Phase-Field Models[43,44]

Fourth-order phase-field models introduce higher-order boundary conditions and require C1-continuous elements for direct solution. They are therefore generally solved using C1 finite elements, isogeometric methods, or mixed order-reduction methods. Despite these difficulties, fourth-order phase-field equations can represent strong fracture anisotropy.

14.1 Governing Equations and Boundary Conditions of Fourth-Order Phase-Field Models

The main advantage of fourth-order over second-order phase-field models is their ability to represent strong anisotropy. In general, the fourth-order phase-field fracture energy density is written as follows[43,44] (note that $\Delta d = d_{,ij}$ is not the Laplacian operator):

$$\gamma(d,\nabla d,\Delta d)=\frac{1}{c_0 l_0}[\alpha(d)+l_0^2\nabla d\bullet\mathbf{L}\bullet\nabla d+l_0^4\Delta d:\mathbf{A}:\Delta d] \tag{14.1}$$

Since the higher-order boundary conditions in [43,44] are overly restrictive, the author provides a detailed derivation here (assuming that tensors $\mathbf{L}$ and $\mathbf{A}$ are symmetric):

$$\delta_d\int_\Omega\gamma(d,\nabla d,\Delta d)d\Omega=\int_\Omega\frac{1}{c_0 l_0}[\alpha'(d)\delta d+2l_0^2\nabla d\bullet\mathbf{L}\bullet\nabla\delta d+2l_0^4\Delta d:\mathbf{A}:\Delta\delta d]d\Omega \tag{14.2}$$

We also have:

$$\begin{aligned}\int_\Omega\frac{2l_0}{c_0}\nabla d\bullet\mathbf{L}\bullet\delta\nabla d d\Omega&=\int_\Omega\frac{2l_0}{c_0}\nabla\bullet\left(\nabla d\bullet\mathbf{L}\delta d\right)d\Omega-\int_\Omega\frac{2l_0}{c_0}\nabla\bullet\left(\nabla d\bullet\mathbf{L}\right)\delta d d\Omega\\&=\int_{\partial\Omega}\frac{2l_0}{c_0}\left(\nabla d\bullet\mathbf{L}\delta d\right)\bullet\mathbf{n}dS-\int_\Omega\frac{2l_0}{c_0}\nabla\bullet\left(\mathbf{L}\bullet\nabla d\right)\delta d d\Omega\end{aligned} \tag{14.3}$$

$$\begin{aligned}\int_\Omega\frac{2l_0^3}{c_0}\Delta d:\mathbf{A}:\delta\nabla d d\Omega&=\int_\Omega\frac{2l_0^3}{c_0}\nabla\bullet\left(\Delta d:\mathbf{A}\bullet\delta\nabla d\right)d\Omega-\int_\Omega\frac{2l_0^3}{c_0}\nabla\bullet\left(\Delta d:\mathbf{A}\right)\bullet\delta\nabla d d\Omega\\&=\int_{\partial\Omega}\frac{2l_0^3}{c_0}\left(\Delta d:\mathbf{A}\bullet\delta\nabla d\right)\bullet\mathbf{n}dS-\int_\Omega\frac{2l_0^3}{c_0}\nabla\bullet\left(\Delta d:\mathbf{A}\right)\bullet\delta\nabla d d\Omega\\&=\int_{\partial\Omega}\frac{2l_0^3}{c_0}\left(\Delta d:\mathbf{A}\bullet\delta\nabla d\right)\bullet\mathbf{n}dS-\int_\Omega\frac{2l_0^3}{c_0}\nabla\bullet\left[\nabla\bullet\left(\Delta d:\mathbf{A}\right)\delta d\right]d\Omega+\int_\Omega\frac{2l_0^3}{c_0}\nabla\bullet\left[\nabla\bullet\left(\Delta d:\mathbf{A}\right)\right]\delta d d\Omega\\&=\int_{\partial\Omega}\frac{2l_0^3}{c_0}\left(\Delta d:\mathbf{A}\bullet\delta\nabla d\right)\bullet\mathbf{n}dS-\int_{\partial\Omega}\frac{2l_0^3}{c_0}\nabla\bullet\left(\Delta d:\mathbf{A}\right)\bullet\mathbf{n}\delta d dS+\int_\Omega\frac{2l_0^3}{c_0}\nabla\bullet\left[\nabla\bullet\left(\Delta d:\mathbf{A}\right)\right]\delta d d\Omega\end{aligned} \tag{14.4}$$

The first integral still depends on $\delta\nabla d$. Further integration by parts requires restricting the variation to the boundary $\partial\Omega$, so we decompose $\delta\nabla d$ into its normal and tangential components with respect to $\partial\Omega$:

$$\delta d_{,j}=Q^j\delta d+n_j D\delta d \tag{14.5}$$

Here, $Q^j=\left(\delta_{jk}-n_j n_k\right)\partial_k$, $D=n_k\partial_k$. Then:

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$$\int_{\partial\Omega}\frac{2l_0^3}{c_0}\left(\Delta d:\mathbf{A}\bullet\delta\nabla d\right)\bullet\mathbf{n}dS=\int_{\partial\Omega}\frac{2l_0^3}{c_0}\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\bullet\left(Q\delta d\right)dS+\int_{\partial\Omega}\frac{2l_0^3}{c_0}\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\bullet\mathbf{n}\left(D\delta d\right)dS \tag{14.6}$$

Letting $\mathbf{H}=\Delta d:\mathbf{A}$, we have:

$$\begin{aligned}&\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\bullet\left(Q\delta d\right)=H_{kl}n_lQ^k\delta d\\&=\left(Q^rn_r\right)n_kH_{kl}\delta d-Q^k\left(H_{kl}n_l\right)\delta d+n_qe_{qpr}\partial_p\left(e_{rsk}n_sn_lH_{kl}\delta d\right)\end{aligned} \tag{14.7}$$

By Stokes' theorem:

$$\begin{aligned}&\int_{\partial\Omega}\frac{2l_0^3}{c_0}n_qe_{qpr}\partial_p\left(e_{rsk}n_sn_lH_{kl}\delta d\right)dS=\oint_C\frac{2l_0^3}{c_0}\left(v_kH_{kl}n_l\right)\delta ddl\\&=\oint_C\frac{2l_0^3}{c_0}\left(\mathbf{v}\bullet\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\delta ddl\end{aligned} \tag{14.8}$$

Here, $v_k=e_{krs}b_rn_s$, and b_r is the unit tangent vector along edge C.

Therefore:

$$\begin{aligned}&\int_{\partial\Omega}\frac{2l_0^3}{c_0}\left(\Delta d:\mathbf{A}\bullet\delta\nabla d\right)\bullet\mathbf{n}dS=\int_{\partial\Omega}\frac{2l_0^3}{c_0}\left[\left(Q\mathbf{n}\right)\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\bullet\mathbf{n}-Q\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\right]\delta ddS\\&+\oint_C\frac{2l_0^3}{c_0}\left(\mathbf{v}\bullet\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\delta ddl+\int_{\partial\Omega}\frac{2l_0^3}{c_0}\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\bullet\mathbf{n}\left(D\delta d\right)dS\end{aligned} \tag{14.9}$$

Then:

$$\begin{aligned}&\int_\Omega\frac{2l_0^3}{c_0}\Delta d:\mathbf{A}:\delta\nabla dd\Omega=\int_{\partial\Omega}\frac{2l_0^3}{c_0}\left[\left(Q\mathbf{n}\right)\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\bullet\mathbf{n}-Q\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)-\nabla\bullet\left(\Delta d:\mathbf{A}\right)\bullet\mathbf{n}\right]\delta ddS\\&+\oint_C\frac{2l_0^3}{c_0}\left(\mathbf{v}\bullet\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\delta ddl+\int_{\partial\Omega}\frac{2l_0^3}{c_0}\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\bullet\mathbf{n}\left(D\delta d\right)dS+\int_\Omega\frac{2l_0^3}{c_0}\nabla\bullet\left[\nabla\bullet\left(\Delta d:\mathbf{A}\right)\right]\delta dd\Omega\end{aligned} \tag{14.10}$$

Then:

$$\begin{aligned}&\delta_d\int_\Omega\gamma(d,\nabla d,\Delta d)d\Omega=\int_\Omega\frac{1}{c_0l_0}[\alpha'(d)\delta d+2l_0^2\nabla d\bullet\mathbf{L}\bullet\nabla\delta d+2l_0^4\Delta d:\mathbf{A}:\Delta\delta d]d\Omega\\&=\int_\Omega\left\{\frac{1}{c_0l_0}\alpha'(d)-\frac{2l_0}{c_0}\nabla\bullet\left(\mathbf{L}\bullet\nabla d\right)+\frac{2l_0^3}{c_0}\nabla\bullet\left[\nabla\bullet\left(\Delta d:\mathbf{A}\right)\right]\right\}\delta dd\Omega\\&+\int_{\partial\Omega}\left\{\frac{2l_0}{c_0}\left(\nabla d\bullet\mathbf{L}\delta d\right)\bullet\mathbf{n}+\frac{2l_0^3}{c_0}\left[\left(Q\mathbf{n}\right)\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\bullet\mathbf{n}-Q\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)-\nabla\bullet\left(\Delta d:\mathbf{A}\right)\bullet\mathbf{n}\right]\right\}\delta ddS\\&+\int_{\partial\Omega}\frac{2l_0^3}{c_0}\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\bullet\mathbf{n}\left(D\delta d\right)dS+\oint_C\frac{2l_0^3}{c_0}\left(\mathbf{v}\bullet\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\delta ddl\end{aligned} \tag{14.11}$$

Using the boundary condition $\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\bullet\mathbf{n}=0$, we have

$$\left(Q\mathbf{n}\right)\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\bullet\mathbf{n}-Q\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)-\nabla\bullet\left(\Delta d:\mathbf{A}\right)\bullet\mathbf{n}=-Q\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)-\nabla\bullet\left(\Delta d:\mathbf{A}\right)\bullet\mathbf{n} \tag{14.12}$$

The phase-field equation is therefore:

$$\begin{aligned}&\frac{G_c}{c_0l_0}\left[\alpha'(d)-2l_0^2\nabla\bullet\left(\mathbf{L}\bullet\nabla d\right)+2l_0^4\nabla\bullet\left[\nabla\bullet\left(\mathbf{A}:\Delta d\right)\right]\right]=-g'(d)\Psi(\varepsilon),\ \text{in }\Omega\\&\mathbf{L}\bullet\nabla d\bullet\mathbf{n}-l_0^2\left[Q\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)+\nabla\bullet\left(\Delta d:\mathbf{A}\right)\bullet\mathbf{n}\right]=0,\ \text{on }\partial\Omega\\&\left(\Delta d:\mathbf{A}\bullet\mathbf{n}\right)\bullet\mathbf{n}=0,\ \text{on }\partial\Omega\\&\mathbf{v}\bullet\Delta d:\mathbf{A}\bullet\mathbf{n}=0,\ \text{on }C\end{aligned} \tag{14.13}$$

Various simplifications of the boundary conditions are discussed below. If $\Delta d:\mathbf{A}\bullet\mathbf{n}=\mathbf{0}$, they simplify to:

$$\begin{aligned}&\mathbf{L}\bullet\nabla d\bullet\mathbf{n}-l_0^2\nabla\bullet\left(\Delta d:\mathbf{A}\right)\bullet\mathbf{n}=0,\ \text{on }\partial\Omega\\&\Delta d:\mathbf{A}\bullet\mathbf{n}=\mathbf{0},\ \text{on }\partial\Omega\end{aligned} \tag{14.14}$$

These are the boundary conditions given in Refs. [43,44].

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For an isotropic fourth-order phase-field model, $L_{ij} = \delta_{ij}$, $A_{ijkl} = \delta_{ij}\delta_{kl}$, and the boundary conditions simplify to:

$$\begin{aligned} d_{,i} n_i - l_0^2 d_{,iil} n_l &= 0, \text{ on } \partial\Omega \\ d_{,ii} &= \mathbf{0}, \text{ on } \partial\Omega \end{aligned} \tag{14.15}$$

These are the boundary conditions given in Ref. [45].

14.2 Strong Anisotropy

The directional fracture toughness of an anisotropic fourth-order phase-field model is derived similarly to Section 8.4, so the details are omitted here; see Refs. [43,44]. Briefly, its fourth-order structural tensor allows a rotational period of $\pi/2$, producing up to four fracture toughness minima.

Figure 20 shows strong and weak anisotropy. A second-order anisotropic phase-field model with a structural tensor can represent only weak anisotropy, whereas a fourth-order model can represent strong anisotropy. Because the strongly anisotropic surface is nonconvex, there are directions in which cracks cannot propagate, allowing a class of strongly anisotropic fracture materials to be represented[46].

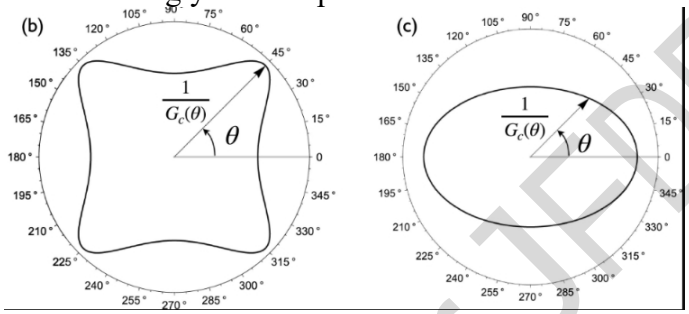


Figure 20. Strong anisotropy (left); weak anisotropy (right)[44]

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